

Improving the Decision Support in Shop Floor Operations by Using Agent-based Systems and Visibility Frameworks

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GLOSSARY

Agent. The agents are pieces of software that includes autonomy skills, social abilities and reasoning capabilities.

Artificial intelligence (AI). The science and engineering of making intelligent machines (McCarthy 1987).

Alignment. The IT alignment exists when the information system and its DSS is in concordance with business organisation's goals.

Computer Integrated Manufacturing (CIM). This is the approach of using computer to control the manufacturing/production process.

Cooperative Distributed Problem Solving (CDPS). A system based on a network of semi-autonomous processing nodes that working together to solve a problem, which typically is agent-based system.

Customer Relationship Management (CRM). This strategy is widely implemented for managing a company's interactions with customers, clients and sales prospects.

Electronic Product Code (EPC). The Electronic Product Code is the Universal identifier that provides a unique identity for every physical object anywhere in the world.

Electronic Product Code Information Service (EPCIS). The goal of EPCIS is to enable disparate applications to leverage EPC data via EPC-related data sharing, both within and across enterprises.

Decision Support Systems (DSS). The Decision Support Systems are computer-based information systems that support business and organizational decision-making activities.

Hardware Abstraction Layer (HAL). This abstract layer allows programmers to write device-independent software by avoiding hardware details.

Internet of Things (IoT). The Internet of Things concept refers to uniquely identifiable objects (Things) and their virtual representations in an Internet-like structure.

Information Systems (IS). The information systems are implemented within an organization for improving the effectiveness and efficiency by using data coming from work systems, people, developments and implementation methodologies.

Key Performance Indicator (KPI). The companies evaluate their success in particular activities during operations by using the KPIs.

Mashup. A mashup is a Web page or application that uses and combines data, presentation or functionality from two or more sources to create new services.

Multiagent System (MAS). A decentralized software system composed of multiple agents, which have a local view of the problem or system, and interaction protocols to generate negotiated proposals.

Multiagent Platform. An agent-based platform is a set of agents in which the agents share facilities as director facilitator and management tools.

Programmable Logic Controller (PLC). The PLC is a digital computer used for automation of electromechanical processes, such as control of machinery.

Product-Driven Approach. This negotiation strategy allows products to manage their destiny during the production/manufacturing.

Radio Frequency Identification (RFID). The Radio Frequency Identification technology uses communication via radio waves to exchange data between a reader and an electronic tag attached to an object.

Representational state transfer (REST). This is a style of architecture that consists in a series of principles between clients and servers based on resources and representations over HTTP.

Return of Investment (ROI). This financial ratio, which is also known as rate of profit, defines the money gained or lost relative to an initial investment.

Markov Decision Process (MDP). MDP is a mathematical framework for modelling decision-making in situations where outcomes are partly random and partly under the control of a decision maker.

Lean Manufacturing. This production practice only preserves the creation of customer value, this ideas initially coming from Toyota.

Mass Customization. This new management strategy is based on the use of technology in the search of systems able to produce custom outputs.

Service Oriented Architecture (SOA). This architecture defines interoperable services that can be used within multiple and separate systems from several business domains.

Small and medium enterprises (SMEs). This definition categorizes companies with fewer than 10 employees as "micro", those with fewer than 50 employees as "small", and those with fewer than 250 as "medium" (Commission 2005).

Software as a Service (SaaS). Sometimes referred to as "on-demand software," is a software delivery model in which software and its associated data are hosted centrally.

Visibility Frameworks. It is an abstraction tool, whose common code providing tracking functionality to others by using public interfaces, which use hardware identification technologies as RFID or bar code.

Warehouse Management System (WMS). This part of the production control is the generation of instruction related with the movement and storage of materials in the warehouse.

ABSTRACT

Recent research on highly distributed control methods has produced a series of new decision-making philosophies based on negotiation, which bring together process engineering with computer science. Among these philosophies, the ones based on MAS have become especially relevant to address distributed decision-making in manufacturing/logistics environments. These agent-based approaches allow including new manufacturing trends in the decision processes because of their scalability, ability to negotiate and distributed capabilities. These new trends are emerging to achieve the current new manufacturing/logistics strategies; some outstanding examples are: the “Common Use Model”, the “Enterprise Integration”, the “Agile Manufacturing”, the “Environmental Alignment” or the “Internet of Things”.

However, these MAS models have the drawback of an excessive dependence on up-to-date field information and on complex interaction protocols. Therefore, this thesis proposes a theoretical and experimental agent-based DSS architecture that is designed and developed to include

these new trends in shop floor operations, but including the RFID information feedback. Based on these automatic product feedbacks generated by the RFID visibility frameworks, the proposed MAS allows defining a competitive space where intelligent products negotiate by using their own knowledge and global/business constraints. Specifically, this product-driven MAS has been structured on a split organization model to enforce the idea of division between physical elements and “Information and Communication Technologies” (ICT). This division into two platforms simplifies the design, the development and the validation of the MAS in shop floor environments, providing a higher level of abstraction and preserving the independence between platforms.

The proposed MAS framework, called MAS-DUO, has been tested in the ground handling operations at the Ciudad Real Central Airport and in a simulated logistics center at Autolog Labs.

Keywords:

Agent-based Systems, Shop Floor Operations, DSS, Identification Technologies, Intelligent Product

ABSTRACT (CASTELLANO)

La investigación sobre los métodos de control distribuido en entornos industriales ha producido una serie de nuevas filosofías basadas en la negociación, que combinan la ingeniería de procesos con la informática. Entre estas filosofías de control destacan las tecnologías basadas en Sistemas MultiAgente (SMA), llegando a ser especialmente interesantes en sistemas de decisión para fabricación y logística. Estos sistemas basados en agentes permiten incluir nuevas tendencias de gestión para la fabricación en los problemas de planificación gracias a su escalabilidad, su capacidad de negociación y su diseño dirigido a sistemas distribuidos. Estos nuevos conceptos/tendencias están surgiendo para cubrir las actuales demandas en fabricación y logística; algunos ejemplos son: el “Modelo de Uso Común”, la “La Integración Empresarial”, la “Fabricación Ágil”, el “Alineamiento con el Entorno” o la “Internet de las Cosas”.

Sin embargo, estos modelos basados en SMA tienen la desventaja de una dependencia excesiva de la información real del entorno y en la definición

de complejos protocolos de negociación. En este trabajo, una arquitectura teórica y experimental para el soporte a la toma decisiones basada en un SMA es diseñada y desarrollada, en la que se pueden incluir nuevas estrategias para las operaciones de planta basadas en RFID. Usando estas nuevas herramientas de visibilidad como RFID, la información generada por el entorno permite un espacio competitivo donde los productos inteligentes negocian usando su propio conocimiento y restricciones globales. Estas nuevas estrategias dirigidas por productos y aplicadas a la planificación han sido estructuradas en un nuevo modelo de organización que mantiene la división entre los recursos físicos de las planta y los sistemas de información. Esta división en dos plataformas de agentes simplifica el diseño, el desarrollo y la validación del SMA, dando un mayor nivel de abstracción y manteniendo la independencia de cada plataforma.

La arquitectura propuesta, llamada MAS-DUO, ha sido analizada en las operaciones en rampa del Aeropuerto Ciudad Real Central y en el modelo de centro logístico desarrollado en las instalaciones de Autolog en la UCLM.

Keywords:

Sistemas Multiagente, Operaciones en planta, Soporte a la toma de decisiones, Producto inteligente, RFID

1 INTRODUCTION

1.1 Motivation

The disaggregation and specialization of customers have changed the way of work in many manufacturing companies, logistics centers and even transport facilities. The current high granularity of customers reflects a product customisation, an increase of the product references and a short product life cycle; together with new cost reduction strategies such as reduction of stocks, the e-commerce and the outsourcing has become a management challenge. Apart from the new environmental sustainability policies, in which the energy efficiency protocols are required. All of them are increasing the complexity of Supply Chain Management (SCM), especially at the shop floor level. The current situation has forced the companies to redefine their business models into complex distributed processes, which will define their future positions in the market.

This complexity becomes especially relevant when dealing with typically big plants or distribution centers where the current control is rigid and hard to manage, maintain and update. Traditional manufacturing/logistics control systems, in spite of having steadily increased in reliability and

significantly reduced production costs, still have the drawback of their reduced flexibility. Initially, it seemed to be good solution but in fact, these systems are not prepared for the present new complexity, which seriously affects their performance under perturbations and new requirements. Traditional hierarchical control systems however, are slow to adapt to these new production requirements and dynamic sequences (Bansiya and Davis 2002). Bussman outlined that the current basis of CIM is inflexible, fragile, and difficult to maintain; CIM systems have a fixed control hierarchy that does not support changes, reconfigurations and extensions (Bussman, Jennings et al. 2004). Most of the current manufacturing/logistics systems are not maintained outside normal operation conditions, the machine data for diagnosis is difficult to access and the control is completely automated as a whole system.

Now, instead of designing monolithic control systems that need to be specifically programmed to solve every single problem they can encounter, new proposals are dividing the control into autonomous functional subsystems with the possibility of interacting between them by using negotiation methods (Zlotkin and Rosenschein 1996). For a modern manufacturing/logistics system however, while finding an adequate response to a broad range of possible new situations business must be as usual; and this should apply even while the system is being modified or updated. Thus, the flexible control becomes standard practice as it allows a division of the problem into smaller and more feasible ones. More so, if these sub-modules are designed with enough autonomy to manage their own tasks while maintaining efficient interacting and exchanging information capabilities (Chan and Chan 2009). The manufacturing control technologies have to be adapted to

these feasible sub-modules, in which the information is automatically accessible by using flexible service-oriented interfaces.

The aim to achieve manufacturing flexibility directly implies the inherent capacity of the required “Mass Customization”, which combines the contrasting elements of mass production and customization. On one hand, “Mass Production” implies cost reduction due to the scale effects and the gained production experience. On the other hand, customization focuses on the individual customer’s requirements, which results in an effective competitive position. “Mass Customization” therefore aims at manufacturing products to meet individual customers’ needs with mass production efficiency. Thus, the customized products are offered at prices comparable to standard products and continuous individual relationships are established between the customers and the manufacturers. One well-known scenario of “Mass Customization” is the pallet picking in distribution centers; the customers demand a specific combination of products for each pallet instead of having complete pallets of a specific product. Therefore, it makes it necessary to refine the control methodology in the way of build mix pallets. The research community remarks advances in information technology as well as new management concepts to be able to achieve this hybrid competitive strategy of “Mass Customization” as usual (Garcia, McFarlane et al. 2003).

In order to achieve Mass Customization in a “business as usual” context, the definition of new control strategies at several levels is required, especially at the shop floor operation level. The implementation of some strategies such as “Just-in-time” (JIT), “Agile Manufacturing”, “Environmental Alignment”, “Common Use Model” or “Enterprise Integration” (Morel, Valckenaers et al. 2007) allow deploying a scenario

able to address this challenge, which will define new accurate and flexible business models (Figure 1). The “Agile Manufacturing” strategy, which is linked to JIT from a manufacturing point of view, reduces inventories and increases the accuracy of internal processes as a strategy to improve productivity (Ramesh and Devadasan 2007). Then, strategies like the “Common Use Model”, which comes from airport management (Bellioti 2008), defines the enterprise space as a shared environment where companies share facilities, equipment and even human resources, leading to a direct cost reduction. The “Environmental Alignment” concept takes into consideration the external disturbances during operations, and assumes that the information system has to integrate features for assessing environmental deviations and disturbances (Avila, Goepp et al. 2009). The “Enterprise Integration” strategy is a new paradigm to organize humans and machines as a whole system, not only at the operation level, but also at the management and corporate levels (Panetto and Molina 2008). Apart from strategies, there are other surrounding concepts that include a set of visions that can be reused in the problem at hand, such as the “Internet of Things” (Atzori, Iera et al. 2010), the “Smart logistics” (Uckelmann 2008) or the “Visibility Frameworks” (Thiesse, Floerkemeier et al. 2009). All of these techniques involve the definition of new rules and protocols to work in shared manufacturing/logistics environments.

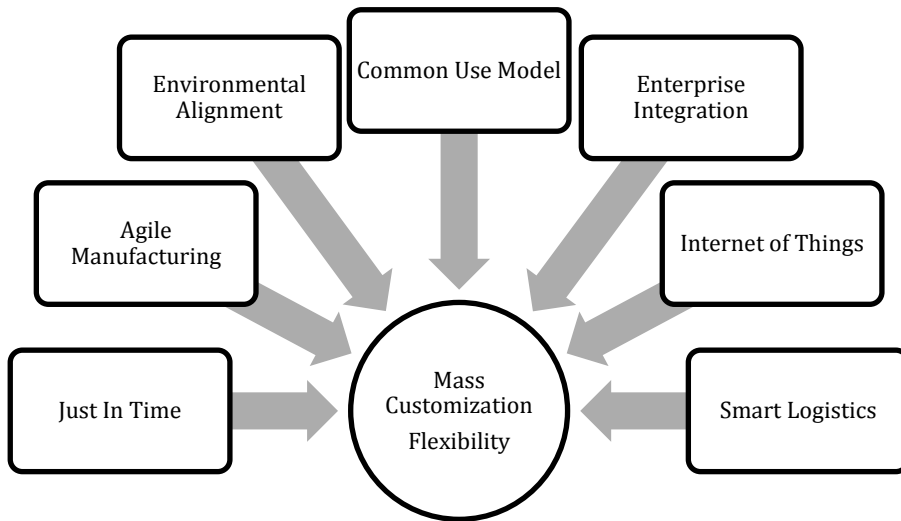


Figure 1 New concepts in order to achieve flexibility and “Mass Customization”

These concepts/visions directly affect the shop floor way of work; the shop floor operations involve the sequencing, the control and the optimization of operations required to assist products or processes at their specific situation in the plant level. The sequencing of tasks for medium and long term scheduling, statically determines the requirements and tasks, which affects the dimensioning of both material and human resources. Then, these resources take short-term planning and even online decisions whenever deviations or disturbances from plans arise (Schönberger and Kopfer 2007), which are managed through an online scheduling control but with a lack of strategy control (García, de las Morenas et al. 2011). This short-term assignment of tasks is made attending to review, control, planning, and information updating with 24-hour-data availability rather than to mathematical models. Nevertheless, this constant information flow is directly affected by the demanded flexibility and subjects to the reality of a stochastic industrial

environment, both problems lead to an environmental uncertainty (Fynes, de Búrca et al. 2004). The flexibility and uncertainty at the shop floor require a constant renegotiation that has usually not been duly parameterized. As an example, airport ground-handling operations are affected by complex uncertainties and disturbances (delays, employees' illnesses, weather conditions, machinery breakdowns...) that produce a high number of deviations from the original schedules. In order to develop these new requirements at the shop floor level, companies encounter several different control/scheduling problems, including:

- The usage of the resources: it is not easy to know when a physical resource is available or busy. The lack of vertical information feedback from the plant leads to a high dependency on the surrounding environment.
- The expected duration of processes and tasks: each task has a different length depending on the external and internal conditions. The process cannot be completely defined as a mathematical static function or in a template; the shop floor situation is in constant change.
- The coordination of the involved elements in the decision-making: it is not easy to define rules of negotiation in high-distributed environments where the decision-making information comes from different sources. In case of shop floor operations, the decisions have to reflect the strategic policies and local constraints in every single operational decision.

The problem is even worse when the operations have to be distributed between different companies or departments in enterprise integration. Currently, the subcontracting and specialization in companies are normal, their closer interaction forces to improve the collaboration. Consequently, a new detected coordination problem is how to share all the sensible information, and thereby create a global picture of the state of the operations in the shop floor. This common blackboard should reflect a picture, in which the state of products and processes are accessible and independent to the business model of each company (Garcia, McFarlane et al. 2003). Nowadays, each company has its own business model; and under this business model there are information system definitions that include databases, Enterprise Resolving Planning (ERP) systems, Customer relationship management (CRM) systems, Intranets, etc. These systems are fully integrated in the companies, and they are already implemented and supported by huge investments. The definition of a global business model in shared industries can be dramatic; this idea requires a complete migration of each information system into a global system definition, which means a full integration into a unique system. These systems are not easy to define or maintain, and the results of these big information systems are low scalability, compatibility risks during the deployment and a not well-defined way to share the information. Following these ideas, “The future of ERP: Why the 'big ERP' approach is dead” by Thomas Wailgum explains that the traditional big ERP paradigm has reached its end (Wailgum 2010). These huge systems tend not to be used because they are difficult to deploy and update due to the breadth of the system, which provokes a long Return of Investment (ROI). Instead, new ERP systems become modular, distributed and a lot more agile as they break apart. These trends include software as services

(SaS), on-demand applications, enterprise 2.0 collaborations, virtualizations or cloud platforms in the new ERP systems.

In this new context, the research community in soft integration and virtualization are becoming significant. A key example is the term “Internet of Things”, which is closely connected to the technologies of ubiquitous computing and especially to Radio Frequency Identification (RFID) (Atzori, Iera et al. 2010). “Internet of Things” stands for the soft connection of computer networks with physical objects, which will improve industrial operations by increasing transparency in supply chains. These virtual products allow for provide service-oriented strategies and event driven controls launched by the feedback coming from the shop floor environment. More specifically, the technical committee of the International Federation Automation and Control in the Manufacturing Plant control (IFAC-TC5.1) has addressed the manufacturing/logistics control challenges and issues which have risen around the Intelligent Manufacturing System (IMS) paradigm as essentials. These challenges are to apply Micro-electromechanical systems (MEMS), Mechatronics, Manufacturing Execution Systems (MES), Multiagent Systems (MAS), Holonic Manufacturing Systems (HMS) and e-technologies to the manufacturing control. Their main objective is to control with more agility the entire manufacturing chain (Abramovici and Filos 2011)

1.2 Situation in specific scenarios

In order to analyse the situation and to validate the results obtained in this thesis, two different shop floor scenarios were selected with specific features. The first scenario is the ground handling (GH) operations in the Ciudad Real Central Airport. This new airport was looking for tools to

help the asset management in order to provide accurate GH services; the airport started to accept incoming flights in 2008. The second scenario is the shop floor operations in a logistics center simulated at Autolog Labs; this miniature model is a test bench based on existing layouts. Both scenarios are including the mentioned strategies as a key factor to exploit competitive advantages in their markets; they have broad common requirements but different details that need to be addressed separately.

The ground handling (GH) operations need to schedule resources in order to attend the incoming flights. It includes the tracking of resources in the terminal, physical constraints, a decision making process and an effective communication with the upper information systems, in the case at hand it is as an ERP from *Société Internationale de Télécommunications Aéronautiques* (SITA). This is a pure scheduling problem in a high dynamic environment where a continual planning is required. Instead, the logistics center scenario apart from the scheduling problem such as the ground handling; also incorporates new specific manufacturing hardware integration as PLC or robots. This scenario is not as high dynamic environment as the GH but it is restrictive in the space of decisions because of the complex constraints and the communication protocols with the machinery. For example, the pallets cannot overtake others in the conveyors but the overtaking is possible in the warehouse or in the picking zones. The simulated logistics center, in which the test is deployed, is based on a real layouts and it also includes the possibility to add new modules in order to analysed future modules or configurations for the companies involved. The next subsection details the current situation of both cases.

1.2.1 Ground handling on airport operations

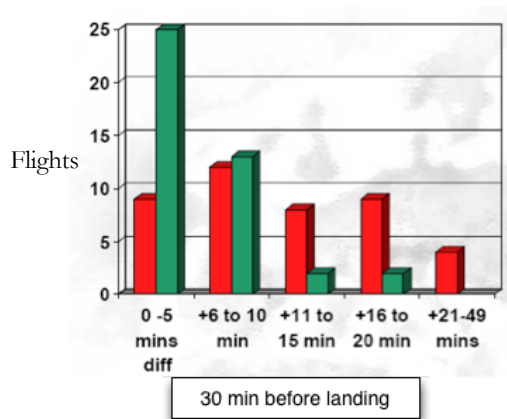
Modern airports can be extremely complex organizations, which often involve governmental organizations, private companies, airlines, aircraft operations and airport operators (Ashford, Stanton et al. 1998). Besides, airport operations are affected by constant uncertainties and disturbances (delays, employees' illnesses, weather conditions, machinery breakdowns...) that produce a high number of deviations from the original schedules. The current situation in the world economic markets have especially forced air transport agents to modify the way they are operating, the results bring to light the need to improve productivity and reduce costs by using new strategies. New ideas such as the "Common use model" are arising in airport operators to gain flexible control over facilities and services, increasing passenger-processing options and acquiring use efficiencies (Bellotti 2008). Airport common usable space is the environment in which any airline may operate, that is, there is no space specifically dedicated to any single airline. In this situation, all airline usable airport space is available for use by any airline. The goal of the full "Common Use Model" consists of minimizing the amount of time given to a specific airline, as well as maximizing the full use of the airports, which improves productivity based on the increased utilization of existing resources. The airlines are assigned with no preferences given to any individual airline, which might suppose conflicts in the schedule of resources in case of disturbances, delays or breakdowns. The figure 2 shows a branch of a Heathrow terminal in which two airlines operate with common facilities and resources (Figure 2).



Figure 2 A branch of the Heathrow Terminal. Google and BAA Copyrights

Nowadays, the “Common Use Model” is usual in many airport operations and equipment; the implementation results of Common Use Terminal Equipment (CUTE), Common Use Self-Service Kiosks (CUSS), information displays, baggage displays or sorting have been successful (Bellioti 2008). A natural next step of this model is to be deployed in the GH operations. It is matching up with the increasing of outsourcing GH operations, especially in Europe, which at least define a new stakeholder in the air transport operations. The definition of the GH operations as a common use facility forces a redefinition in work habits, especially in the allocation of resources. In order to schedule shared resources, the interrelations between all the GH elements and the constant the information flows from the air transport agents are critical; the current air transport agents are divided into: airports, airlines, Air Traffic Controls (ATC) and Central Flow Management Units (CFMU). Nowadays, GH management is an additional player that must participate in the re-negotiation and in the dynamic scheduling to provide accurate global

charts show what are the average errors/deviations in the Estimated in Block Time (EIBT) per flight in a big airport, with 3 cases: 30 minutes before landing, 10 minutes before landing and 4 minutes before landing. The EIBT is the expected time when the aircraft is in front of the gate ready to be attended by the GH resources (Figure 4). These tests were launched at the London Heathrow Airport in 2008 with 48 flights, and this information introduces the problem of the constant disturbances and deviations at airports. The red bars represent the error that was measured before the deployment of the Airport collaborative decision-making (ACDM) project by Eurocontrol and the green bars details the error after the use this ACDM technology (Eurocontrol 2003-2009). In both cases, the estimated error is very big and the expected times are volatile, some remarkable ratios are: 30 minutes before landing 50 % of the flights have an average error of about ± 10 minutes. Then, 10 minutes before landing 80% have an error of around ± 3 minutes and even 4 minutes before landing, there can be an error of around 21 minutes.



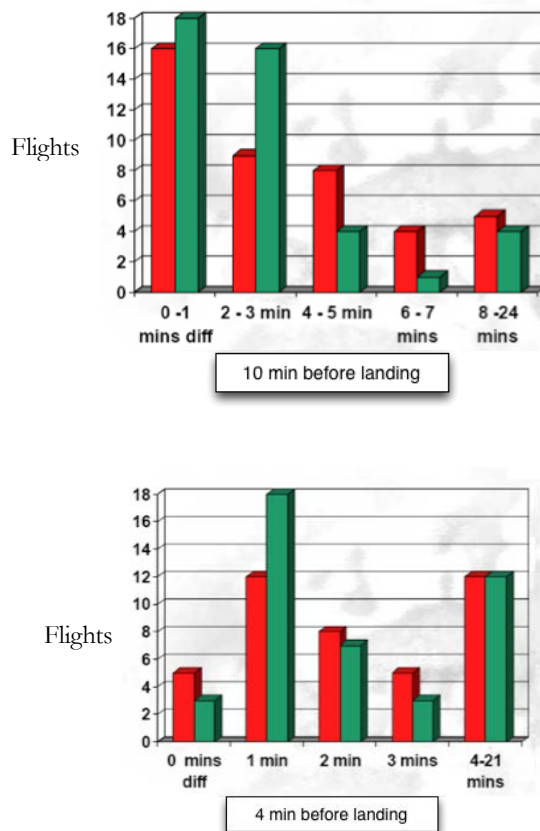


Figure 4 The errors in the Expected in Block Time (EIBT). CDM
Copyrights

Therefore, these service companies need to share real time information to be able to define a shared dynamic picture of the airport, which should be made up by all these Key Performance indicators (KPI) (Thorne, Barrett et al. 2007). The amount of the available data between air transport agents is huge and its flow is quick and non-stopping; the Target Off-Block Time (TOBT), the Estimated In-Block Time (EIBT), the Estimated Landing Time (ELDT), the gate allocation, the taxiway or the flight plans are some examples (Figure 5). Looking at these complex information flows, there are Eurocontrol projects like Single European Sky ATM Research Programme (SESAR) or Airport Collaborative Decision Making

(CDM) that are spending huge investment in this global integration of data, which also define rules and protocols for sharing data between air transport agents, governments and regulatory associations (Eurocontrol 2007). Apart from the company approach to the problem, there are also several academic scheduling DSS close to the handling operations requirements, which also propose the integration of the existing information flows to establish accurate estimations in in-block/off-block time, gate assignment or taxi times (Lim and Wang 2005; Atkin, Burke et al. 2008).

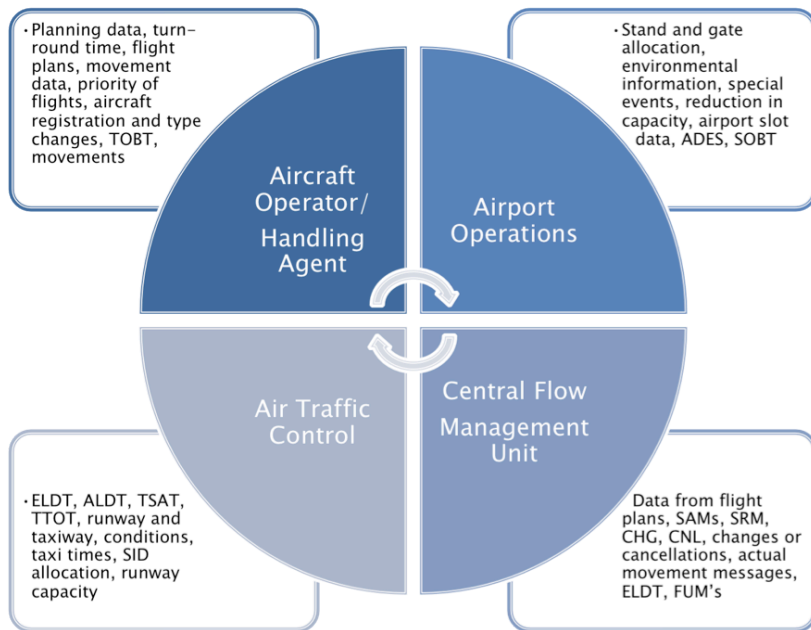


Figure 5 Airport operations information flow

Even if all the information is accessible, the GH planning needs complex decision support software to perform complex calculations and estimations, in order to allow real time negotiations, to monitor the resource usages, and to provide status reporting (Eurocontrol 2003-2009). In the “Common Use Model” setting, this GH planning becomes

complex since they have to provide service to several different types of incoming flights, airlines, passengers or cargo services, which are affected by uncertainties and disturbances that produce a high number of deviations from the original schedules, as is detailed by Dorndorf (Dorndorf 2002). To get an idea about the complexity of the scheduling and control problem faced by the ground handling management of any airport this thesis highlights two of them. The first control problem is the large list of tasks that have to be performed for each combination of incoming-outbound flight. The main functions are: guiding the aircraft into and out of the parking area, towing with pushback tractors, lavatory drainage, water cartage, air conditioning, airstart units movements, luggage handling, air cargo handling, refuelling, ground power supply, passenger stairs movements, wheelchair lifts, hydraulic mules supply, deicing, catering, passenger service, check-in counter services, gate arrival and departure services, staffing the transfer counters or field operation service. Therefore, a huge set of corresponding physical and personal resources have to be controlled by the system, some of these resources are: chocks, bag carts, dollies for containers and pallets, refuelers, tugs and tractors, ground power units, buses, container loaders, transporters, air starters, drinking water trucks, lavatory service vehicles, catering vehicles, belt loaders, passenger boarding stairs, pushback tugs and tractors, de/anti-icing vehicles, operations (load control) agents, warehouse agents, crew chiefs, ramp agents, transfer agents, inbound runners, lavatory agents, mail/freight agents, bag room agents or drivers.

Besides, apart from working in a dynamic and shared environment, the ground handling companies are obliged to reduce costs while fulfilling the service contract with the airlines in order not to pay fines because of delays. The way to reduce costs is to improve the productivity by properly

scheduling and track the resources that attend the incoming flights; this improvement also helps to reduce carbon emission by sustainability policies, which will be an obligation in green airports.

Therefore, the situation of the GH control/planning is complex because there are huge quantities of resources and the expected times are very volatile with long delays in short periods. Besides, the GH companies need to estimate the Target off Block Time (TOBT), this time specifies when the aircraft is ready to depart and to take the taxiways. With this information, the Air Traffic Control (ATC) configures the slots and the assignment of gates. If the TOBT estimation is wrong, the next flight has to be reassigned to another gate and the GH resources that attend this flight are not ready for this new configuration. The next list enumerates the most important problems during the control of GH operations:

- The uncertainty in arrival times and gates, especially with the EIBT and with the last minute gate assignments.
- Each flight has different needs and requirements depending on the number of passengers, the cleaning, the loading of fuel, the catering or on the required maintenance. Therefore, the mass customization also appears in the GH operations, which require accuracy in the attending time of these specific GH processes.
- The accurate estimation of the target off block time for the new ATC gate assignments.
- The very short transfer or turnover time which is around 30-45 minutes.

When analysing the possibility of applying these ideas to the real situation of the Ciudad Real Central Airport in Spain, the airport management staff become supportive, although a word of caution arose regarding possible implementation problems. For that reason, the decision was a pre-validation environment that made it possible to test these new concepts before the real deployment, which were presented in a demo show. The target at hand was to achieve an effective mass customization in the ground handling that habilitates the implementation of individual services to the passenger based on a system able to plan ground-handling resources while they attend to incoming flights. This system included decision-making software as well as hardware identification technologies.

1.2.2 Logistics centers

Nowadays, the distribution centers operate in a highly competitive environment, which forces them to constantly improve in terms of quality, response, agility and flexibility. These new requirements become especially hard to tackle when dealing with typically big distribution centers where the usual control is rigid and hard to manage, maintain and update. For flexible systems however, finding an adequate response to a broad range of possible new situations where business must continue as usual; and this should apply even while the system is being modified/updated. Several authors remark that the distributed control in logistics centers becomes standard practice as it allows a division of the problem in smaller and more feasible tasks (Encinas, García et al. 2010) (Valckenaers and Van Brussel 2005). More so, if the parts are endowed with enough autonomy to manage their own tasks as functional units. The next figure details a common initial division of tasks during the

product life in the logistic centers that divide the current control into sequential steps (Figure 6).

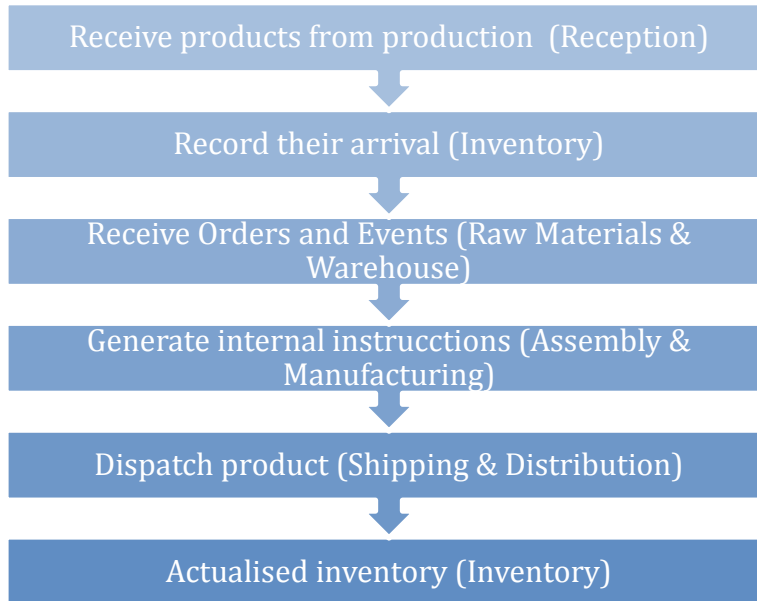


Figure 6 Most common tasks in the logistics centers

Nevertheless, the interrelations of these tasks are being modified given the new demands explained in the previous section; the logistics sector introduces the proximity of the client as a competitive advantage against the competition based on the “Mass Customization” concept. In the logistics centers, the results of this proximity are: the short product life cycles, the increase of product references, the necessity of faster responses when introducing new products or the preparation of mixed pallets. Especially dramatic is the preparation of mixed pallets because they tend to be assigned before they are even completed and any disturbance is a problem for the client; this is the shift to the multilevel pallets, which are made up by several different layers of products, instead of unique product pallets. This mixing of different products in the same pallet is mainly provoked by the fine granularity of the current

heterogeneous clients. This has direct consequences on material manipulation systems and therefore in shop floor control, which takes charge of the materials and product movements. In order to address this situation, the logistics centers have detected two interconnected problems at hand: the number of references and the fine granularity of the final clients. One example in Spain is J.C. Garcia Carrion, which is a leading dairy product company in Spain. This company, which has the main logistics center in Daimiel, has increased the number of reference in more than 150 different products during the last 6 years and they are not prepared for a full customization pallet process. The current situation of the company is that the productivity has been reduced in a direct relation with the number of references (Figure 7). In this company, the seven existing production lines need to be flexible in order to manage all these references and prepare the customized pallets for each client.

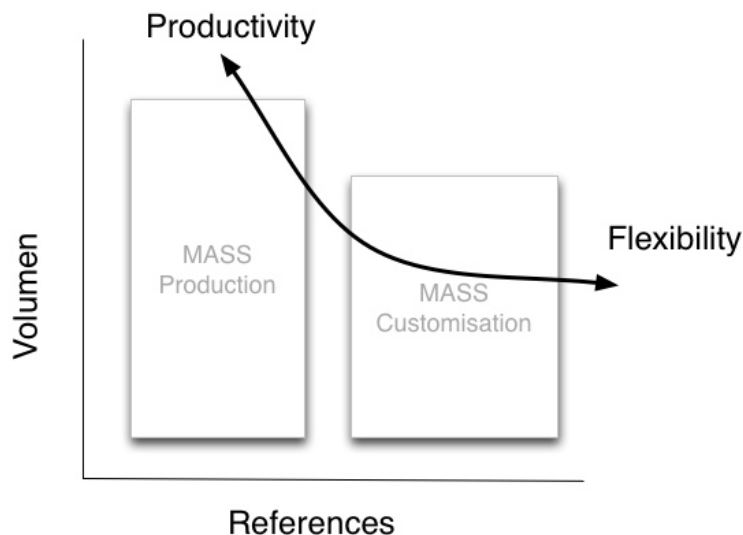


Figure 7 Productivity in logistics centers

This intensification of accurate movements complicates the schedule, in which different rollers and chain conveyors, trans-elevators and operators work together to manipulate a large quantity of specific references, not only a few line of generic products. It supposes a challenge in the way of work and this mass customization is directly affecting the logistics centers. In these cases, the use of control and tracking techniques in distributed information systems are crucial to tackle errors and complex scheduling that depends on orders, disturbances and the state of the warehouse (García, García et al. 2011). Identification technologies such as RFID provide accurate visibility in this distribution systems, which allow the application of the before mentioned. However, to test these new technologies and concepts, it is necessary to build an experimental platform, which simulates the facilities of a real distribution center because of the impossibility to test in real installations (Thorne, Barrett et al. 2007). Therefore, an experimental platform was designed in order to support research, development, and implementation of DSS enhanced with RFID and visibility frameworks; it has been assembled at the Autolog Labs of the University of Castilla-La Mancha (UCLM) in Spain. The purpose of this model system is to provide a test-bench to experiment with multidisciplinary theories, techniques, and algorithms that can be grouped into four areas: automation and robotics, instrumentation and control, production planning, and logistics and distribution. The platform is made up of a physical part, which represents the loading and unloading docks of the distribution centre and a 3D visualization of specific warehouse layouts (Figure 8). Furthermore, to increase the similarity of this platform with the real systems, a PLC network controls the plant and a commercial ERP manages the orders.



Figure 8 The simulated Logistics Center in Autolog Labs

1.3 Objectives

The committee of the "Council of Supply Chain Management Professionals" (CSCMP) highlighted the collaboration between enterprises based on new information systems as the most relevant activity within the new developments in SCM, with a consensus of 80%. This implies the identified necessity of developing systems for the dynamic interconnection and flexible introspection between business models as well as inside them. The argumentation of the committee also corresponds with the requirements in the analysed scenarios. All these ideas are part of the "e-business" (Johnson and Whang 2009) and the "Business to Business" (B2B) concepts (Dai and Kauffman 2002), in which the online communication between business units improves the decision processes.

The aim of this thesis is to achieve an accurate decision-making/control in the shop floor levels of the Supply Chain Management (SCM) by using agent-based software, visibility frameworks and new manufacturing concepts. Specifically, the proposal that is highlighted is an intelligent product architecture based on a MAS that processes the information coming from the sense levels but aligning the decision making with the strategic policies. This intelligent product vision supposes a shop floor control specialization of the “Internet of Things” concepts.

These short-term assignment decisions have to be made attending to review, control, planning, information updating and 24-hour-data availability rather than to mathematical models (Garcia, McFarlane et al. 2003). The proposal is to take short-term planning and even online decisions whenever disturbances arise, which provoke complex deviations from the original schedules that can produce an out of control situation (Wong, Leung et al. 2006). This is mainly produced because the lack of communication between the supply chain levels causes an inaccurate reflection of the strategic decisions in the shop floor level (Schönberger and Kopfer 2007). Apart from the main objective, this will include the explanation of the proposal from several viewpoints: components, models and interactions. Then, two main branches appear as sub-objectives: firstly, the integration of field information into the decision process by using visibility frameworks. Secondly, the interaction protocols and information processing in order to align the decision process at the shop floor levels of the supply chain (Figure 9). Both objectives are taking into account the involved events and their effects on the rest of elements in the supply chain.

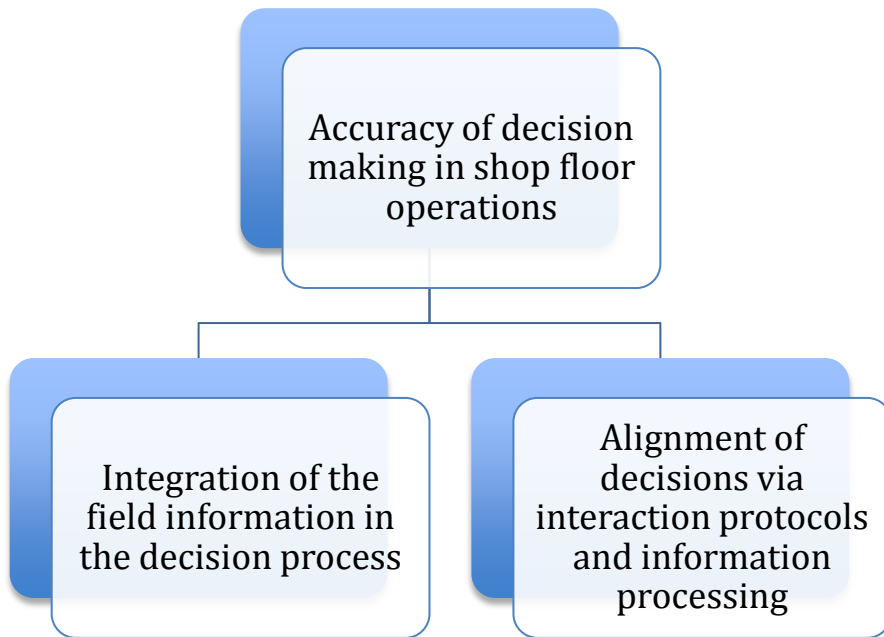


Figure 9 Objectives

Based on these main two sub-objectives, other technical surrounding objectives need to be achieved in the search of the main aim:

- The generation of a constant real-time picture about the shop floor environment that dynamically shares the circumstances of the physical resources to the involved stakeholders. It has to communicate the environmental uncertainty produced by the disturbances at the field levels but abstracting the upper levels about the physical layouts and configurations.
- The definition of dynamic negotiation protocols in order to share the information among the affected elements in the operational supply chain but maintaining the independence between them.

The global alignment during the decision-making in high distributed and heterogeneous environments is complex.

- The application of information systems that are geared to discover information and services, which has to allow flexibility in the definition of tasks, goals and plans.
- The incorporation of these new features and technologies to the existing information systems in the companies; the integration of the proposal with the current information systems as the ERPs, the CRMs or the Warehouse Management Systems (WMS) and also with the hardware devices as Programmer Logic Controllers (PLC), machinery or robots is essential.

Then, an application objective, which is inherent to the main objectives, is to detect the implementation and deployment problems during the start-up of the proposal in real scenarios. These problems are specially discussed during the application of the proposal in the test bench scenarios.

1.4 Thesis organisation

This thesis is structured as follows: the contents are divided into the current introduction that brings together the current situation and the objectives. Then, the state of the art introduces the involved technologies related with the manufacturing and shop floor control in terms of visibility, software and decision-making technologies. The third chapter introduces the proposal MAS-DUO that includes the definition of the proposed framework and the development methodology. The fourth chapter shows the deployment in the specific scenarios and a discussion of

the results, in which models, interactions and components are detailed in each case. Then, the contribution and conclusion chapter shows the outcomes in both scenarios and a common argumentation of the contributions. Finally, this chapter also enumerates the conclusions of this thesis and new topics that can be addressed in future developments.

2 STATE OF THE ART AND ITS APPLICATIONS

The new definition of DSS at the shop floor levels includes a multidisciplinary research where physical elements, information systems and artificial intelligence techniques have to work together to tackle the best decision-making during operations in plants, terminals or logistics centers. This problem is the basis of the cooperative distributed problem solving (CDPS) with a multi-scale decision-making, which requires fusing several technologies together in operations research.

The main research areas, based on the surrounding objectives, are the data acquisition and the information processing, both applied together in order to achieve the new management and manufacturing strategies. The next mapping chart shows these two main branches and the new manufacturing concepts such as primary concepts; these branches are sequentially divided into end-nodes that present the involved technologies in this thesis (Figure 10). Firstly the next section introduces the current surroundings manufacturing concepts that are being introduced by several authors from the International Federation of

Automatic Control (IFAC), the Intelligent Manufacturing System (IMS), the Foundation for Intelligent Physical Agents (FIPA) or the IEEE-IES Technical Committee on Industrial Agents in order to establish solutions to the current problems of the shop floor companies. These concepts define the strategies that deal with the current shop floor situation and establish the basis of the requirements that need to be resolved by the proposed technology.

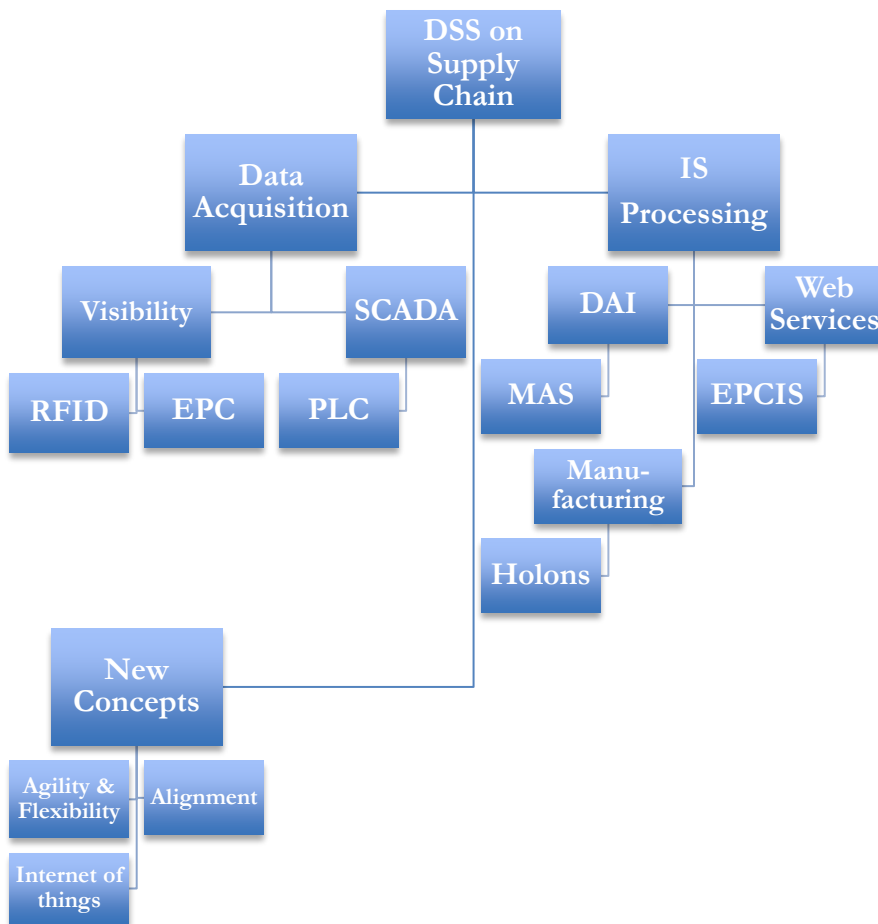


Figure 10 Map of involved technologies in this thesis

2.1 Agile and flexible manufacturing trends

The “Mass customization” combines the contrasting concepts of mass production and customization. Mass production implies the cost reduction from scale effects and the gained production experience. The concept of “Mass customization” is attributed to Stan Davis in “Future Perfect” (Davis 1989) and was defined by Tseng and Jiao as "producing goods and services to meet individual customer's needs with near mass production efficiency" (Tseng, Jiao et al. 1996). However, from a management point of view, the definition that best fit in this context is that realized by Chase, Jacobs and Aquilano (Chase, Aquilano et al. 2000); they define the mass customization as the method of "effectively postponing the task of differentiating a product for a specific customer until the latest possible point in the supply network". Therefore, an effective customization requires moving processes to the last steps of the supply chain, which increases the stress because the customers are closer and any disturbance is sensitive. It supposes a challenge in the design of the production and in the operation control, especially on the shop floor because the number of processes has increased and the disturbances are more sensitive for the clients. Therefore, the introduction of new customization processes is obliged to include agile and flexible features in the shop floor control, while continuing business as usual.

The last available point to customize the products in a normal supply chain is before distribution. The product customization has forced to move the production and the warehouse closer, storing in the warehouse a generic product, which is differentiated in the last moment (Figure 11). Therefore, the processes after the warehouse are not only delivery; there are customization processes as picking, packaging, kitting or any specific production processes as painting or drilling. This movement of activities

defines a new set of requirements in the shop floor operations: quickness (the client is already waiting for the order), accuracy (any error will be a problem for the client), flexibility (the customization is changing depending on each client), agility (the capacity of quick responses to customer needs) and productivity (to improve the final ROI). The forwarding of customization processes also allows reducing complexity in the warehouse because of the reduction of product references in a smaller line of generic products. Then, it also supposes a simplification of the WMS because the forecasting systems are easier and all the products in the warehouse have a high turnover compared to storing all the possible references. All the before mentioned “Mass customization” concepts directly generate great savings to the companies, but new accurate control and scheduling techniques need to be addressed after the warehouse.

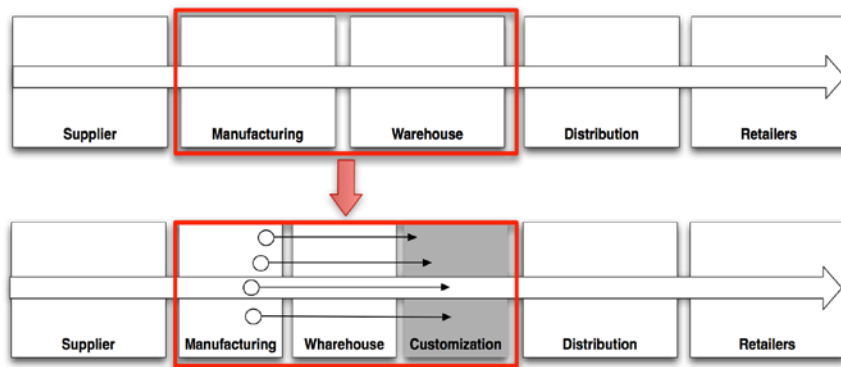


Figure 11 Customization processes in the SCM

Besides, these new last minute processes require flexibility and agility because the product life is shorter and the granularity of the clients is very high to define fixed solutions to each configuration; the capacity of quick responses to customer needs must be usual (Valckenaers 1994) (Chan and Chan 2009). In terms of manufacturing, the agile vision is the next step of

lean manufacturing, which defines processes, tools and trainings to quickly respond to customers' and market needs while still controlling cost and quality (Ramesh and Devadasan 2007). The second related manufacturing concept is flexibility, which means the ability to deal with slightly or greatly mixed parts, allowing variations in the assembly and alternatives during the sequence of processes. The flexibility allows to change the production volume and to modify the design of a certain product being manufactured. Both terms begin to introduce the concepts of parts or modules, which outline the necessity of new control techniques based on heterarchical models. These models divide the system into independent subparts that make decisions depending on internal and external conditions. This division, which is made up of autonomous production units, obliges defining distributed cooperating facilities among themselves. As an initial division, the next figure shows the fractal layers proposed by Leitao that divide into levels the distributed manufacturing environment (Leitão and Restivo 2006) (Figure 12). Firstly, the cell level executes the low level actions in which the machinery has its own control with its specific constraints. Above the cell level, the shop floor is the lowest level that introduces the flexibility requirements; in this level the orders are broken into single instructions executed by the cells that directly manipulate products and resources. This level supposes the first co-operation between elements, it schedules the instructions and controls the operational disturbances between operational cells; shop floor control is often referred as Manufacturing Execution Systems (MES).

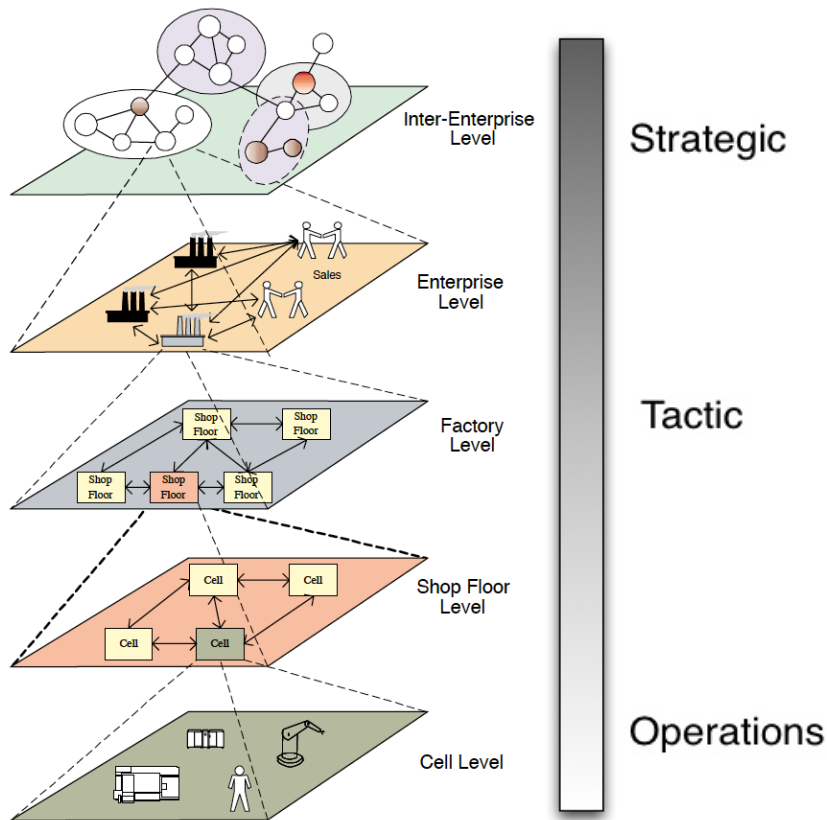


Figure 12 The layers of the distributed manufacturing (Leitão and Restivo 2006)

At the shop floor level, the phase of product scheduling is where the plan gets operational and is divided in executable packages as it breaks the long-term policies and orders into scheduled operational decisions at shop floor level. This instruction breaking is based in its own criterion fed by the information coming from the closer levels. Therefore, a specialization of these dispatching processes can define a flexible and agile decision-making during operations because of its central position (McFarlane, Sarma et al. 2003); this DSS uses information coming from the both adjacent levels: the enterprise level (information systems) and the cell level (physical resources). More specifically, in the manufacturing/production control proposed by Macfarlane and Wong

(Wong, McFarlane et al. 2002), the product scheduling is an interface between the equipment status and the order status (Figure 13). This is the upper layer in the manufacturing control that is coupled with the physical layouts and with the hardware details.

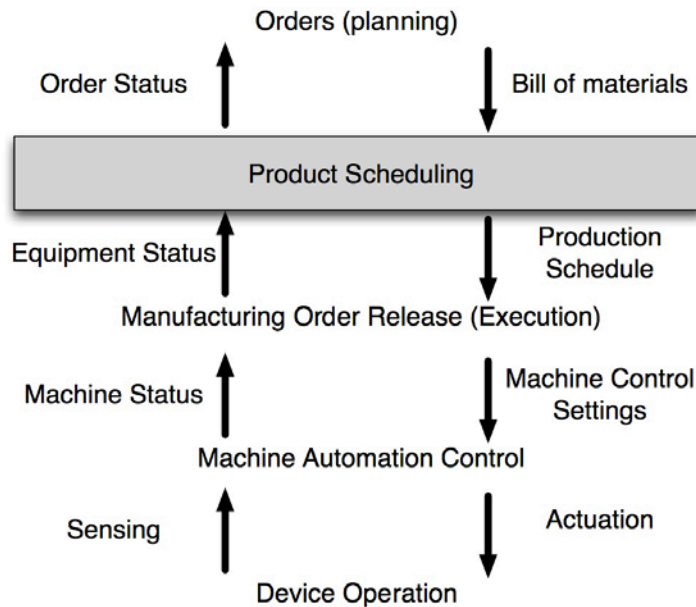


Figure 13 The manufacturing Control (McFarlane et al. (McFarlane, Sarma et al. 2003))

Once the information about the orders is available, the factory level manages the collaboration between shop floors as the first step to introduce the strategic controllers over the production. Here, the production plans include the long horizon of orders and this information use to be managed by ERP and CRM systems. The factory level sends complete orders to the product scheduling at the shop floor level and it waits for results. Finally, at the top of the distributed manufacturing, the enterprises and inter-enterprise levels control the collaboration between companies and departments in the search of common objectives.

This organization of manufacturing control starts at the top with a summary of the production schedule and sequentially distributed details appear at each level until the instructions reach the executable actions are completed. In particular, each level of the hierarchy creates instructions for its subordinate units with ease feedback from the lower levels. Looking at the specific case of shop floor control, Macfarlane et al. detail several layers: planning, schedule, execution, machine control and device operations (McFarlane, Sarma et al. 2003) (Figure 13). With these hierarchical layers, the faults that occur in the implementation of the planning of production are fixed in the next planning cycle as a synchronous protocol.

The position of the shop floor level is the interface between the tactical operations of the factory level and the executions of instructions at the cell level. This interface makes the product scheduling important so as to allow proper vertical information sensible to changes and disturbances. The distribution centers are a direct and clear scenario, in which the product scheduling breaks orders in scheduled input/output pallet instructions at the warehouse area. The product scheduling prepares and sends a series of instructions to the different resources, which are hardware devices or even employers. In the case of hardware, the specific machine control reads and understands the instructions and also prepares the setting up (Figure 13). Once the configuration of the machine is stored in the local control, the device executes the selected instruction. After that, the information feedbacks are required in the other way, which makes data available to check the correct state of the operations to the upper levels. The sensors in the hardware devices inform the machine status and the surrounding elements about the state of the instruction,

this make possible compose the global status of the machine and the status of the instruction that initially was sent by the product scheduling.

Looking at the previous description, the product scheduling is a main part of the manufacturing control since it has to pre-schedule and control the atomic instructions that it has previously broken based on the horizon of orders and the machine status. The current capacity of the product schedule is fixed and it is hard to define discovering techniques, simulation processes or even re-planning during operations. Nowadays, the product scheduling is efficient in static situations and fixed layouts but cannot achieve the agile and flexibility requirements, especially in complex and compound processes (McFarlane, Sarma et al. 2003). Traditional hierarchical control systems, however, are slow adapting new production requirements and sequences (Caridi and Cavalieri 2004). At this point, the flexibility and agile requirements oblige defining a planning system able to dynamically calculate priorities, dynamic schedules, assignments, estimations, simulations and discovering of functions; most of these features have a great dependency on the physical configurations and layouts. An alternative to this hierarchical control appears in the heterarchical approach based on distributed control using negotiation methods (Frayret, D'Amours et al. 2004); the redefinition of the product scheduling control problem requires the application of these new philosophies at the shop floor level.

Now, instead of designing monolithic control systems that need to be specifically programmed to solve every possible problem, the control can be divided into components, or units, with their individual control and an inherent interaction with others. Dean and Wellman defined production control as the process of choosing, initializing, and monitoring the actions in a production system with the aim of achieving an optimal production

plan, which is a scheduling problem in nature (Dean and Wellman 1991); it introduces the transition states of the products during the manufacturing process. In a finite set of states of the products, the control schedules and monitors the instructions that provoke the transition events between the states until the goal state. Specifically, the product schedule controls the sequence of instruction that performs a task, in which there are state transitions and that are directly affected by the specific layout of the plant (Equation 1). The next equation lists the sequence of states until the state 'n', which is the objective of the task or goal state.

Equation 1 The product state transitions

$$\textit{Product state transtions} = \{S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow \dots \rightarrow S_n\}$$

Dilts et al. consider four types of possible control architectures to supervise this sequence of processes (Dilts, Boyd et al. 1991). Firstly, the Centralized control defined a unique node that takes the decisions in a concentrate capacity. Secondly, the hierarchical control describes a layered control level where the decision capacity is divided in specialized steps. Then, based on the previous, the modified hierarchical architecture maintains the layered modules but adding capacity of interaction in both ways, which provides a result in case of disturbances. Finally, the last architecture is the heterarchical, which is based on autonomous nodes without a global vision of the environment. This architecture has the drawback of the lack of global alignment during the decision process and hard interaction protocols and this independence of agents tends to make the control of the system difficult to predict (Sun, Hsu et al. 2009).

At this point, the division in more feasible problems that divide the problem in layers and functional units is clear, which can be seen as specialized autonomous intelligent blocks as logical units in a heterarchical system. These logical units have their own objectives, social capabilities and intelligent features (Wooldridge, Weiss et al. 2002). Therefore, distributed heterarchical control approaches are ratified by manufacturing trends; these heterarchical systems must include interaction protocols and AI techniques between these autonomous units by using heuristics methods in the search of the best plan (Duffie 2008). Nevertheless, heterarchical distributed systems have the inherent drawback to highly distributed systems: the difficult integration and the lack of a global point of view. A global policy needs to be reflected in every single operation, which could be dispersed in heterarchical systems where subtasks and services are autonomous and self-deliberative. In order to fix this problem some authors propose the alignment, which exists when the information system is in concordance with business organisation's goals (Morel and Pereira 2007). However, to achieve an upper level in the organisational decision making, a shared value function and the constant feedback of information in both ways are needed, which reach the second level of alignment, called "Environmental Alignment" (Camponovo, Pigneur et al. 2004). This second alignment level takes into consideration the external disturbances and assumes that the information systems have to integrate features for assessing this environment (AVILA, GOEPP et al. 2009). The next section details the concept of alignment and how it can be attainable with the existing technologies.

2.1.1 Alignment

Given the actual high level of the operational disaggregation in worldwide companies, the management staffs need to drive their business decisions

to a common point of view where all their parts support the global objectives of the company. This necessity of integration brings forward concepts such as alignment that has become particularly important in the field of business integration and interoperability. Luftman et al. point out that firms need to align their IS strategies with the rest of strategies in the business (Luftman and Kempaiah 2007). Much of the new advances on IT alignment were supported on the work of Henderson and Venkatraman who developed a strategic alignment model using two dimensions: the strategic integration and the functional integration, taking also into account what they call cross-domain (Venkatraman, Henderson et al. 1993). Camponovo and Pigneur suggested the necessity of three levels to make possible the complete alignment of the IS (Camponovo, Pigneur et al. 2004). The first one is the strategic alignment, related to the internal configuration between the IS and the enterprise components. The second level, which is aligned with the environment, takes into consideration external disturbances (AVILA, GOEPP et al. 2009). Finally, the third level deals with how the system has to evolve over the time in order to maintain the alignment.

These new alignment concepts use technologies coming from distributed artificial intelligence, and specially propose their application in order to achieve environmental alignment. On one hand, new interaction protocols can face the strategic alignment. On the other hand, the environmental alignment is achieved by getting an accurate picture from the environment; this information needs to be up-to-date and linked to the physical world through trace and tracking services. Many industries can be benefited from the advantages provided by alignment; the case of airport operations is one of them. These operations involve the cooperation of international regulation bodies, private companies,

airlines, aircraft operations and airport operators. All of them in the same decision process but with common and individual objectives; this situation presents a decision making process hard to align, especially when disturbances occur (Ashford, Stanton et al. 1998). At this point, new concepts as “Internet of things” are dealing with the information generated by these disturbances, the objective of “Internet of things” is to automatically capture the information of products when new events occur. The next subsection details this vision and the involved technologies.

2.1.2 The intelligent product vision from the Internet of Things

In practice, the environmental alignment requires effective methods to access to the external and internal surrounding information. This information feeds the internal beliefs of the distributed intelligent units that compound the heterarchical control. In the manufacturing research community, as a new logic interface between components, the service vision arises as a tool to provide a dynamic access to information between modules (Cavalieri, Gaiardelli et al. 2007). The most important example is the new “Internet of Things” vision, which provides new mechanisms based on a virtualization of physical resources to provide services between the real world and the computers. “Internet of Things” virtualizes every physical thing in this world, which can also become a computer that is connected to the Internet. The “Internet of Things” adds another data dimension, it allows the physical world, things and places, to generate data automatically; it is about sensing the physical world. These smart things communicate among each other and with computers in a machine-to-machine way on Internet. However, it only recently became relevant to the practical world because of the progress

made in hardware developments were in the last decade. Once the physical world can include software avatars, an essential research is to apply this vision to the shop floor in order to integrate the virtual and the physical parts of the products in a logical unit, which is called intelligent product. In the next list, Wong enumerates the requirements to define a intelligent product (Wong, McFarlane et al. 2002):

- The product possesses a unique identification.
- The product is capable of communicating effectively with its environment.
- The product can retain or store data about itself.
- The product deploys a language to share its features, production requirements and plans.
- The product is capable of participating in relevant decision-makings to its own destiny.

Therefore, this intelligent product presents autonomous capabilities and services to the rest of shop floor elements in a flexible way to define plans; the intelligent product can be the logical unit that builds the required heterarchical control. The intelligent product service-oriented approach provides a way to create a new architecture that reflects products as components toward to autonomy and heterogeneity features. These autonomous services help to manage human control activities related with assets, spare parts, tools and their relationships during the product lifecycle. In this vision, the humans delegate the control activities to the smart products, which qualify their functionalities as services. It

increases the return on investment based on an organisational agility and interoperability, because it allows dynamically discovered functionality with minimum coupling (Gebauer, Fleisch et al. 2005). Besides, the service-oriented approaches allow loose coupling between technologies and underlying applications, which is useful in the manufacturing environment because of the heterogeneity of elements. From a distributed manufacturing point of view, when the product takes the initiative during the plan, it is known as product-driven (Wang, Huang et al. 2004). This shop floor control is a service approach where the products search and compete for services to assist them during the production process along the life cycle.

The division in functional product intelligent units at the shop floor operations can achieve agility and flexibility requirements but a dynamic interface with physical resources needs to be addressed at the shop floor. The next section explains the EPC visibility framework from the data to the service definitions in order to clarify how this interface can be tackled. Therefore, the EPC visibility framework is taking charge of feeding the intelligent product with information captured from the shop floor.

2.2 Visibility frameworks

The intelligent product is based on the constant information flow between the physical elements and their virtual avatars, however to achieve this interface in real time an actualised, uncoupled and independent picture of the shop floor is important (Thorne, Barrett et al. 2007). At this point, hardware, software and data interfaces work together to provide accurate visibility frameworks as the capturing part of the IoT. A visibility framework must contain:

- A real time data acquisition mechanism.
- A conversion of the acquired data into relevant information by using standardized and public representations.
- A continuous access to the information, which requires standard interaction protocols.

This is the case of the EPC Global Architecture Framework, which was initially supported by the Auto-ID center. EPC Global has been used as a guide for industry and for provider companies involved in visibility frameworks and RFID developments (Brock 2001). The aim of this framework is to enumerate the standards, the design of services, the data models, the core services implementation and the guidance for deployment visibility frameworks in SCM. On October 2001, the Auto-ID Center and its supporters began the first practical test of large-scale technology in a shop in Sam's Club in Tulsa, Oklahoma. The Auto-ID Center, now Auto-ID Labs, is developing, constructing and validating the global infrastructure that will instantly identify any object anywhere in the world from a network computer. The Auto-ID Center was officially closed in October 2003 and the technology was transferred to EPC Global. EPC Global is the leading definition that is providing visibility rules in supply chains using RFID. Many companies and governments make up this organization, some examples are: Johnson & Johnson, Kimberly-Clark Corporation, LG Electronics, Procter & Gamble or Sony Corporation.

In order to achieve the presented functionality several processes divide the visibility frameworks: the identification, the capture and the exchange (Figure 14). In the case of EPC global, the identification process is

tackled by using the RFID tag and readers. The capture process is the lower level that provides business context to the identification. Finally, the exchange process is responsible for carrying out the publishing and communication of the EPC data to the involved partners and stakeholders.

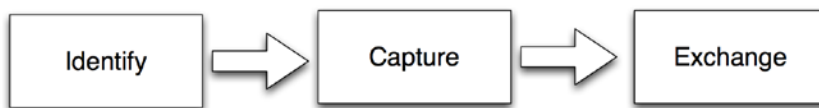


Figure 14 Main processes in the visibility frameworks

However, nowadays the most common technology of automating identification is the barcode; one key example is the Universal Product Code (UPC), which can be found in almost every manufactured product. This technology initially consists in a set of printed parallels bars and spaces, in which the read distance is about one meter. Then, newer versions of the bar code based on data matrix appeared in the market. A data matrix code is a two-dimensional matrix barcode consisting of black and white "cells" or modules arranged in either a square or rectangular pattern. The greatest improvement of the data matrix is that the encoded information can be text or numeric data and the usual data size is around 1556 bytes (Ohbuchi, Hanaizumi et al. 2004). Currently one of the most popular data matrix is the Quick Response (QR) code, which is a type of matrix barcode that was first designed for the automotive industry but now is adopted in many others areas (Gao, Prakash et al. 2007). Nevertheless, the barcode has great drawbacks with the current identification requirements in a supply chain, which initially are: the limited data stored, the necessary orientation of the product during the reading (line of sight), there is no possibility of embedded products, the

security because of the ease of copy (counterfeit) and sometimes the required use of human operating. In this situation, the appearance of new identification technologies based on radio frequency is essential in the automation of the identification in the supply chain (Michael and McCathie 2005).

2.2.1 The identification process: RFID

RFID systems are a generic term of smart cards in the identification processes, where radio waves carry the stored information using several radio frequencies. The data is stored on an electronic data-carrying device, which is called transponder or tag, and is located on the object that to be identified. The tag consists of a receiving antenna and transmitter, a coil and a chip (Figure 15). The second element is an interrogator or reader, which can read/write the tag. The communication between both is in a preselected frequency of the electromagnetic wave, in RFID the available frequencies are: Low Frequency (LF 30-300KHz), High Frequency (HF 3-30 MHz) and Ultra High Frequency (UHF 300MHz-3 GHz). The EPC Global Class-1 Gen2, as the most common, details the communication protocol between the hardware and software of the reader and tag, which is the air interface (Krishna and Husalc 2007). The main benefit to have the use of this technology instead of bar codes is the “non line of sight” nature of the radio. The radio nature allows to a non-contact identification, a read/write capability, a multiple reading capacity, higher reading distances and security techniques at the tag level. The chip/tag includes bytes of information; even with a customization of the technology different performances can be achieved, an initial option is the use of active tags, passive tags or a mix of both. The passive tags do not require an internal power supply, the power is supplied from the

incoming signal and the same wave transmits the outbound signal through the antenna. The passive tags have a short read range compared with active tags, which have their own internal power source. The active tags can transmit at a higher power and a longer range, as well as have larger memories than passive tags.

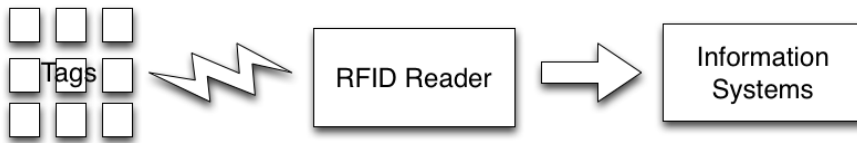


Figure 15 Simple RFID architecture

The benefits of the identification of products/resources through RFID are clear, but some disadvantages need to be detailed. Firstly, a key point is the high cost of passive tags and readers, in which there are working several research centers because of the exponential increase of the budget in global solutions (Roberti 2006) (Subramanian, Chang et al. 2005). Secondly, the customization of the hardware (tags and readers) because of the radio-frequency effects in metal and liquid materials (absorption, detuning and reflection), the reading collisions or the counterfeiting need to be addressed individually (Finkenzeller 2010). Therefore, the problem is the investment in the hardware equipment and the specialized consultancy during the deployment compared with the barcode, there are several works that are researching the ROI in RFID deployments (Sarma, Weis et al. 2003).

2.2.2 Capture and exchange

Once the identification process is realized between the RFID reader and the tag, the information submitted by the tag is stored in the memory of the reader. Therefore, the next steps are the correct capturing of the

information and the exchanging with the rest of interested stakeholders. The first step is to analyse the information that has been built, which is the core of the EPC Global data definition. The incoming data is called EPC, which is a set of identification coding standards that are stored in the tag. The familiar EAN/UPC bar code, which encodes the Global Trade Item Number (GTIN), provides identifying information for only the class of the item. EPC goes beyond this when the GTIN is encoded the EPC but adding an individual serialized item or instance of the item (sGTIN). Two units of the same product will have the same GTIN, but will have different EPCs (sGTIN) (Brock 2001). The next table details an example of a pure identity URI that is mapping the EPC data structure (Table 1). The header, the filter value and the partition identifies the length, the type, the structure, the version and the generation of the EPC. EPCglobal assigns this data and the company prefix to the company. EPCglobal is the organization set up to achieve worldwide adoption and standardization of the EPC technology. Then, the item reference identifies the object class of the product and the serial number is the instance of each product, the company assigns both.

Table 1 EPC Structure

<i>urn:epc:id:sgtin:0037000.112345.400</i>		
<i>Header</i>	<i>8 bits</i>	<i>sgtin</i>
<i>Filter Value</i>	<i>3 bits</i>	
<i>Partition</i>	<i>3 bits</i>	<i>5</i>
<i>Company Prefix</i>	<i>20-40 bits</i>	<i>003700</i>
<i>Item Reference</i>	<i>24-4 bits</i>	<i>11234</i>
<i>Serial Number</i>	<i>38 bits</i>	<i>400</i>

This encoding includes the unique identification code for products, not only for classes or references that supposes the instance of the product; it means an available classification of the products by reference, serial, class or company in the tag. The classification at this level is useful during operations because it offers a quick search for specific elements that can be grouped by company, class or even individually.

Once the data of the tag is formatted, there are lower level protocols in the EPC Global stack that control the reader management, the filtering and collection before sharing the information with the stakeholders (Figure 16). The Low Level Reader Protocol (LLRP) defines the control and the delivery processes of the tag between the hardware devices and the filtering module. After that, the Application Level Events interface (ALE) filters, collects and prepares the tag reads to the capturing application, which is the uploading interface to the EPCIS. At this point, firstly, the EPCIS takes charge of the standardization as the interface to

The EPCIS architecture includes three main modules, in which there are two applications and a repository and therefore two interfaces: the capture interface and the query interface. The next enumeration explains the role of each part in the EPCIS stack:

- The capturing application receives all the information submitted by the hardware through the RFID middleware, which are mainly filtered EPC data, but even integration of existing identification systems like bar codes or even manual input could be included. This process begins to attach business information to the EPC event by using adding information coming from the context. These events include the location, the time or the contained objects of the identified product. In a software point of view, this is a real time “push IT service” to deliver EPC events into a repository. Most of the existing implementations are based on web services ready to generate XML messages based on the standard and send them to the repository.
- The EPC repository is the database able to store the EPC events submitted by capturing applications, the uploading is made through the query interface. The objective is to provide a database accessible by using the Web services defined by the EPCGlobal standards in order to be accessible from different suppliers in a secure way during the product life-cycle.
- The accessing application is responsible for carrying out the interaction of the EPC information with the rest of the business stakeholders, even for trading partners. The interaction can be on-demand by using polling or subscriptions, both are based on asynchronous callbacks. In the polling case, the stakeholder

makes a request through the query control interface and receives a response immediately. In the subscription mode, the stakeholder establishes a subscription for a periodic or triggered query; once the query is executed, the results are asynchronously pushed to a recipient via query callback interface. Both combined with appropriate security mechanisms satisfy the communication protocols of the involved companies and upload online information to the existing information system of the companies. However, this information needs to be automatically published to the interested partners in order to achieve an effective communication. On this problem, the services layer defines the query interfaces: the query control interface and the query callback interface. Most of these definitions are published through an Extensible Markup Language (XML) schema via Web service Description Language (WSDL) over the Hypertext Transfer Protocol (HTTP). The service definitions use XML schemas, XML schema definitions (XSD) and Services-Oriented Architecture (SOA) approaches based on a WSDL. The objectives of these interfaces are to improve the fast development, the flexibility and the integration with the existing information systems of the companies. Besides, these protocols include authentication and authorization mechanisms to provide secure transaction. The next figure shows a proposal of query control interface as a WSDL, in which methods and arguments are described as services to allow automatic interaction with the EPC Repositories (Figure 17).

```
<<interface>>
EPCISQueryControlInterface
---
subscribe(queryName : String, params : QueryParams, dest : URI, controls :
SubscriptionControls, subscriptionID : String)
unsubscribe(subscriptionID : String)
poll(queryName : String, params : QueryParams) : QueryResults
getQueryNames() : List // of names
getSubscriptionIDs(queryName : String) : List // of Strings
getStandardVersion() : string
getVendorVersion() : string
<<extension point>>
```

Figure 17 Example of an interface definition

The complete standard definition is accessible on the EPCGlobal Web site (EPCGlobal 2007). Looking at the EPC Global architecture, this visibility framework is layered, extensible and modular (Figure 16). In order to support broad cases, the EPCIS provides abstract data models and data definitions, and flexible services organized in different service layers. The data layers specify a generic syntaxes and semantics of the data, in which the result of this definition is the Core Event Types (Figure 17). The initial event type is an abstract event definition called EPCIS Event, which arises as the parent definition and it is the only one not customizable. This class includes the time of recording and the event time; then, the extended classes include variables such as the EPC, the business step, the reader point, the location or any specific field defined by the stakeholders. If the stakeholders require more information in the EPC events, more information can be added to this initial definition of events by extending the classes.

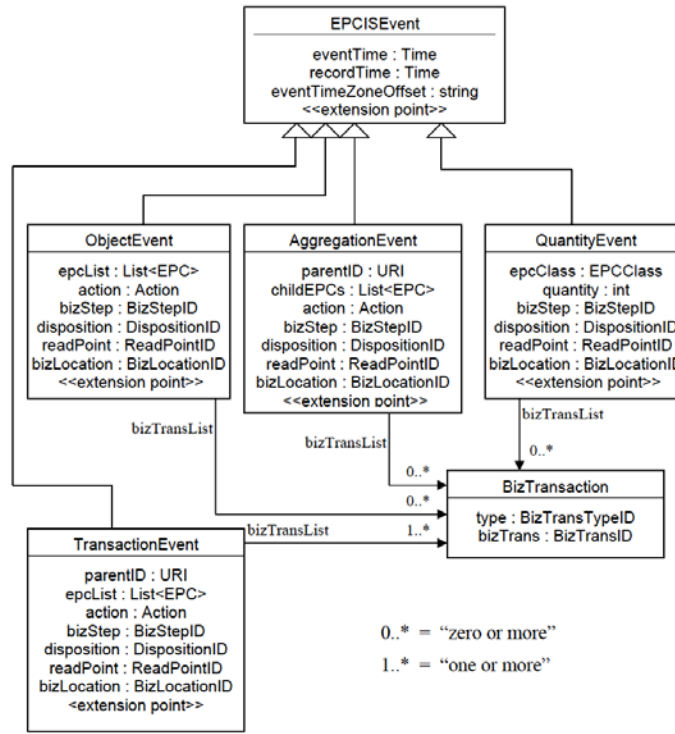


Figure 18 UML Diagram of The core events. EPCGlobal Copyrights

The next list enumerates the events that represent the core events of the EPCIS, they are also detailed in the figure 18 as classes in a UML class diagram (Figure 18):

- The ObjectEvent represents events produced by a product or a list of products.
- The AggregationEvent represents events that one or more entities have been made up together at the same element, as in a pallet or in a box.
- The QuantityEvent represents events that include a quantity of elements that are sharing a common EPC.

- The TransactionEvent represents events that denote an association or dissociation of one or more business transactions.

Inside every of the core event types there is useful information that can be used during the decision making processes in logistics and shop floor, especially in planning and asset management. This information abstracts the upper levels of the IT systems about physical details and specific layouts. This is the case of the “What?, When?, Where? Why?” information about the product; the next enumeration explains each component of these four dimension data that are included in the EPC events (Figure 19):

- What: This represents the subject of the event, which is the EPC of the product or products that are involved in the action.
- When: The date and the time when the event occurs, it also stores the record time and the time zone. In both cases, the data representation is a time stamp.
- Where: The location where the event has occurred, it is the physical read point, which defines the business location in the plant or shop floor where the event occurred.
- Why: The business context of the event, this information allows to identify the business step in the supply chain where the product is involved.

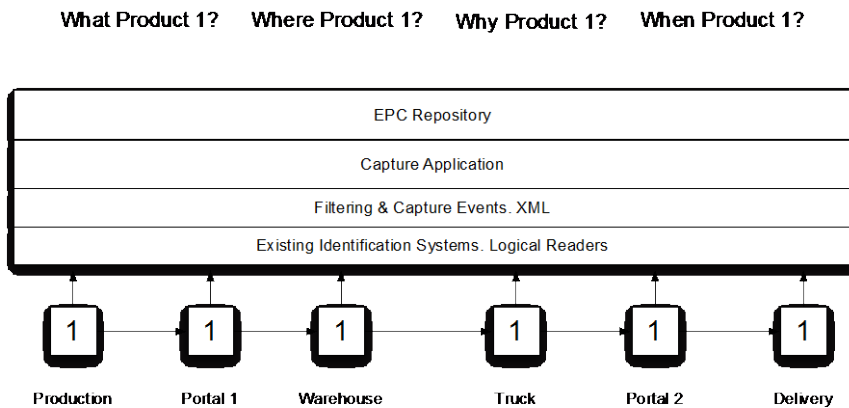


Figure 19 EPC Answers events during in the product life cycle

At least, this definition of events represents measured points of the states of the products during operations. The “What? When? Where? Why?” information allows maintaining a tracking process that is able to provide uncoupled information to the business intelligence, which can be applied in order to plan, to detect inefficiencies and even to forecast events in SCM (such as shipment delays, inventory shrinkage or out-of-stock). In this way, two companies/spin off related with the Auto-ID Labs are providing solutions layering the information flow into specialized ones; these are the cases of OATsystems and Redbite, which are spin-offs in the MIT and in the University of Cambridge. Especially interesting is the case of OATSystems, they propose a middleware in order to connect the ERP systems with the identification devices; this middleware is divided into the OAT device manager and the OATxpress. The first one connects hardware devices with managers that receive RFID information and the second one, once the RFID information is processed, offers services to the upper systems as ERP or WMS (Figure 20).

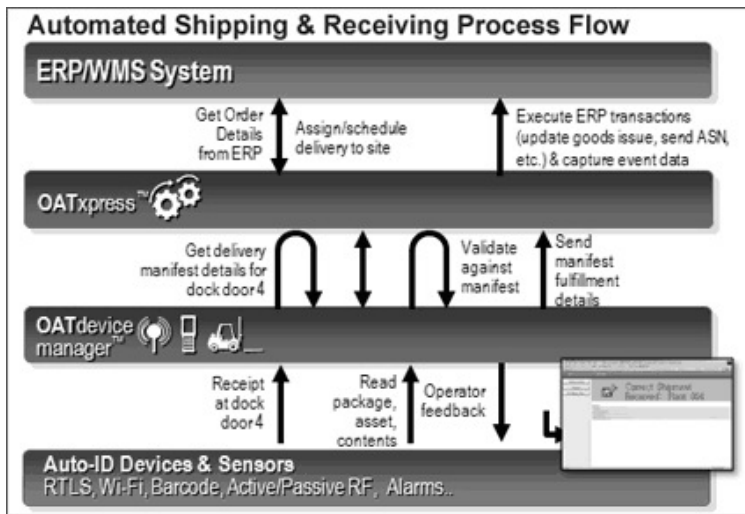


Figure 20 OATSystems information flow. OAT copyrights

The results of these visibility systems are clear, they allow sharing the right information with independent business models and external information systems. Instead of having a global business model, the EPC structure can be perfectly adapted to individual business models or even logical units that require feed field data from composed business models. The inter-organizational data exchange based on the product life cycle supposes a great advantage in small and medium enterprises (SME) in order to improve their competitiveness because of the global integration, which improves the competitiveness for system components. The figure 21 shows an example of how the companies that are involved in the same supply chain can obtain the right information about products through queries; either for polling or for subscriptions by using the defined public interfaces. In the example, the blue company is waiting to start the process “3” when the red company finishes the product process “2”. The subscription to this event generates a call-back to the company “4” when the process is finished. This process is transparent to both information systems and adding real time features (Figure 21).

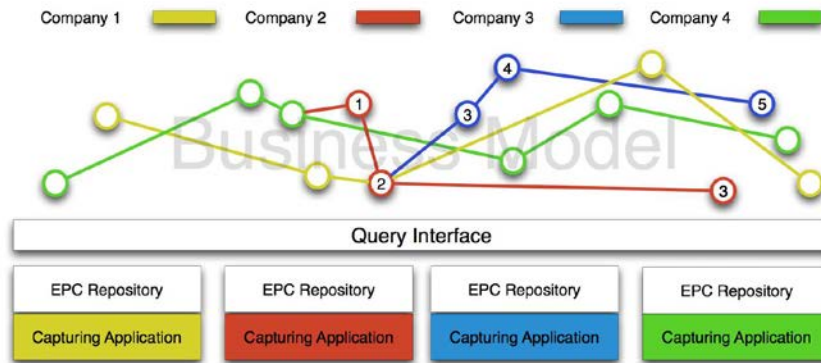


Figure 21 Interaction of business models by using EPC data

After the description of the new visibility framework current features, the next subsection details the agent-based concepts coming from the Distributed Artificial Intelligence (DAI) as a way to process all the information coming from these visibility frameworks. The manufacturing agent-based architectures, methodologies and frameworks that are presented in the next section are lean on the Agent Oriented Software Engineering (AOSE), as a main field in the computer science. The AOSE argue the analysing, designing and implementing software as a collection of interacting autonomous agents based on the previous programming paradigms (Jennings 1999). For example, the AOSE can include additional extension of the Unified Modelling language (UML) called Agent UML (AUML) (Bauer, Müller et al. 2001).

2.3 MAS in Manufacturing/Shop floor

Once the information about the environment, which is made up of the state of products and resources, is accessible to the product scheduling control through the visibility frameworks, the next step is to process the information in order to support an effective and flexible product decision-making. New service based discovery within distributed artificial

intelligence are appearing to fit into those requirements, among them agent-based looks on of the most appropriate (Wooldridge, Weiss et al. 2002). From a manufacturing vision point of view, several authors present Multiagent systems (MAS) and Holonic Manufacturing System (HMS) as the leading technologies to achieve the current control requirements (Leitão 2009). Most of the basis of the existing MAS and HMS manufacturing control proposals are a generalization of the hierarchical and heterarchical controls based on distributed decision-making, which contain specialized software autonomous units called Holons, and agents interact. Initially, Holons were motivated by the inability of the existing manufacturing systems, mainly on the outdated concept of CIM. The CIM is inflexible, fragile and difficult to maintain, which is caused by a centralized control. Firstly, the Holon concept was initially presented by Arthur Koestler based on social and biological systems (Koestler 1969), and its forward extension concepts have been focused in research committees such as: the Intelligent Manufacturing System (IMS) program, the IFAC technical committees in manufacturing and logistics systems (TC 5.X) or the IEEE IES technical Committee on Industrial Agents. The HMS vision consists of self-reliant units that are able to control and execute their own plans as a part of a flexible manufacturing set (Bussman, Jennings et al. 2004). Bussman defines the Holon architecture as a layered unit divided by three specialized operational levels: the physical processing, the physical control and the information processing part (Figure 22). The information processing combines the interface with Holons and humans, and the self-reliant capacity of decision-making in a unique element.

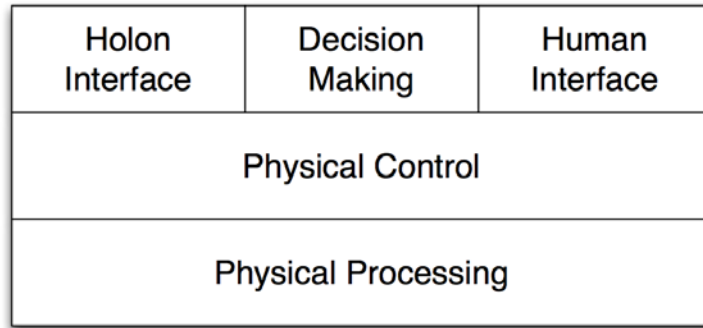


Figure 22 Holon architecture

More specifically from a manufacturing point of view, Marik and MacFarlane address the Holon as a logical unit with hardware integration, high level information processing and communication capabilities (Marik and McFarlane 2005), they consider three communication channels (Figure 23):

- The Intraholon communication, which is between the low-level part and a pure software agent, senses and acts between the hardware elements and the agent part.
- The Interholon communication among agents, this is the horizontal information flow used by the agents during the negotiation process.
- The direct communication between the real time-control parts of neighbouring holonic agents. It is the horizontal communication used by the low-level elements in order to provide direct real time communication.

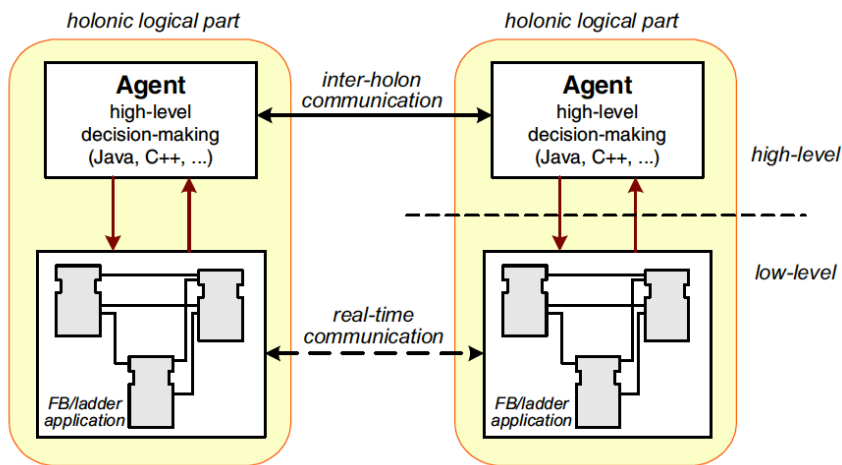


Figure 23 Decision making levels (Marik and McFarlane 2005)

Secondly, instead of the pure manufacturing vision as the Holons, the agents have arisen as one of the main fields that mix computer science and artificial intelligence in several fields. One of these fields is manufacturing control, where the agent software paradigms can implement concepts developed in the holonic vision. The obtained result can thoroughly satisfy the initial holonic requirements by using the existing agent-based developments as: interaction protocols, role assignments, behaviours, self-organizations, communication languages or performatives (Vladimír, Vrba et al. 2005). These approaches are similar and complementary, in which the Holons can be implemented by agent-based software frameworks. Some authors present Holon concepts as a vision by the research community to reach the new manufacturing requirements and the agent technology systems as one of the possible implementations of it (Giret and Botti 2004). The agent-based systems are an extensive major field of research in computer science and artificial intelligence; it is not only a manufacturing vision. This technology provides frameworks, protocols and software techniques to simplify the

design and implementation to address complex distributed systems (Bellifemine, Poggi et al. 2001).

The agents are entities of the computer system capable of undertaking autonomous actions to achieve self-interested and cooperative goals (George 1995). The specification of agent is based on techniques of modularity, distribution, abstraction, and intelligence. Once the agents manage the information, the system incorporates functionality as services that can be directly plugged to support new layout configurations; this supposes flexible interaction. This agent interaction is based on an architectural style that defines an interaction model between two primary parties: the service provider, who publishes a service description and provides the implementation for the service, and a service consumer. Both can either use the Uniform Resource Identifier (URI) for the service description directly or can find the service description in a service registry and invoke the service. This way of work defines the last programming paradigm, which is currently being adopted by the software providers. This is the service approach, which adds new levels of adaptability and functionality (Figure 24). The service provides a framework in which programmers can work in a variety of styles and freely intermixing modules from different paradigms but sharing common interfaces. Previous paradigms, which are currently used in business, are designed by the industry as components but a lack of flexibility is remarkable; especially denoted is the cases of the object oriented paradigms (García, de las Morenas et al. 2011). Initially, programming paradigms were procedural, structural and assembled by using memory addresses; these paradigms are currently included in the hardware or in the low level manufacturing elements such as the PLC. The next figure shows a brief shorter list of the programming paradigms described before (Figure 24).

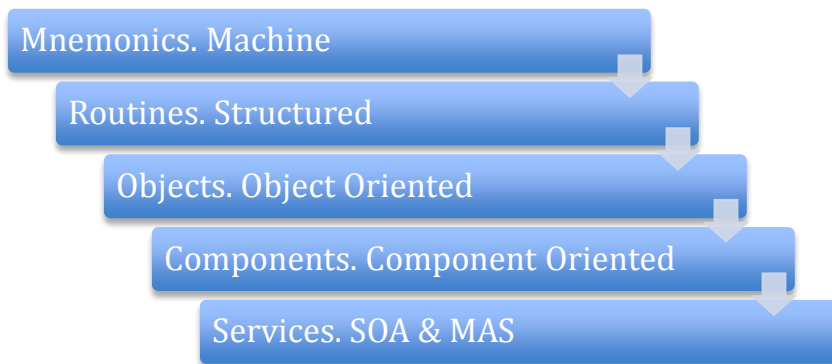


Figure 24 A Chronogram of paradigms in programming

2.3.1 Agent-based technology

Agent technology is an area of distributed artificial intelligence (DAI) that has already been successfully applied to industrial environments (Shen, Wang et al. 2006). It is hard to find a definition of the term ‘agent’ to which all authors can agree. Fisher defined an agent as an encapsulated entity with traditional AI capabilities (Fisher 1995), while Jennings defined an agent as a self-contained problem-solving entity (Jennings 1999). In general, it can be said that the agents are entities of the software system, which are able to perform autonomous actions in a flexible way in pursuit of established objectives. The characteristic of flexibility implies that these agents have a reactive character – as they can perceive the environment and act to meet the changes in conditions -; they also have a pro-active character – as their behaviour is oriented towards the achievement of predefined objectives -; and they have social capabilities – as they can interact with other agents through negotiation processes in pursuit of their respective objectives - (Wooldridge, Weiss et al. 2002). Wooldridge and Jennings describe the agents by enumerating several features, which are (Jennings and Wooldridge 1998):

- **Autonomy:** The agents have the control over their own actions and internal states. Their behaviour is totally predictable or controllable by themselves; they can act without direct human intervention.
- **Social ability:** The agents act with other agents (artificial or human) to achieve their goals, complete assignments and to help others. This may require communication, coordination, cooperation, competition and conflict management capabilities.
- **Reactivity:** The agents can perceive their environment, which may be the physical world, a user behind a graphical interface or other agents, and respond appropriately to changes brought about in it.
- **Initiative / Proactivity:** The behaviour of the agents is determined by the objectives or goals to pursue and therefore no action can occur only in response to the environment.

In addition to these attributes, there are others authors that remark more features such as (Jennings and Wooldridge 1998):

- **Reasoning and learning:** The agents are able to change their behaviour through experience and intelligence.
- **Transportation:** Mobile agents are able to move between nodes/computers in a network and run on different platforms.
- **Credibility:** The agents have a valuable reputation that influence in the behaviour of the rest of the agents, in their thoughts and feelings.

- Temporal continuity: The agents are continuously running processes and behaviours.

The breadth of the agent concept directly obliges to focus the target of the application; firstly, it is necessary to know what the application domain is in order to define the better internal structure. To be able to address some of the previous features, most of the research community divides the internal agent into three architectures (Nwana 1996). The first one is the reactive architecture where the reasoning process is a perception-action cycle: stimulus and response (Figure 25). In a reactive process, there is no explicit representation of the environment or modelling, the decisions do not take into account the past or the possible future. In more complex reactive agents, as coordinate or cooperative agents, the reasoning process only takes in account the instruction history as perception or beliefs (Demazeau and Costa 1996). The main problem of this agent architecture is the lack of information about the environment to respond appropriately. Therefore, if the reasoning is based in only local information it decreases the learning capabilities because the development of the behaviours is complex and the inputs are reduced.

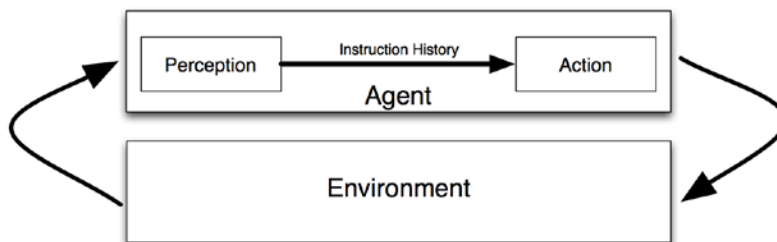


Figure 25 The reactive architecture schema

The second agent architecture is the deliberative, which situates two processes between the perception and the action: the deliberation and the reasoning. The deliberation decides the objectives that need to be addressed, and the reasoning how to achieve those objectives previously selected. This is the base of the practical reasoning, which is based on the constant decisions for what action has to be taken in order to facilitate the objectives. The third architecture is a mix of the previously presented; this combines reactive and deliberative components in a hierarchy of interacting layers and where each layer defines an abstraction level.

Once the agents and its internal behaviours are clarified, the next step is the explanation of MAS as a set of agents. Therefore, a MAS can be defined as a set of specialized agents, which are capable of communicating and interacting in order to achieve their individual and cooperative goals, even when they do not have enough knowledge and/or skills to achieve individually their objectives (Sandholm 1999). These properties make the MAS structure applicable to highly dynamic situations, which turn them into promising candidates in providing a management solution. Moreover, software agent technology can monitor and coordinate events and disseminate information improving visibility and creating organizational memories (Zlotkin and Rosenschein 1989). An important issue is that a set of agents can learn from their own experience, receive information from others about their environment and adapt them to be closer to the solution at every instant. Therefore, specialised agents can take unexpected fluctuations or variations in the environment into account immediately; thus creating an autonomous system that is able to operate without user intervention. The inherent communication between agents is based on the speech acts, which have the characteristic of actions in their utterance, including illocutionary,

locutionary and prelocutionary aspects (Austin 1962). Some examples are: asking, making a request, describing, subscribing or proposing. In order to organize a standard definition of the speech acts and the agent communication, the Foundation for Intelligent Physical Agents (FIPA) arises in 1995; they developed the Agent Communication Language (ACL), based on what was learned on the Knowledge Query and Manipulation Language (KQML) (Poslad, Buckle et al. 2000). In this standardization, the performative defines the type of speech acts or communicative acts; some of the most common performatives of FIPA are: accept-proposal, agree, cancel, call for proposal (CFP), confirm, inform or failure. Besides, this interaction needs patterns of message exchange, FIPA implements several interaction protocols but it is not restricted from using or defined new, in order to simplify the design of the programmers (Vidal, Buhler et al. 2001). The ContractNet protocol, the request interaction, the brokering, the subscribing or the auction protocol are some of the predefined interaction protocols included in the FIPA standard. The next figure shows the contract net protocol, which is one of the most used interaction protocols; it negotiates the acceptance or rejection of predefined proposals between agents (Figure 26). The parameters that are needed for communication vary according to the situation; the only parameter that is mandatory is the performative, although it is expected that most ACL messages will also contain sender, receiver and content parameters. This kind of communication supposes a dynamic way to synchronize the functionality based on the SOA paradigms; instead of the object-oriented programming that only can invoke previously compiled functions. Apart from the syntax, there are several communication technologies that are supporting the MAS information flow; some examples are the Common Object Request

Broker Architecture (CORBA), the Remote Method Invocation (RMI) or the Distributed Component Object Model (DCOM).

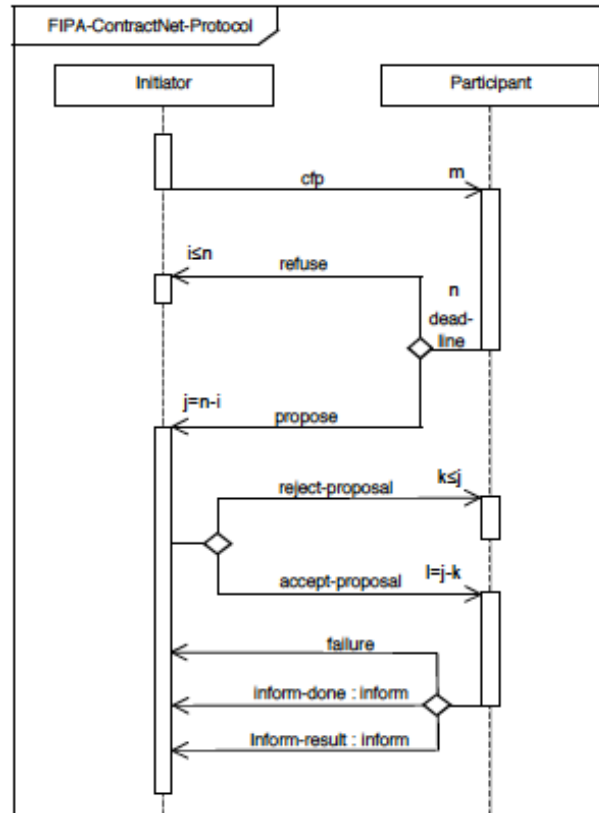


Figure 26 The contract net interaction Protocol. FIPA Copyrights.

Once the protocols and technologies allow a distributed and transparent communication between agents with an internal reasoning, a first organization to classify the MAS features is essential. The first classification of MAS is modelled in layers, which mainly includes two types of layering: the vertical and the horizontal (Figure 27) (Di Marzo, Foukia et al. 2004). In the horizontal layering, every layer can act as an independent agent, it means that the agent is directly connected to the perception and to the action as the touring machines. In the vertical approach, there are interfaces between the agents as perception/action

communications and only the last layer is connected to the real world. The vertical layering is the most common in order to specialise the reasoning of each step and clearly divide the functionality and knowledge. Specifically, the MAS vertical layer architecture and the deliberative agent architecture have been the better extended in the research community (Ouelhadj and Petrovic 2009), mainly because of the application of BDI reasoning.

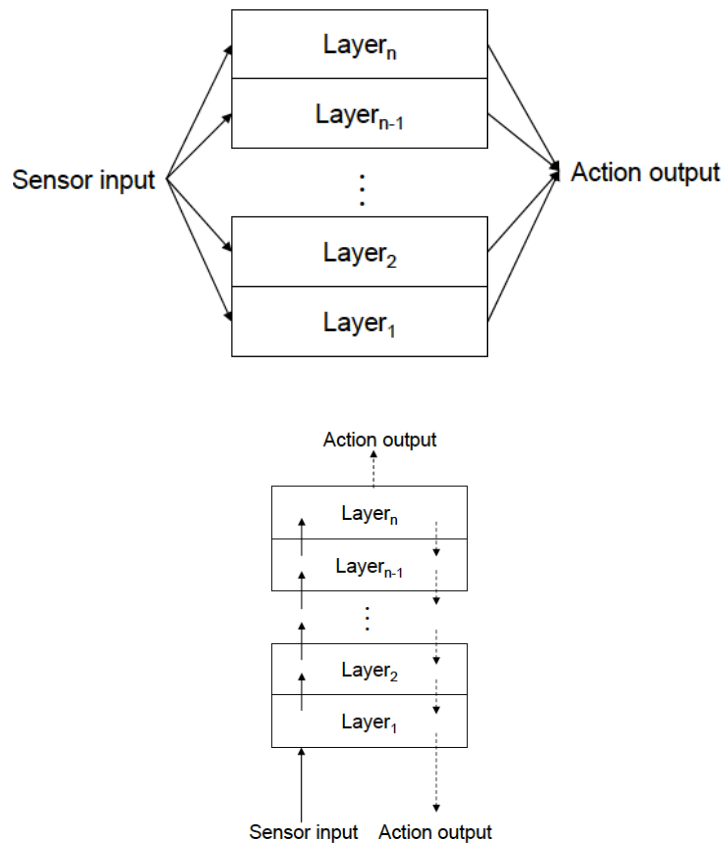


Figure 27 The agent architectures: Horizontal and Vertical

To implement these agents in a very direct and relatively easy way, the agents can be endowed with associated information defining their beliefs,

desires and intentions (BDI); which structures their interaction capabilities (Georgeff and Rao 1998). The basic components for BDI agents are the data structures, which represent their beliefs, and the desires, the intentions and the functions represent the deliberation process that specify the outcomes/instructions. Several authors have extended the initial BDI architecture but the main concepts are explained in the next list (George 1995):

- The beliefs represent the internal perception of the state of the agents and their environment, which is made up by using the incoming information captured in the perception process. The term belief rather than knowledge recognizes that the beliefs may be not necessarily true.
- The desires are the objectives that the agents would like to accomplish or bring about in the search of the goals. A goal is a desire that has been adopted for active pursuit by the agent as a final step. The term goal adds further restrictions because the set of active desires must be consistent.
- The intentions represent what actions the agents have chosen to do. The intentions are the desires to which the agent has committed as instructions; in implemented systems, it means that the agent has begun to execute a plan.
- The plans are sequences of actions that agents perform to achieve one or more of their intentions; these plans can include other plans.

The intentions play a leading role in the BDI model as they determine the practical reasoning of the agent. Thus, the BDI agent-based architectures lean on practical reasoning; so that their decision process is similar to that followed by humans. The process of plan computing is as follows: a sensory input receives an event and a belief revision function modifies the beliefs. Then, the option generation function evaluates the beliefs in order to generate the possible options, at this moment the desires are established. After that, the filter function considers current beliefs, desires and intentions and produces new intentions. Finally, the action execution function receives the intentions and decides what action performs (Figure 28).

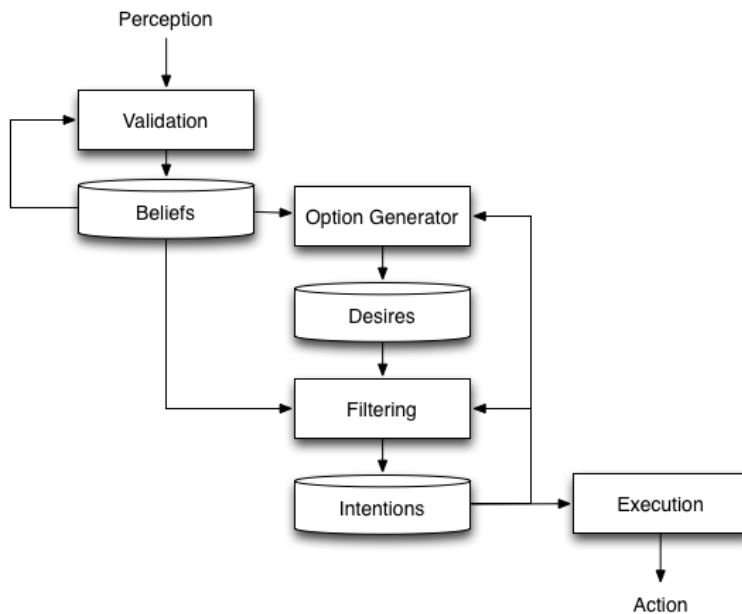


Figure 28 BDI Reasoning

In the BDI reasoning, the events active the agents by updating their beliefs, trigger plans or modify goals. The events may be generated externally, received by sensors or even produced by them. The next subsection explains the most important research impacts of the agent-

based technology in the manufacturing area related to the problem at hand, in which the BDI is well accepted. One example of the existing successful acceptance of the BDI vision is for example the development of customized BDI agents with the Jadex tool (Pokahr, Braubach et al. 2005). Jadex addresses the object-oriented representation of the BDI concepts with XML descriptions, including the goal conditions, the condition events and the environment feedbacks (Pokahr A 2003); it is a reasoning engine developed with the Java Agent Development Framework (JADE).

2.3.2 Survey of Holons/Agents in manufacturing

More than 100 agent toolkits were detected as possible solvers in several manufacturing domains; many of them were focused on scheduling and control problems (Lee and Kim 2008). The research community has developed customized holarchies as architectures or methodology approaches during the last decade (Leitão 2009). In the beginning, the research works introduce the “ContractNet” protocol to the community, which supposed the first interaction protocol between agents that include advances in negotiation; this protocol initially described the automated competitive negotiation through contracts, which has been studied and redefined by several authors (Smith 1980). This early definition of interaction protocols between software units allowed to begin the involved researches to design architectures to support distributed manufacturing negotiations. One of the earliest approaches to agent/Holon control were proposed by Shaw, this control was based in the subcontracting of tasks between manufacturing cells (Shaw and Whinston 1988). At the same time, Parunak proposed the first pure agent-based manufacturing, called YAMS (Parunak 1987). This

approximation includes the representation of the manufacturing elements using software units with a “ContractNet” interaction protocol.

Then, pure manufacturing agent-based systems were developed as an extension of previous object oriented philosophies, with a high degree of autonomy and an inherent capability of specialization. Along this line of thinking, there were introduced complete manufacturing control systems as that proposed by Burmeister (Burmeister, Bussmann et al. 1998), the Product-Resource-Order-Staff architecture (PROSA) methodology by Van Brussel et al. (Van Brussel, Wyns et al. 1998) or the ExPlanTech by the Gerstner Laboratory (Pechoucek, Vokrinek et al. 2005) that provide support for decision making using distributed reasoning systems. In 1998, as one of the key references in the holonic manufacturing systems, PROSA includes the definition of a three-dimension specialization of agents: product, resources and orders. This concept belongs to the Manufacturing Control Systems Capable of Managing Production Change and Disturbances Project (MASCADA), in which a pro-active reasoning takes outcomes in order to reduce the effects of disturbances in the manufacturing schedule (Kollingbaum, Heikkilä et al. 2000). Other surroundings works are adding new elements to a distributed control, the ADActive holonic COntrol aRchitecture for distributed manufacturing systems (ADACOR) adjoins a supervisor Holon, which introduces coordination, monitoring and optimization to the holonic-distributed negotiation process (Leitão and Restivo 2006). This supervision works as an expert system, in addition to the PROSA approach.

Other works such as the Plant automation Based on Distributed Systems (PABADIS) project of Lüder et al. (Lüder, Peschke et al. 2004) outlines the use of new types of agents and topologies in control systems. PABADIS and Product Oriented Manufacturing Systems for Re-

Configurable Enterprises (PROMISE) propose a Manufacturing Execution System (MES) level and a Field Control level completely decentralised; both were EU projects funded by the 6th Framework Programmes. At this point new visibility technologies are introduced; the products or resources are located by using smart identification tags, which are attached directly to the product as mobile software agents. Once the product arrives at a specific reader station, the agent detects the RFID tag and initialises a negotiation to ensure a proper product processing; then, further negotiations based on AI techniques produce the necessary instruction to process the products. This work introduces the idea of the junction between the RFID technologies and MAS information processing. At this point, the RFID is presented as a visibility service for the agent manufacturing applications, in which Auto-ID Labs is the research reference. This leading global network of academic research laboratories works in the field of networked RFID. The labs comprise research universities to architect the “Internet of Things” together with EPCglobal. Auto-ID Labs apart from research in the application of RFID in the industrial spaces and in supply chain management are involved in the integration of RFID with intelligent decision-making techniques. One key idea is to develop the agent-based manufacturing control as an intelligent product-driven negotiation in the supply chain (Wong, McFarlane et al. 2002). Auto-ID Labs proposes RFID as an information service, which gives an industrial and business context to the products; RFID is additional sensors that close control specific products (López 2011). This allows an agent-based control to every product with regard to the logical state of the product and the involved identifiable resources; together mixed within the mobile software agents define the intelligent product. The join of RFID with new paradigms of information

management as MAS has been also used for short term planning by several authors (Pechoucek M. 2007) (Vladimír, Vrba et al. 2005).

In this line of thought but including BDI features, JACK is another reference in manufacturing agent system; it is a framework in Java for MAS development (Howden, Rönquist et al. 2001). It defines an environment for building, running and integrating commercial JAVA-based MAS using a component-based approach. JACK is based on the BDI model dMARS developed at the Australian Artificial Intelligence Institute (d'Inverno, Luck et al. 2004). One of the new ideas of Jack, which can be applied to shop floor scheduling, was the task manager role in the implementation of agents. It can provide several notions of time such a real time clock or a fast-forwarded, slowed or even stopped. JACK remarks the use of an expert system in the scheduling of the agent intentions, and the development as an extension of object-oriented development based on the BDI architecture. The JACK development environment can be divided into three main components:

- The JACK agent language, this is a superset of the Java language, and introduces new semantic and syntactic features, classes, interfaces, and methods to deal with agent-oriented concepts.
- The JACK compiler, which compiles the JACK Agent Language down to pure Java, allowing the resulting agents to operate on any Java platform.
- The JACK agent kernel, which is the runtime program, within which JACK agents operate, provides the underlying agent functionality that is defined within the JACK agent language.

A key example of the implementation of the JACK concepts in manufacturing environments was the Cambridge packing cell (Fletcher, McFarlane et al. 2003). This demo demonstrated that mass-customisation with distributed control that exhibits a high potential for responsiveness in customized packaging. JACK agents execute their reactive and deliberative behaviours through pre-compiled plans, which feed information from three interfaces:

- A common blackboard, which is connected to a PLC that handles the communication with hardware, exchanges the sensor data and the commands with the shop floor machinery.
- An ID system that sends identification messages via Simple Object Access Protocol (SOAP) to gather RFID data on which shuttle has just passed a reader. This system was called Savant, which was the initial version of the EPCIS.
- A server that provides knowledge in the orders that is being placed via SOAP.

Once the information about the shop floor environment is accessible to the MAS and effective interfaces are defined, a gain of knowledge based on simulations and estimations is essential to provide accurate decisions. Manufacturing Agent Simulation Tool (MAST) introduces simulation concepts to the handling of manufacturing materials as manufacturing cells, conveyor belts, diverters or AGVs (Vrba 2006). In the simulation phase, the agents carry out the transportation of discrete entities among manufacturing cells, which gain knowledge before the real deployment in the manufacturing cell. This research group called “The Agent Technology Center” also drives other interesting agent project as the

AgentFly: Towards Multi-Agent Technology in Free Flight Air Traffic Control or task oriented agent architecture as abstract solvers to a broad set of problems (Šišlák, Pěchouček et al. 2008).

Getting more focus on industry applied manufacturing agent proposals, Bussman proposes production 2000+, which probably was one of the first long scale agent-based system implanted in a manufacturing process (Bussmann and Schild 2001). The target area was the production system for manufacturing cylinder heads in the Mercedes-Benz engine plant in Stuttgart. The agent-based control system consists of three types of agents: work piece agents, machine agents and switch agents. The control algorithm takes charge of the loading, the searching, the allocation and the processing tasks, which are scheduled by using the physical distribution of agents; this proposal also presents the tracking of the resources by using RFID. Looking at the deployment in real scenarios, different researchers and companies have developed industrial applications of manufacturing MAS with satisfactory results (Glaser 2004; Leitão 2008; Xia 2009).

More agent-based systems appeared as extensions of the knowledge oriented methodologies, such as: the MAS-Common KADS developed by Iglesias et al. (Iglesias, Garijo et al. 1998); or the CoMoMAS proposed by Glaser (Glaser 1997). There are agent methodologies based on roles such as that proposed by Moulin and Brassard (Moulin and Brassard 1996); GAIA by Wooldridge et al. (Wooldridge, Jennings et al. 2000); or MaSE by DeLoach et al. (DeLoach 2001). And other agent methodologies are system oriented such as: MASSIVE, proposed by Lind et al. (Lind 2001); Tropos by Bresciani et al. (Bresciani, Perini et al. 2004); or MESSAGE by Caire et al. (Caire, Coulier et al. 2002). Another example proposed by Zhang et al. comes from the supply chain

environment and is based on simulation and decision making tools (Zhang, Anosike et al. 2006).

Other interesting works in the surrounding research of the agent/Holon are based on server facilities and hardware communication standards. As service facilities, there are some examples like: CIIPLEX (Labrou, Tolone et al. 1998) or mediators such as MetaMorph that provide administrator and registration services to the agents (Maturana, Shen et al. 1999). The results of these ideas were that management services and director facilitators are required. Looking at this problem, FIPA took in charge of the standardization in order to integrate these works, FIPA defines protocols and languages to allow interoperability in the agent spaces (Poslad, Buckle et al. 2000). FIPA includes definitions in several areas of the agent-based technology such as agent communication, agent transport, agent management, agent architecture and applications. These definitions not only include architecture or design proposals; most of them already include the definition of frameworks and toolkits. It produces an abstraction in which common code providing generic agent functionality that can be specialized or improved by user code, thus providing specific functionality. These agent toolkits include several layers of functionality as transportation, ontologies or discovery services that cannot be implement in every development. Some examples as ZEUS include this abstraction, which defines low level services as message exchange using the TCP/IP protocol and high level services by using management agents (Nwana, Ndumu et al. 1999).

Following the toolkit developments, one of the most well-known toolkits that is used in many of application fields is JADE, which is not a manufacturing approach in nature. JADE is a FIPA compliant agent

open source project distributed by Telecom Italia Labs (TILab) that has been under development since 1999 at TILab (Bellifemine, Poggi et al. 1999). The JADE platform is made up of a number of containers that operate on individual machines (Figure 29). The containers are the running instances where several agents share computer facilities; a container cannot be divided into two machines. A platform can be thought of as defining a common application domain, and agents within this platform have access to the common infrastructural services. JADE includes the most common facilities in the definition of agents. Each JADE agent has access to an Agent Management System (AMS), which manages the platform and supervises access to it as well as provides white page services (Bellifemine, Poggi et al. 2001). The Agent Management System (AMS) provides the naming service (i.e. ensures that each agent in the platform has a unique name) and represents the authority in the platform (for instance it is possible to create/kill agents on remote containers by requesting that to the AMS). Then, Directory Facilitators (DF) offers Yellow Pages services; for that reason, a unique Universal Discovery Description and Integration (UDDI) identifies every agent in the platform. Therefore, the DF provides the yellow page services, which mean that the agents can directly search for services in order to achieve their goals, not for agents. Apart from JADEX or ZEUS, there are runtime frameworks for agent development such as the Cognitive Agent Architecture (Cougaar). Cougaar is an open source runtime framework that includes a variety of demonstration, visualization, and management components to simplify the development of complex distributed applications based on agents (Helsinger, Thome et al. 2004). Apart from the toolkits, all of agent development frameworks include information exchange technologies and some approach to offer a semantic interoperability of agents in order to facilitate the development. The

toolkits include some technologies such as: Agent Communication Language (ACL), DARPA Agent Markup Language (DAML) and other standards of XML like the Web Ontology Language (OWL) (Haarslev and Måller 2003).

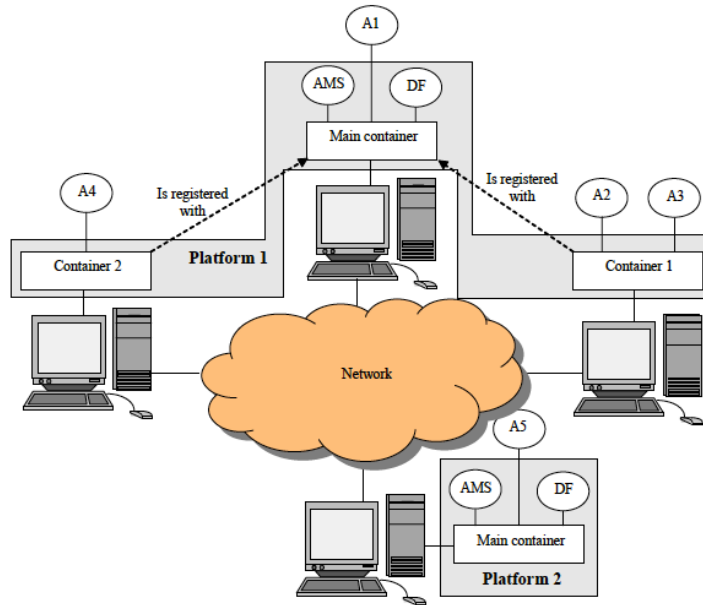


Figure 29 The JADE Architecture. TILAB copyrights

2.3.3 Planning Techniques on MAS

In general, a MAS shop floor problem can be defined as a stochastic problem of planning a set of products and resources. In MAS planning, the solution is a plan, which is a partially ordered sequence of actions that when they are successfully executed, the results are a set of achieved goals for some of the agents. In the shop floor planning approach, the agents are strongly related and cooperatives during the execution of the plan, any disturbance is sensitive to the MAS and therefore to the collaborative negotiation during operations. In general, the agent-based plan phases can be distinguished in the next list (DesJardins, Durfee et al. 1999). The

planning is a combination of phases 2 and 3, which are often interleaved; a subset of agents could perform planning by executing the next phases:

1. To allocate goals to agents.
2. To refine goals into subtasks.
3. To schedule subtasks by adding resource allocation (possibly including the agents) and timing constraints.
4. To Communicate planning choices (of prior steps) to recognize and resolve conflicts.
5. To execute the plans.

There are, however, many other methods to assign tasks in a more distributed way, allowing the agents a higher degree of autonomy and privacy. For example, complex task-allocation protocols may be used, such as auctions and market simulations. The market simulation, which is a common task-allocation technique for distributed systems, has implementation problems. These problems have been detected in the integration with the existing systems, the impact when modifications appear in the model and in a non-feasible complexity in the design. In these high heterarchical techniques, the role of every element is not easy to put in concordance with the global decision. The MAS implementation of these techniques require deep knowledge of technical details in agent architectures, agent development tools, MAS concepts; especially if the complexity of the schedule algorithm arise to levels such as market simulations (de Weerd and Clement 2009). For distributed planning problems, auctions are a more simple way to assign tasks to the agents that attach the best rewards but quantifiable instructions need to be

addressed. This way of work requires a value-oriented environment, as the proposed by Markov. The Markov Decision Process (MDP) is not only a planning value oriented technique; it also represents a way of schedule in uncertain or partially observable spaces. The implementation of planning MAS based on MDP has been analysed by several authors (Spaan and Melo 2008).

MDP is a discrete time stochastic control framework where the probability to choose from an initial state to another state depends only on the current state and in the selected action, not in the prior history; this is known as the Markov property. MDP is useful for studying a wide range of optimization problems solved via dynamic programming and reinforcement learning. In a game or in a perfect world the actions are deterministic, but the shop floor and the manufacturing environments are stochastic in nature because they are affected by constant disturbances, which cannot be calculated as a mathematical model. The next list explains these types of actions (A), which produces a transition between the state (S) and the expected state (S'):

- Deterministic Actions: $T: S \times A \rightarrow S'$. For each state and action, the inference specifies a new state.
- Stochastic Actions: $T: S \times A \rightarrow Prob(S')$. For each state and action, the inference specifies a probability distribution over next states, which is represented by using the distribution $P(s' | s, a)$. The figure 30 shows a directed graph with probability transition between states (Figure 30).

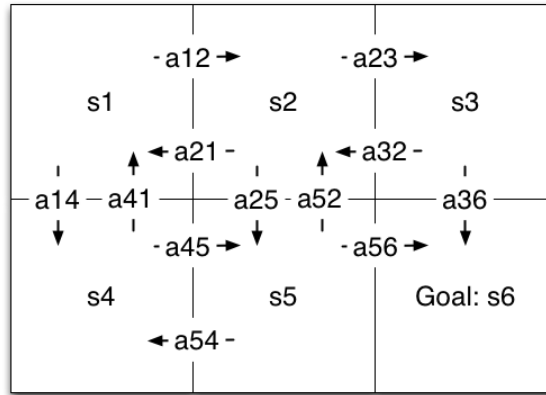


Figure 30 A MDP transition states example

The MDP objective is to achieve a policy of actions (π) by mapping from the state (S) to the target state (S'). Specifically, the core problem of MDPs is to find a policy for the decision maker: a function π that specifies the action that the decision maker will choose in every state. Once a Markov decision process is combined with a policy, both fix the action for each state and the resulting combination behaves like a Markov chain where the discount rate is typically close to one. The discount rate is a concept is coming from the idea of a reward in the future is not worth quite as much as a reward now because it is in constant change; the shop floor environment is highly active. The goal is to choose a policy π that maximize an accumulative function of the rewards (R) with expected discounted sum over a potentially infinite horizon (γ). The next equation defines a policy by choosing the best reward between the states in the search of the goal state during a task (Equation 2).

Equation 2 The policy equation

$$\sum_{t=0}^{\infty} \gamma^t R_{at}(S_t, S_{t+1})$$

There are several techniques to solve the problem at hand as: value iteration, policy iteration, modified policy iteration or prioritized sweeping (Puterman 1994). However, if the transition probabilities are not priori defined or even the rewards the problem is one of reinforcement learning (Sutton and Barto 1998). For this purpose, it is useful to define a further function that corresponds to taking the action and then continuing optimally or according to the policy as the $Q^*(s, a)$; the optimal solution in the $V^*(S)$ function is the maximum reward in the state transition (Equation 3). While this function is initially unknown, the experience during learning is based on (S, A) pairs that means "I was in state S and it tried doing A and S' happened" feed the function during operations. Thus, one has an array Q and uses experience to update it directly. This is known as Q-learning; the power of reinforcement learning lies in its ability to solve the Markov decision process without computing the transition probabilities (Littman 1994).

Equation 3 The MDP function

$$Q^*(s, a) = R(s) + \gamma \sum_s P_a(s'|s, a) V(s')$$

$$V^*(s) = \max_{a \in Ap(s)} [Q^*(s, a)]$$

However, manufacturing complex planning requires rethinking traditional approaches to AI planning in the treatment of disturbances and tracking. The resulting plans in a dynamic environment require a continual approach in which planning and execution are interleaved. The uncertainty is recognized and handled appropriately when re-planning can be performed while the situation changes or planned actions fail. Furthermore, the complex planning and the execution problems require

multiple computational agents and human planners that continually collaborate in the search of the best-fitted solution. Durfee et al. argue that developing distributed continual planning systems will be necessary in order for planning applications to be successful in these stochastic planning (DesJardins, Durfee et al. 1999). The shop floor systems are subject to disturbances that need to be faced through constant negotiations among the involved elements, and having the previous problems in mind. Therefore, it is not possible to obtain static mathematical models that are reliable enough even for short term planning. The resulting classic schedules usually incur in delays, over-costs, waste of time, unproductive resources or out of control situations. The new software techniques related to artificial intelligence, as MAS (Tu, Lin et al. 2009), are better suited for solving this kind of problems than the classical tools, especially when they are subordinated to a real time feedback of information (García Ansola, García Higuera et al. 2011).

2.4 Key notes

Several concepts, technologies, methodologies and real deployments have been presented in this chapter, most of them related with visibility frameworks and agent-based proposals in shop floor operations. Shop floor operations involve the sequencing, the control and the optimization of the operations required to assist products in the supply chain, upon which concepts such as the “Common Use Model”, the “Alignment” or the “Internet of things” define new strategies to manage these operations. However, in order to include these strategies in shop floor operations new technologies have to be deployed and adapted. Looking at the requirements, the selected technologies are RFID as a visibility framework and agent-based systems such as distributed information processing. As the new leading technology for obtaining shop floor

information, this work proposes RFID to support the uptake of information generated in highly dynamic environments; RFID allows feeding data to the shop floor operations in an “awareness” way. This technology relies on radio-frequency tags and antennas or sensors to read or write tags. In the market, the Electronic Product Code (EPC) is being imposed as the standard for coding and sharing the RFID. Once the information is well adapted and available to generate an independent picture of the shop floor environment, the agent-based paradigm is an expert area of the AI to process this distributed information. Agent-based systems provide service-oriented facilities to generate a flexible decision-making, which is currently useful to achieve agility in the manufacturing/logistics decisions. However, the agent-based problems arise in the integration and customization of these systems when they have to be deployed in shop floor specific scenarios. The research community has not addressed fast deployments based on MAS in manufacturing/logistics scenarios and a usual application of the agent-based systems in shop floor operations seems to be far off.

3 THE PROPOSED DSS FRAMEWORK: MAS-DUO

In many of the previous manufacturing MAS research works, there is a detected superstructure that is mainly made up by products and resources agents representing assets as well as upper information systems as expert systems. Besides, during the current daily operations of the companies, these two main control parts are also detected: the physical plant and the information system. The physical plant control is very focused on the PLC technologies and hard industrial technologies in a pure manufacturing control, instead of soft tools as ERP systems or Web systems used by information systems providers. The next shop floor controls should maintain at least this specific service architecture division, which covers the required specialization to feed field data, to share information with stakeholders and to maintain their independence and decisions. The last two requirements are achieved in many cases, but otherwise the business as usual integration between agents and the visibility frameworks could be far. Many of the agent-based existing manufacturing controls try to address these features in several ways; however, most of them are mainly based on an agentification of the

manufacturing elements, in which a not well-described flexible negotiation based on the field information feedback is denoted. If the field information is not automatically fed by the agent-based system, the maintenance is hard and the results are not satisfactory because the decision-making and protocols need to be adapted to each modification (Pechoucek, Vokrinek et al. 2005). This work proposes the junction of automatic EPC data feeding and the flexible agent-based information processing that will allow defining a product-driven DSS in a new product scheduling at the manufacturing control. This junction tries to fix continual planning problems in shop floor environments where there have been detected three specific information deficiencies, especially remarkable in high volatile industrial environments like big industries, factories or distribution centres. These information deficiencies at the shop floor are divided into three main branches, which are:

- The usage of the resources: The information systems cannot directly access in order to detect when a shop floor resource is available or busy. The information feedback about the state of the resources is complex with a high dependency of the surrounding environment.
- The expected duration of processes or single tasks and their flexible definition: each task has different lengths depending on the external and internal conditions. The processes cannot be completely defined as static mathematical functions or in simple templates. Consequently, it obliges defining a constant flexibility based on the dynamic redefinition of tasks and their interrelations. This simplifies modifications in the business

models but exponentially makes the product scheduling more complex.

- The coordination of the involved elements in a task or process: it is not easy to define the rules of negotiation in a high collaborative environment and to reflect the strategic policies in every single operational decision; the shop floor environment trends to the un-alignment in the decision process, especially when disturbances occur.

This work proposes not only the use of visibility for track and trace control applications as well as the MAS to make possible a decision support. The EPC atomic business information with proper information systems can be used to create business intelligence in several ways: forecasting, simulation, reinforcement learning, KPI detection, etc. This will potentially improve the shop floor operations, which are currently in underway by the leading providers of retail management such as IBM, SAP, T3Ci or BEA Systems.

3.1 The selection of technologies: EPC and agent-based systems

The visibility frameworks can help deficiencies about the shop floor information feedbacks; nevertheless a well-defined integration with the decision-making systems is poignant, which is currently not well achieved (Garcia, McFarlane et al. 2003). The new visibility frameworks based on RFID are powerful tools in distributed continual planning because of the continuous track and trace information about the physical resources. The information package that is coming from the RFID readers has the identification and extra information about the product/resource like the

physical position, the timestamp or the business context. This information can place requests for questions such as: What entities were the subject of the event?, When did the event occur?, Where did the event occur? and Why did a particular event happen?. This four dimension data make possible the cohesion as well as the uncoupling of specific physical layouts to the upper systems. These four answers can response to most of the information required in the product scheduling. For example, in the airport GH operations scenario, this information can respond to what GH resources are the subject of the event, the time when it took place, the location where the resources were situated on the terminal and what was the state or the business context of the resource, whether the resources is free, busy or in transit. The minimum coupling is achieved because the dependence about the shop floor elements is taken in charge by the visibility framework, which in the case at hand uses the EPC Global definition.

The EPC visibility framework provides accurate, fast and automatic identifications. The next question is how to integrate it with the existing information systems or interfaces. This launches new interesting lines of business intelligent integration, using distributed artificial intelligence techniques to dynamically plug information-processing techniques with visibility frameworks. In the case at hand, the EPC infrastructure provides Web services to generate fast integration. Nevertheless, the integration of the current manufacturing/ logistics shop floor environments include several manufacturing elements apart from the information systems as actuators like PLCs, robots, machines or simple operators. To be able to provide this integration in on-line scheduling situations, the agent-based systems offer key reasons instead of other techniques. The most important motives are:

- High capacity of negotiation: the functionality of the agents can be published as services in public director facilitators, which allow dynamic interaction driven by discovery services in the new Software as a Service (SaaS) trends. The SaaS is a software delivery model in which software and associated data are centrally hosted on the cloud and served to clients on demand (Gold, Mohan et al. 2004).
- Quick learning and deployment of decisions: each entity (agent) can apply internal pre-planned intentions to the global negotiation and it usually adds necessary information to the global negotiation processes in a complete Beliefs-Desires-Intention (BDI) reasoning.
- The MAS are designed for distributed environments: the agents can be placed in any computer, device or industrial hardware connected to the net, even in different areas, departments or companies.
- Easy to get on-line real data from the environment: the agents can be implemented to feed environmental information via hardware connections, Web services or in a specific middleware.
- Scalability: the MAS can implement multiple agents in several layered logic platforms; this allows for specializing reasoning to each of them.

Looking at these features, this work uses a horizontal layered MAS control approach in order to abstract the details and offer interfaces between layers as dynamic services. These functional subcomponents

maintain the competitiveness of the existing IT products because the layered control maintains the autonomy and the operations of each existing software module; apart from divides the responsibility of the product scheduling control into the technologies supported by different companies. A full heterarchical control cannot be implemented in the existing companies because the integration of hardware elements with the IT as a whole product is hard to implement and the current systems are not ready, even it is not business interested. The restrictions and dependences or even unsafeness decisions cannot be determined by a diffuse negotiation, the big ERPs need to be divided in interconnected packages. The next figure shows the information flow that is coming from different elements of the company/factory/logistics center to the proposed control (Figure 31).

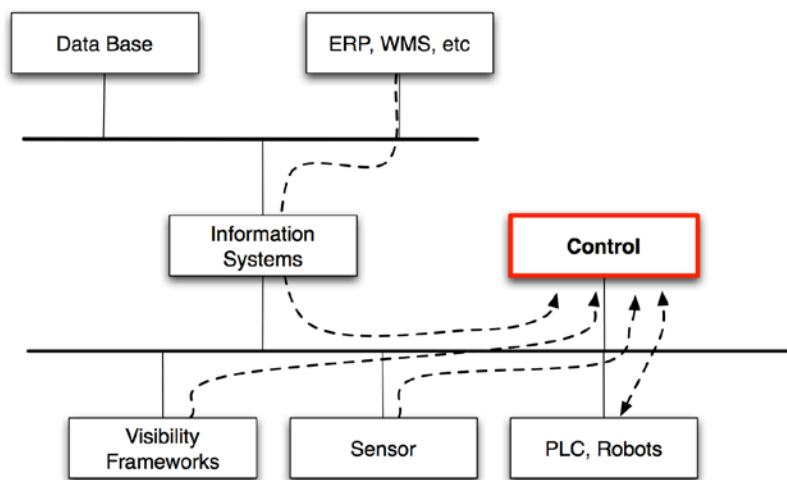


Figure 31 The shop floor control position

These flows are defined by their own characteristics and technologies, which really make the integration difficult. This difficulty requires the use of standard service approaches as the used by the AMS of the agent

paradigm. The next section details how to design the MAS in an organization model able to adapt the existing information flows at the shop floor level.

3.2 Layering the agent-based control to the current shop floor needs

Several problems in the implementation of the MAS vision in the real environments have been detected by several authors (Babiceanu and Chen 2006). Marik and MacFarlane enumerate more than a few barriers, such as the cost of the investments, the operational guarantees, the performance, the scalability, the commercial platforms and methodologies, the engineering education, and the unclear standards (Marik and McFarlane 2005). Currently most of the previous barriers are closer approached with the existing standardization organizations as FIPA, the existing methodologies and the commercial solution systems as JADE or JACK. Then, newer works like that done by Leitao detect that the demand of clients require an implanted solution as proven products, this require a new way of thinking in order to be able to reconfigure the MAS as a interoperability pluggable system based on agent-based components, not only deployed as a unique solution to each problem (Leitão 2009). Other barriers detected by Leitao are the necessity of supervisory tasks that provide robustness, the interoperability based on ontologies or the hardware integration. In this line, Lopez highlights robustness and control of emergence of MAS as key points to integrate MAS as usual (Sánchez López, Brintrup et al. 2010), since the independence of agents tends to make the control of the system output difficult to predict.

Among these well-known problems recognized by the research community, during the interview with the IT staff during the definition of the requirements in the analysed scenarios, the software and hardware companies involved were reluctant to integrate the system as a whole product scheduling, which is the most common approach defined by MAS/Holon researches. In the situation of a unique system, the roles of companies need to be fixed, in terms of modelling, responsibility and management. The companies directly rejected a unique global system; they put forward that there are not business interest in a whole integrated IT product in shop floor operations. Therefore, this work detects that the minimum possible hierarchical control requires at least two horizontal layers with the same level of importance in real scenarios, in which the decisions and responsibilities are split. On one hand, the first system is the high-level information system that is currently called business intelligence, and it is composed by software modules such as: Enterprise Resource Planning (ERP), Customer relationship management (CRM), Warehouse Management Systems (WMS), Retail and Point-of-Sale (POS), etc. Most of them correspond with a soft centralized information system with a database that receives external data like orders or customer constraints, which directly affect the product scheduling. Some existing well-known companies with different proposals in this line of solutions are SAP, Oracle, Navision, SITA or SAGE (Figure 32). On the other hand, there is information processing closer to the devices and the hardware of the company, which is more related to PLC, hardware, robots, CIMs, machinery, etc. This hard information directly affects the product scheduling with physical constrains and field layouts; some examples of companies are ABB, Siemens, Festo or Omron. Then, the conclusions during the interviews in the test scenarios also corroborate the division of the product scheduling into two main systems as well as

this division is also sensitive in the researches of the agent-based manufacturing control where there is detected an agent superstructure where two types of agents emerge. In these research control approaches, it is usual to have agents closely related to the physical world representing assets, products or resources, as they command machinery or moving objects while others are more related with management strategies, expert systems or ERPs, some examples are PROSA o PABADIS.



Figure 32 Analysed information flows in manufacturing/shop floor

After the analysis of the current information flows, the research trends with the appearance of technologies such as the RFID or the agent-based paradigms in the product scheduling. The design of manufacturing MAS of a whole control system is a clear barrier in order to provide agent-based decision-making in the companies. The system as a whole still has been no interesting to the manufacturing IT providers because it is very hard to maintain and to develop with a broad range of technologies. Besides, some interesting results during the test case interviews were related with the responsibility, which need to be divided in expert areas,

not in a unique system. The conclusion of the organization model was clear; the integration in unique decision maker is not been feasible by the current industry. It is not a problem of the available technology, in a near future the technology will fully address this problem. However, the main problems of a whole MAS are the scalability and modularity and the supplier companies are not ready to develop these full systems; it is not business interesting and functionally complex. Then, and specifically looking at the agent-based product scheduling as whole, the next list details the detected barriers:

- The development and maintenance of a complex and large system in a unique control; in case of error, the debugging in agent-based systems requires expert programmers, especially if the system is integrated with specific hardware or expert systems.
- There is non-division of the responsibility during operations in high performance or reliability shop floor processes. There is not direct way to address what is taking the action in charge, making responses or even launching errors.
- The autonomy of every part is essential; the physical part, which needs to be robustness, has to work even if the information system is down.
- The companies involved are expert in their area, the acquired knowledge is complex to implement and share at the same decision level with other companies, even most of them are reticent to share their know-how.

- The required scalability and modularity, especially in stochastic problems where the new visibility frameworks as RFID appears, need to be integrated.

3.3 The basis of the proposal: a division into two agent platforms

Therefore, based on previous handicaps, a horizontal layered MAS as a new product scheduling seems adequate with the current requirements instead of a unique agent platform, but a full design and validation need to be addressed. The next figure, based on the manufacturing control, details the layering of the product scheduling (Figure 33). This layering proposes a division of the agent space into two platforms to be able to make fully implemented MAS in the current manufacturing/logistics scenarios. Therefore, this work introduces an adaptation of the holonic concept in two separated levels, which are developed by using agents. The first platform is an agent structure related to the physical elements of the plant and is directly connected to the physical world, which is called “Physical Agent Platform”. The second platform is an interface with the upper information systems of the company, which is called “IS BDI agent platform”. This idea comes from the Holon concepts of communication in manufacturing control but using a service-oriented division based on protocols and decision-making techniques, instead of unique Holon that incorporates these two expert systems.

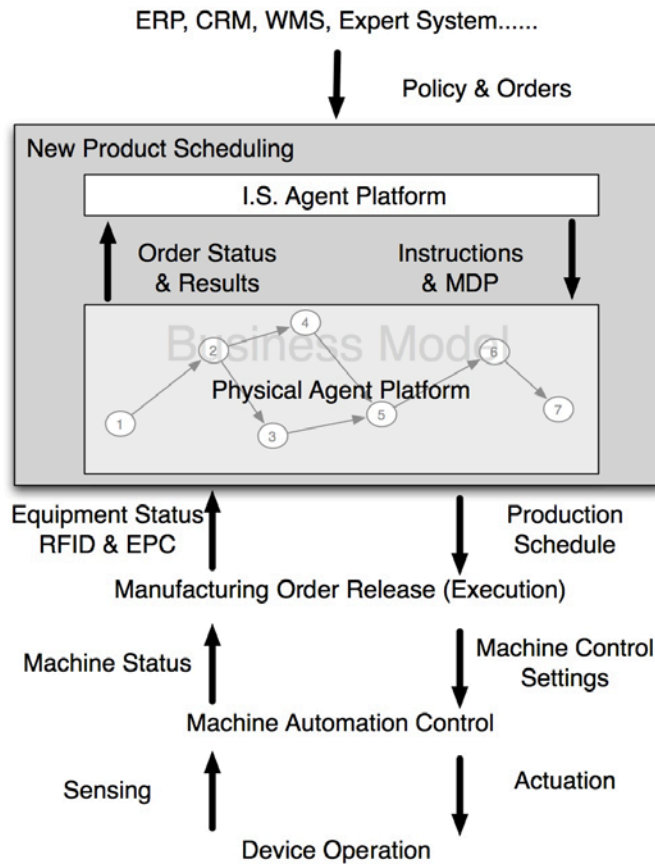


Figure 33 The MAS-DUO Proposal: the new Product Scheduling

This division puts special emphasis on applying strategy policies (IS platform) while attending to information feed from reality (Physical Platform) that includes constant deviations, disturbances and restrictions from the initial schedule. It means that an independent low-level real-time control is connected to the hardware and a deliberative intelligent system provides a high-level decision-making, but both layers are not integrated or coupled in a unique Holon. The physical platform takes decisions when disturbances occur, which are not well defined at the strategic level and produces un-alignments in the decision making process. This situation can only be addressed when the correct information feedback

related with disturbances is guaranteed; specialized visibility frameworks such as the EPCIS can achieve these requirements (Figure 34). This characteristic goes in the way of the well-known problems of lack of field information in the upper logistics and manufacturing levels. The division allows defining specialized software units that better feed its internal beliefs during the reasoning in several ways depending on the surrounding features. It supposes an adaptation of the agents to the real information flows in the manufacturing/logistics environment; this is treated by using the BDI reasoning and other MAS planning techniques. However, new problems arise in terms of integration of both platforms; these problems are addressed in the next subsections.

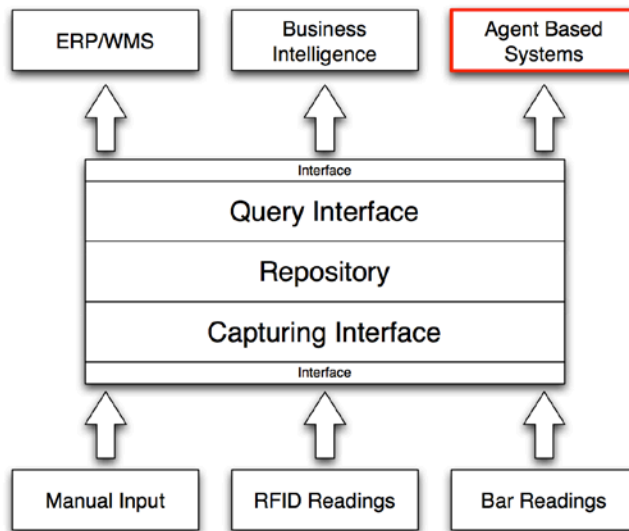


Figure 34 The EPCIS Architecture connected to the Agent-based Systems

The BDI reasoning can be a feasible way to implement both types of agents, which can specialise two different agents: physical agents and information system agents. The MAS can incorporate multiple containers for agents in computing system units, which together made up a

platform. Each platform shares services, technology and data definitions as the AMS and the DF that provide these management facilities by using agent services. The platforms can also establish communication with other platforms but not sharing resources and services, only the selected information in pre-established packages. There are some previous manufacturing MAS models as PROSA that allow a common definition of all these customized agents as products, resources and orders at the same platform. This is an effective initial approach, but for the fact that, after that, the implementation needs a specialization of the agents and their organization to support interaction with the existing physical machinery and information systems; it means that customized agents in specialised layers o platforms are essential. Especially clear is the real time communication layer with the plant in order to be able to automatically take into account the disturbances for the initial planning. It supposes a great specialization that cannot be integrated in a unique logic element. This can be achieved because of the adaptability features of the BDI reasoning and the information coming from visibility frameworks. The BDI paradigm provides a mechanism for separating the activity of selecting a plan and its execution from obtaining beliefs of the environment. The BDI approach allows including different configurations of reasoning depending on its separated functional modules: beliefs, intentions, desires and plans. Therefore, based on this capacity of customization, a division of the manufacturing agent space is proposed based in the customized deliberation capacity of the BDI reasoning and the feedback of the visibility frameworks.

The next table details the BDI characteristics of both agents: the Physical BDI agents and the IS BDI agents (Table 2). Inside the Physical BDI agents there are two subclasses: firstly, the resource agents represent

elements that are attending or manufacturing the product such as robots, converses, fork lifts, carts, etc. Secondly, the product agents are the goods that are being processes as pallets of bricks or cargo flights, looking at the proposed scenarios. This is only a functional division because during the development both types of physical agents include the same software modules. In the Physical BDI agents, the beliefs are fixed; it means that their view of the environment is as plain as if the element is ready or busy, which is often based on their specific position in the plant or if a function block is active in the PLC. Then, the desires of the Physical BDI agents are simple; the objective is to finish processes or operations depending on the policy parameters. The resource agents take simple desires during the processing of the products; the main objective is to finish the process in time. In the product agent case, the desires are to achieve the goal state, which means that the instruction has to be fully executed. Finally, the intentions of this type of agents are machinery movements or physical actions that can be done as single operation during the plan. Most of them are direct actions that can be done in a hardware instruction, in a simple activation or in a direct human processing. The physical intentions are atom actions closer to the desires and almost correspond with the tasks in operational levels of the SCM.

On the other hand, the second type of agents represents the information system elements; these are complex functional components as the ERP systems, WMS systems or POS systems. The beliefs of these agents are complex because they are feeding data from several sources: databases, other IT systems, physical elements, human constraints, etc. The desires, which are currently addressed by business intelligent system, are dynamic and flexible; the desires/objectives change frequently. Finally, the intentions are complex, supposing a plan of low-level intentions; they are

composed in order to define the tactical decision making level. The intentions of the IS BDI agents are in charge of the tactical/strategic operations of the SCM.

Table 2 The proposed classification of the BDI agents

<i>Type of Agent</i>	<i>Subclasses</i>	<i>Beliefs</i>	<i>Desires</i>	<i>Intentions</i>
Physical	Resource	Fixed	Simple	Low level (Operations)
	Products	Fixed	Simple	Processes
Information System		Complex	Flexible	High Level, Composed

This division organization model, which is the basis of proposed MAS architecture, must include the definition of an interface in order to negotiate between both platforms. This new organization model therefore affects the behaviour of the rest of definitions during the development of the MAS, which is software-based in nature. Before the description in detail of the proposed MAS model organization, some systems functional pre-considerations need to be addressed to carry out the agent-based requirements in order to provide an effective scheduling on the shop floor. These considerations include:

- Every agent has the same visibility of the environment; it means that there is a unique visibility system. The beliefs of the agents have to be equal for every product, resource or event.

- The decision parameters during the definition of policy are global and equal for every agent. The weight of the decision policy taken by the upper IT needs to be equal for all the elements, otherwise out of control situations can appear; different global policy parameters can unbalance the decision making process.
- The communication environment is not hostile; it means that the communication is fluid during the execution. This approach is not designed for disaster scenarios or hostile scenarios.
- The rewards when a new action is taken are always positive or 0.

These pre-considerations are essential in MAS-DUO, especially for the Physical BDI agents that are detailed in the next section. These agents require common visibility, common decision parameters and effective communication in order to define accurate rewards in every action.

3.4 Physical BDI Agents: A proposal of intelligent product

A straightforward online feedback of data was emphasized as one of the benefits of using agent-based systems. This characteristic can be applied to fix the well-known problem of the lack of field information in the upper logistics and manufacturing levels. This subsection explains how to integrate the product/resource events with MAS. The agents closer to the physical environment, which are called Physical BDI agents, can be most likely directly benefited from visibility frameworks through a software interface, which in the case at hand is the standard EPCIS. The EPCIS specification helps the definition of this dynamic interface by using the existing services, which automatically publish the real-time information coming from the plant and its circumstances, reporting treated

information and abstracting upper IT levels. In the BDI point of view, a well-formed reasoning requires a constant information feedback from the environment to constantly feed their beliefs. MAS-DUO proposes the EPC subscriptions to connect the physical products/resources with the internal reasoning of the affected agents; the beliefs of these agents are updated through the standard EPCIS XML specification (Figure 35). This communication between products/resources can implement several access methods supported by EPCIS. Apart from the subscription to receive events from the tracked products, there are more protocols such as direct queries, specific events or alarms generated by triggers in order to obtain more specific information about products.

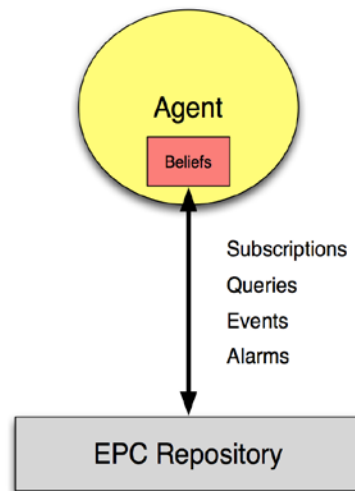


Figure 35 Updated beliefs by the EPCIS

The junction of BDI and EPC in the Physical BDI agents constitutes a Procedural Reasoning System (PRS) with an actualized RFID event database. It is a framework for constructing real-time product-driven reasoning, which can perform decision-making in dynamic shop floor environments. From a coordination point of view, the physical agent

platform is a product-driven approach where the products compete for shared resources or processes; this competition requires a negotiation using global policy parameters. Otherwise, if there are no common parameters, un-alignment decision problems can appear in this high heterarchical system. The un-alignment decision problems reflect different responses between the strategic decisions and the product decisions. If the products drive the negotiation, they need to refresh their internal state to be able to trigger instructions looking at their real state and the state of the plant. However, to be able to identify the state of each product in the supply chain two features need to be addressed: firstly, a previous definition of the states based on the available EPC information. Secondly, an asynchronous interaction protocol, based on the beliefs, that every time identifies the product state. Therefore, the proposal is based on a subscription to the incoming EPC object events and an internal modification of the BDI reasoning to process these events, which occur when the products cross business steps or physical locations (Figure 36).

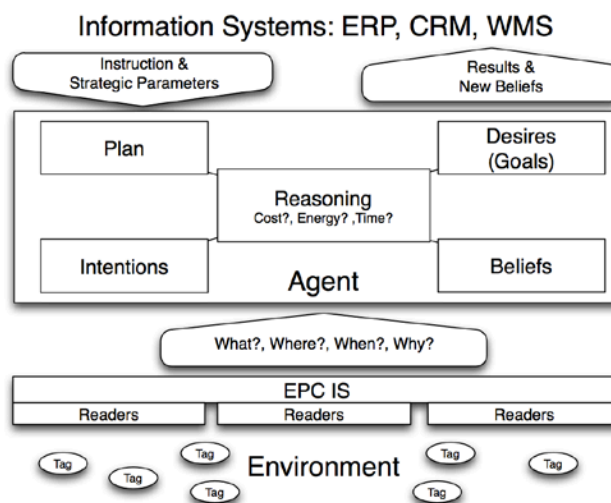


Figure 36 The reasoning process in the Physical BDI agent

The states of the product are defined and detected by the “Where?” (Read point) and “Why?” (Business step) EPC information. In detail, this information, which is defined in the EPC vocabulary, describes the physical and business location of the products. This work considers this junction the best way to define the state of the product; this data takes into account business details and the physical position, both are the bigger constraints during operations on the shop floor (Equation 4). Therefore, the number of possible states is the combination of the physical locations and business steps during the life cycle of the product. In the next equations, the p represents the products, which have a set of Reader Points (RP) and Business Steps (BS). If there are more business steps and read points, the precision during the decision-making is bigger because the increase of the number of states. The definition of states is flexible, so new read points and business step can be added during operations.

Equation 4 The definition of the Physical BDI states

$$ReadPoint(p) = \{RP_1, RP_2, \dots, RP_n\}$$

$$BusinessStep(p) = \{BS_1, BS_2, \dots, BS_m\}$$

$$States(p) = \{RP \times BS\}$$

$$States(p) = \begin{bmatrix} S_{00} & \cdots & S_{0n} \\ \vdots & \ddots & \vdots \\ S_{m0} & \cdots & S_{mn} \end{bmatrix}$$

For example, if there are 4 reader points and 3 business steps in the shop floor scenario; there are 12 product states during operations. Some examples of product states defined in the scenarios as business steps

product at reader points are: a pallet in transit at the picking zone, a flight in block at the terminal 2 or a busy electrical stairs at the hangar 3.

Once the states are identified by using a simple inference of BS and RP, the current state (S) of the product and the desirable state (S') define the transactions between states $S \rightarrow S'$. The full transitions do not have to be direct in only one step; there can be intermediate states, which have to be identified during the reasoning process in a dynamic programming process. This is the definition of task, which means the sequential definition of states until the goal state (S_n) (Equation 5); in which every instruction corresponds with a unique state transition. In the problem at hand, there are available technologies to solve state transition problems as the MDP techniques but need to be integrated in the BDI reasoning. The objective of these techniques is to obtain the best transitions from the product state S to the product state S' depending on configurable rewards.

Equation 5 The tasks as state transitions

$$Task(p) = \{S_0 S_1 S_2 \dots S_n\}$$

The state transition problems require calculating a reward in every transition in order to schedule the plan. Therefore, the generation of new beliefs such as costs, times or energies consumed that can be valuable are useful in the search of the best-fitted schedule; these new beliefs and the reward function define the decision policy. The next subsection explains the algorithm that includes these beliefs in the reasoning process of the Physical BDI agent.

3.4.1 Customized BDI Reasoning

The BDI reasoning of each agent has to select its intention by using these new generated beliefs. This really supposes a BDI customization in the physical agent; the figure 36 details the situation of the Physical BDI agents that need to pick up information from the EPCIS, update their beliefs, generate the new beliefs, plan a interaction to compete for products and resources, and validate the results with the upper information systems (Figure 36).

In detail, the figure 37 details the customized algorithm as a sequence of instructions during life cycle of Physical BDI agent; it is the core of this customized reasoning (Figure 37). Firstly, the initialisation occurs when new products or resources arrive at the shop floor or new events as orders are launched at the shop floor. These initial trigger events used to be taken by the ERP, this is the first step in the customized BDI reasoning: the initialisation (Point 1). After that, the agent establishes a subscription to track the avatar product/resource by using the EPC that was sent by the ERP. The ERP is the first element that controls the product incomings and the order recipients in the strategic/tactic levels (Point 2). Once the subscription is validated, the agent is waiting for any event of the product/resource coming from the EPCIS query module or an instruction from other product agents or even any triggered internal event generated by the proactive behaviour of the agent (point 3).

1. Initialize-state ();
2. Subscription (Electronic Product Code);
3. Repeat while new EPC event or new task or proactive trigger
 - a. Options: option-generator & state identification (instructions-queue)
 - i. Reception of new Instructions
 - ii. Identification of the state ->Where? When? What? Why?
 - b. Selected-options: deliberate (options);
 - i. New Generated Beliefs -> Cost? Energy? QoS? Safety?
 - ii. New Reward Selection -> Markov Parameters & New Beliefs
 - c. Update-intentions & Confirmation (selected options);
 - d. Execution & Results (selected options);
 - e. Get-new-external-events ();
 - i. Where? When? What? Why?
 - f. Drop-unsuccessful-attitudes ();
 - g. Drop-impossible-attitudes ();
4. End repeat

Figure 37 The customized BDI Reasoning in the product scheduling.
(BDI planning)

In the case of EPCIS, any event of the tracked resource in the shop floor is submitted to the involved agent by the subscription service. Once an event has occurred or a new task is required, the captured “What? Where? When? Why?” information allow detecting the state of the product. This action is taken in charge by the option generator in the search of the desires by using the available instructions (point 3.a); therefore a set of possible instructions, which are the available transition states in the search of the goal state, are generated. Then, the deliberation process selects the best-fitted intention depending on global policy parameters and new beliefs generated by the agent during the reasoning; the selection depends on a reward calculated by MDP, which will be detailed in the next section (Point 3.b). This division between the option generator and the deliberation allows separating plan from the current state by using the new generated beliefs. Therefore, the Physical BDI agent takes three tasks

in account during the reasoning process before the execution and validation. These tasks are: the identification of the current state, the generation of focused beliefs and the selection of the instruction by choosing the best reward (Figure 38).

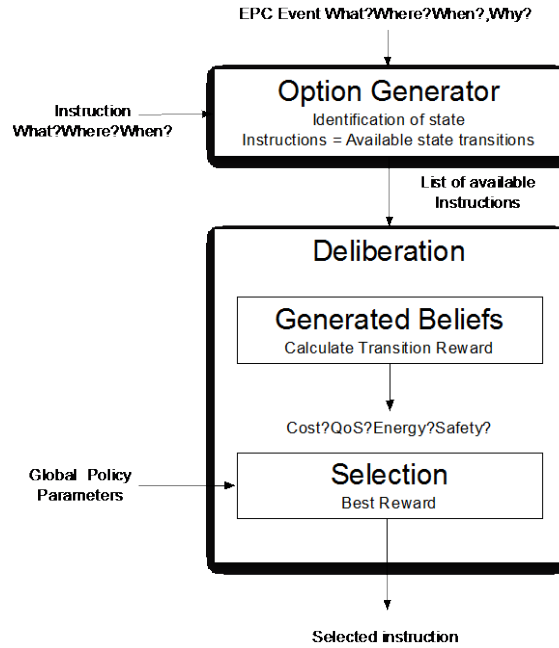


Figure 38 The deliberation process in the Physical BDI Agent

Therefore, as is detailed in the points 2 and 3 of the algorithm, the reasoning is divided in three parts that will be detailed in the next list:

- The identification of the current state and the expected state of the product by using the incoming EPC event and the selected instruction.
- The generation of valuable reward beliefs in the search of the goal state by using the state transitions and the global policy parameters.

- The selection of the best instruction depending of the reward obtained in the state transitions beliefs by using the Markov framework.

Once the state of the product is identified, the own knowledge of the agent generates new focused beliefs depending on the information coming from the EPC event. For example in the airport GH scenario, if an automatic escalator is in the hangar 3 and the automatic mode is enabled, it can infer focused beliefs of every intention such as costs, energy consumed, safety, reliability, etc. This knowledge is achieved and gained by the agent; this inference supposes a learning process during the life cycle of Physical BDI agents. These new beliefs provide useful information focused in the search of possible solutions, even when using data coming from other operators. Furthermore, these revised beliefs simplify the following deliberations because they are business oriented and they do not depend on physical details; it goes in the way of uncoupling systems. Some examples of possible internal beliefs are the cost of the instruction (financial results), the energy consumed during the selected execution (sustainable policies) or the client valuation based on estimated Quality of Service values (QoS). Both parts, option generator and deliberation, can be included in the planning module, which are the point 3.a and 3.b.

After the reasoning process, the confirmation and execution of the selected options generate the updating of intentions. These physical agents report to the IS the results of their plans (point 3.c), which is specially denoted in products agents that need to validate their movements by the upper IS platform. Then, the instruction is sent to real resource through the PLC module or published in a common blackboard in order to be executed by the hardware (point 3.d) and new responses

can be directly captured (point 3.e). Finally, the unsuccessful and impossible attitudes that cannot be reused in the plan are dropped from the option list of instructions in the next loop (point 3.f and point 3.g).

This definition of Physical BDI agent achieves the second level of product intelligence defined by the Auto-ID labs, which supposes to get the next features:

- It possesses a unique identity, which is achieved through the uses of the EPC subscription services that define direct communication with specific products/resources. This also goes in the way of the “Internet of the Things”; the Physical BDI agent allows defining a decision-making avatar of the products.
- It establishes an effective communication within its environment; in this case, the Physical BDI agent provides negotiation protocols to compete for products and resources in a non-hostile communication area.
- It stores data about itself; the EPC subscriptions send to the agents the information about the product, and even new beliefs are generated during the BDI reasoning as cost, energy or quality of service.
- It deploys a language to display its features or production requirements as services to the rest of shop floor elements. FIPA provides interaction protocols able to publish and share information to subcontractors in a standard way.

- It participates in making decisions relevant to its own destiny. The own knowledge of the agent can balance the decision-making by itself and the presented reasoning process supposes enough independence to select its best options. As well as, the decisions are also be supervised by a global policy in order to maintain a decision alignment in this high distributed environment.

This Physical BDI agent is a suitable option in the implementation of a product-driven control, in which products and resources start and launch instructions in a distributed decision space. The product-driven control makes it possible for manufacturing companies to meet business demands more quickly and effectively. The products interact in the decision-making by using their gained knowledge and specialization, which make a really distributed and flexible decision-making process. Even more, the Physical BDI agents can start or drive the interaction protocols with the rest of elements and plan the assignment of resources during their production or delivery. This is the core of the product-driven negotiation, in which the products, only looking at their internal beliefs, trigger the control. For example, if a mixed pallet needs to be ready at a specific time, the instructions will start when the pallet agent considers it by using its BDI reasoning. These low-level actions taken by the Physical BDI agent define a plan in a competitive coordination; these reactive actions have a constant great dependency of the state of the products. This plan is independent of the upper information systems but can be driven by using their global decision parameters.

At this point, the only differentiation between products and resources in the software development is that the products have an extra interface with the IS BDI agent platform. This provides a service in order to provide an information protocol to support the ordering, but there are no

functional differentiations. The reasoning is that products start the interaction protocol trying to get attendant resources, so it defines proactive or active behaviours by the option generator as trigger events. A definition of the Physical BDI agents requires the next data:

- A set of quantifiable instructions those are able to execute a plan, in which subsets are goals in the search of a defined final state. These instructions only represent the available state transitions.
- The knowledge that is able to generate the valuable beliefs from the EPC data. This will be used in the search of the best-fitted instruction at every moment. It means that the capacity to calculate the reward in transition states; it requires an acquired learning, this work proposes as reward matrixes.
- The interaction capacity that is able to send and to receive instructions and events with the physical elements, this work introduces an example with the PLC technology.

Once the instructions are defined, the beliefs can be calculated and the interaction capacities are achieved, the next step is the selection of the best instructions. The next subsection details the proposed techniques that include this capacity in the Physical BDI agents.

3.4.2 Markov Decision Process (MDP) on the Physical BDI agents

The proposed BDI algorithm divides the reasoning in three parts: the option generation, the generation of focused beliefs and the selection of the instructions; these three processes require a flexible intelligent decision technique such as Fuzzy logic, Neural networks, Bayesian

Networks or MDP. There are several reasons to select MDP instead of the others: firstly, the states and transitions are inherent to the MDP, there is a direct modelling between Markov states and product states and between product instructions and Markov state transitions; other techniques such as the Neural networks or Fuzzy logic are more complex to model in a problem, the states and actions need to be individually addressed to be included in the decisions. Secondly, algorithms such as the Q-learning recalculate the complete policy in every action; this means that any modification is automatically updated in the new calculations. At this point, MDP is an exploration strategy, therefore it is design to dynamic spaces in which the model is in constant change; specifically, the Q-algorithm is implemented in flexible environments because it recalculates all the transition probabilities, and any disturbance is directly addressed. These recalculations require more computational requirements than other techniques such as Neural Networks or Fuzzy logic. However, the number of states and resources in shop floor operations is not high and the impact is minimum. Thirdly, the visibility framework feed new probabilities, rewards and state transition to the MDP during operations in a transparent way to the system. The system can learn with the results of the previous instructions or state transitions. MDP is a theoretical framework that outcomes a sequential decision processes, in which states and transitions are quantified in calculated rewards. Therefore, the accurate identification of the current state of the product is essential to be able to generate accurate rewards during the transitions to the next states; the Physical BDI agent calculates the reward during the transition between two Markov states (Figure 39).

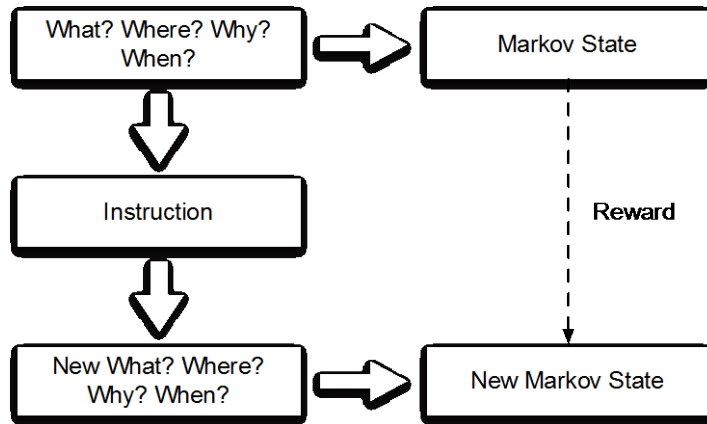


Figure 39 The Markov state transitions of the product by using the EPC data

In order to achieve global reward in the selection of the instructions, two points of view were detailed in the introduction as essential to establish a valuable reward. Firstly, the point of view of upper information system that is able to maintain a strategic alignment in the decision-making, which is especially important in highly distributed system such as the shop floor. Secondly, the physical independence and knowledge gained by the physical resources during the product life cycle, in which any disturbance or physical constraint is taken into account. Therefore, the proposed reward function (R) is obtained by using decision parameters submitted by the IS BDI agent platform (w) and the continuous rewards generated by the beliefs of the physical agents that represent the real situation of the shop floor in a valuable state transition by using rewards matrixes ($x(s,s')$) (Equation 6). In the test cases of this thesis, there were defined 4 parameters in the decisions (Delay, Cost, Quality of Service, Energy) generated by the physical agents and the respective parameters generated by the IS BDI agents (A, B, C, D) (Equation 6). Then, the next equations show as reward matrixes the expected reward during the state transitions in each of the decision parameters. More parameters can be

added but during the interviews with the staff of the test scenarios these four were the most appreciated. A well-formed value function called admissible heuristic can speed up the convergence to the optimal solution and the modification of the parameters can obtain better solutions. For example in the airport GH scenario, a electrical escalator can obtain a better reward than manual stairs because of the time spent is shorter and the cost is adequate to place an incoming Boeing 737 in the parking 1. This decision directly follows the physical constraints and the global alignment of the company.

Equation 6 The equations to calculate the reward in the state transitions

$$R(s, s') = \sum_{i=0}^P w_i x_i(s, s')$$

$$R(s, s') = (A \ B \ C \ D) \begin{pmatrix} Delay(s, s') \\ Cost(s, s') \\ QoS(s, s') \\ Energy(s, s') \end{pmatrix}$$

$$Cost(s, s') = \begin{bmatrix} 0 & C_{01} & C_{0...} & C_{0m} \\ C_{10} & 0 & C_{1...} & C_{1m} \\ C_{...0} & C_{...1} & 0 & C_{...} \\ C_{n0} & C_{n1} & C_{n...} & 0 \end{bmatrix}$$

$$QoS(s, s') = \begin{bmatrix} 0 & Q_{01} & Q_{0...} & Q_{0m} \\ Q_{10} & 0 & Q_{1...} & Q_{1m} \\ Q_{...0} & Q_{...1} & 0 & Q_{...} \\ Q_{n0} & Q_{n1} & Q_{n...} & 0 \end{bmatrix}$$

$$Delay(s, s') = \begin{bmatrix} 0 & D_{01} & D_{0...} & D_{0m} \\ D_{10} & 0 & D_{1...} & D_{1m} \\ D_{...0} & D_{...1} & 0 & D_{...} \\ D_{n0} & D_{n1} & D_{n...} & 0 \end{bmatrix}$$

$$Energy(s, s') = \begin{bmatrix} 0 & E_{01} & E_{0...} & E_{0m} \\ E_{10} & 0 & E_{1...} & E_{1m} \\ E_{...0} & E_{...1} & 0 & E_{...} \\ E_{n0} & E_{n1} & E_{n...} & 0 \end{bmatrix}$$

The reward transition matrixes are fulfilled in the initialisation of the agent; then, a learning process can feed these matrixes with actualized information but a reinforcement process is required. The relation between the reward of the state transitions and the valuable weight decisions of the company provide an alignment in every disturbance or modification from the physical layouts. During the initial deployment in test cases scenarios, the management staff manually fills the reward matrices per each class of the Physical BDI agent in order to submit an initial reward, which supposes an initial extra effort.

Once the reward is available by using the reward matrixes and the state transitions; the Physical BDI agents, which are in a constant product-driven negotiation, take the outcomes using the MDP. The schedule of tasks is based in a negotiation where products and resources compete to obtain the best MDP reward during operations. This reward function reflects a strategic decision-making in every single operational decision that takes in charge physical constraints. The value function $V^* : \mathbf{S} \rightarrow \mathfrak{R}$ specifies the value of the policy Π in the state S . This policy will define the best instruction in every state of the system, even when disturbances occur in order to achieve the environmental alignment. MDP is based in set of states (s), a set of actions (a) that allow to states transactions, a collection of probability functions of successful state transaction (P), the discount rate (γ) and the reward function (R) defined by the company (Equation 7). The discount rate needs to be defined depending on the stochastic degree of the shop floor. This rate is the sum of future rewards

but using a discount factor “ γ ” that increases during the plan. The discount factor can be controlled by the internal beliefs of the physical agent; it means that the agent has an approximately knowledge of its situation in the future. Then, the Q function represents the reward value of the chosen options in the space of possible actions and in all the previous selected steps. The V function defines the selection of the maximum reward in the space of possible options obtained by the Q function (Equation 7).

Equation 7 MDP functions

$$Q^\pi(s, a) = R(s) + \gamma \sum_{s' \in S} P(s'|s, a) V^\pi(s')$$

$$V^\pi(s) = \max_{a \in Ap(s)} [Q^\pi(s, a)]$$

$$R(s, s') = (A \ B \ C \ D) \begin{pmatrix} Delay(s, s') \\ Cost(s, s') \\ QoS(s, s') \\ Energy(s, s') \end{pmatrix}$$

Combining the BDI and MDP techniques, the internal beliefs situate the agent in one of the defined states in the MDP. The intentions of the agent define the actions that will define the state transitions; it allows the generation of a plan based on the Markov parameters depending on the weights of the rewards. The desires are the goal state or final state on the MDP that the agents try to achieve. Therefore, there is an internal MDP in each physical agent, which chooses the actions with the maximum reward trying to reach the goal state. The reward function of the MDP can be customized to the needs of the company, which can be based on the importance of the delays, on the costs of the process, on the importance of the clients, on the power consumption or a mix of all. In

the proposed scenarios, the reward function only depends on the value of 4 parameters defined by the involved company but need to be adjusted. The average reward is defined as the limit of the sum of rewards divided by the number of states, in order to not unbalance the weight of the rewards (Equation 8).

Equation 8 The reward functions

$$R(s, s') = A \text{ Time}(s, s') + B \text{ Cost}(s, s') + C \text{ QoS}(s, s') \\ + D \text{ Energy}(s, s')$$

$$\text{Limit Reward} = \lim_{T \rightarrow \infty} \frac{\sum_{t=1}^T R(t)}{T}$$

Looking at the global decision parameters of the company, if the firm wants to reduce cost it will increase B and it will give more importance to the cost in every single action, instead of focusing the actions in reduce the delays (A) or improving the results with the clients (C). With a correct customization of the parameters, the company can define a dynamic decision policy during the operations. This way of work allows a well-formed decision in the operational level with a limited communication between the neighbour nodes of the supply chain. The BDI resource agents respond with their best rewards to the product requests depending on their policy, and the BDI product agents select one of these requests that best fits with the policy of the product. The BDI product agents do not participate in the selection of the intention by the BDI resource agents, only the confirmation to execute the instruction is done by the interested BDI product agent. The BDI resource agents select the decision applying the global policy, the BDI product agents require the same way of work but they also need the confirmation to execute the instruction by the upper IS platform. This upper validation of expert

systems is important in order to maintain a minimum hierarchical with the strategic requirements of the company. Every single product decision is validated by the information system platform by using an interaction protocol that is detailed in the next section. This is based on sharing the results, which are the final reward and the new beliefs generated in the state transitions, both are sent to the IS platform waiting for a validation by using a FIPA proposal message. After that, if the confirmation from the IS platform is positive the physical agent executes the instruction.

3.5 Information System agents

Apart from the Physical BDI agents, on the upper agent platform, the BDI Information System (IS) agents are closer to the management of the company and to the business intelligent. This agent platform represents the second level in the proposed MAS architecture. Its main concerns are not as much related to the physical performance of the plant as with business decisions and strategies. These agents are related to the fulfilment of orders, including order preparation, plant supply, client management and even asset management. The IS platform receives information supplied by management information systems and processes it together with the results of to apply the global parameters coming from the Physical BDI agents. The physical information are only the results, there are not physical constrains, layouts or restrictions, therefore the uncoupling is achieved. The objective is not to replace the current IS systems, this platform works as the interface between the physical MAS and the existing manufacturing/logistics information systems. The most common existing IT management systems on the shop floor are the Enterprise Resource Planning (ERP), the Customer relationship Management (CRM), the Database, the Warehouse Management System (WMS), the Expert systems, the Enterprise Asset Management (EAM),

the Retail and Point-of-Sale (POS) systems and even the Business Activity Monitoring (BAM) dashboards. These agents work as an interface between the existing IT systems and the agent facilities, it outcomes as a middle step prior to the completed integration in a full agent system. This edge includes agent interfaces able to participate in the current IT shop floor communication within agent software module. The current IT systems are not ready to implement these communications and agent facilities. However, there are new developments that are including Web services to provide effective tools for distributed and even cloud-computing communications. The Web services technology is currently deployed as business usual in many industries, with is not the case of the MAS systems; one good example is the Representational state transfer (REST) technologies.

The Physical BDI agents take the outcomes based on decisions formed by strong relations between agents with self-interested desires. However, the IS BDI agents control a set of decisions in the search of the global requirements of the company with complex and soft interrelations; they also have to serve as an interface with external stakeholders, clients or providers. Once the results of the instructions are accessible to the IS platform, the proposal is negotiated between the agents of the IS platform by using interaction protocols as market simulations. In the test scenarios, there were three IS BDI agents: the ERP agent, the CRM agent and an expert system agent that act as interface with the existing IT systems. If the proposed solution is accepted, the physical agents execute the plan. Otherwise, if the proposal is rejected the physical agents have to recalculate the plan with the new global policy parameters provided by the IS platform during the negotiation; this information flow between platforms is detailed in the next figure (Figure 40).

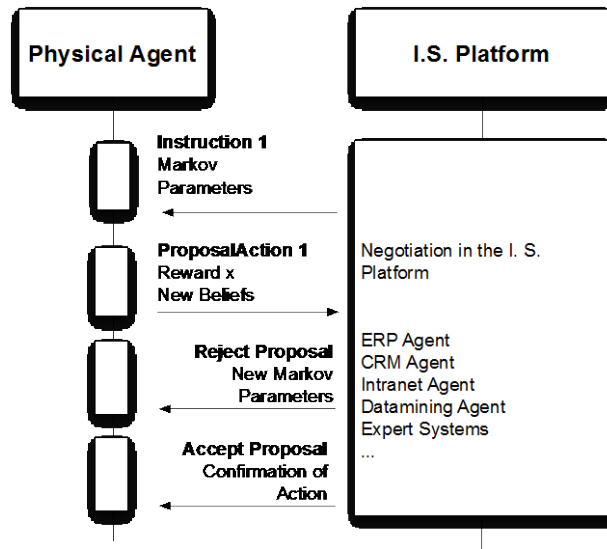


Figure 40 The interaction protocol in the IS platform

In the IS platform the decision process is not as close as is defined in the physical platform, this is heterarchical where the role of the agents is not clear and the coordination is important. The coordination is a property inherent to the agents, not to the system itself but the actions of the agents may interfere; therefore, some kind of dependency must exist among agents to meet these situations. Then, if there are collisions with dependences in this platform, the moderator agent negotiates it, consulting to superior layers but this is not able in the current level because there are not superior layers but some planning methods have emerged to encapsulate this behaviour. In the case at hand, given the different local plan abstractions, each agent is able to identify common goals to which the local goals of the agents contribute. The common goals are called partial global goals as they may be only partially known to the agents. Once a partial global goal has been identified, the local plans can be integrated into partial global plans but this process, by which a joint decision is made by two or more parties, is a negotiation. The parties first inform contradictory demands and then change towards agreement

by a process of allowance in the search for new alternatives by using interaction protocols. Then, the alternative is a solution when the parties reach a successful agreement at least with a minimum consensus. In the case at hand, when there is a minimum consensus between the IS BDI agent about the results the IS platform approve the execution.

Therefore, the main function of the IS platform is to negotiate if the plans proposed by the physical agents fit with the requirements of the company. If they fit, the IS platform allows the instructions to be executed and if there is any inconvenience the only way to maintain the independence is to change the global policy parameters or the instructions. The necessity of an updated global policy in a distributed shop floor system is essential to reflect the objectives of the company at every moment. These global policy parameters are controlled by the business intelligence of the company, which can be implemented by using expert system, KPI and dashboards. A clear feedback from the test scenarios was that the information coming from the physical agents can feed the expert systems in order to provide more knowledge as well as improve the generation or fitter global policy parameters but independent to the physical layouts. The next list describes what are the events of the IS platform that can modify the value of the global policy parameters and the instructions during the negotiation. These variables are not representative in the operational levels because they are tactical and strategic in nature, some examples are:

- The sourcing contracts and other purchasing decisions.
- The production decisions, including contracting, definitions, new processes and available timings.

- The inventory decisions, including quantity, location, and quality of inventory.
- The transportation strategy, including frequency, routes, and contracting.
- The benchmarking of all operations against competitors and the implementation of the best practices throughout the company.
- The milestone payments.
- The customer demand and habits
- The strategic network optimization, including the number, location, and the size of warehousing, distribution centers, and facilities.
- The strategic partnerships with suppliers, distributors, and customers, creating communication channels for critical information and operational improvements such as cross docking, direct shipping, and third party logistics.
- The product life cycle management, so that new and existing products can be optimally integrated into the supply chain and capacity management activities.
- The aligning overall organizational strategy with supply strategy; it is for long term and needs resource commitment.

3.6 Global policy: static and dynamic

The proposed MAS-DUO describes how to obtain a decision making policy even disturbances occur; in shop floor operations it means the list of actions that have to be taken by the resources during the attending of products looking that the policy can change depending of the global parameters of the company. A policy is a mapping from states to actions for every configuration or layout; there always exists an optimal policy. However, the problem is when the reward, which is calculated by the Physical BDI agents, is dynamic depending on the real state of the environment or by the parameters defined by the company; in the problem at hand, the optimal policy is also in constant change. A company can establish several operation modes looking at the requirements needed by only modifying these parameters; the shop floor can dynamically achieve decision-making configurations depending on the requirements of the company. For example, two types of policies can be defined in the decision-making depending on the negotiation in the IS platform (Equation 9).

Equation 9 Two operations modes based on different global policy parameters

Global Policy Parameters in time $T = (A\ B\ C\ D)$

Global Policy Parameters in time $T' = (A'\ B'\ C'\ D')$

A simple modification of the global policy parameters affects the decisions at shop floor level, but a sequential definition of a policy need to be addressed in order to provide effective deployments in real companies. During the initial steps during the deployment of MAS-DUO in shop floor scenarios, the phases without field information were

necessary in order to test and teach the system. A static policy is an initial step in the shop floor DSS implementation, in which there is not actualized field information coming from the real state of the plant through the visibility systems. It means that the state of the Physical BDI agents is not updated with the field disturbances; at this point the system only achieves the strategic alignment. It is useful during trainings because it performs simulation and learning processes, in which the acquired knowledge can be transmitted to the management staff by visual interfaces. These static policies can also be functional in very static environments or when the ROI does not permit including visibility systems such as RFID in the shop floor.

However, in very high dynamic systems the visibility frameworks have to feed information in order to provide a continual decision-making, which is the focus of the environmental alignment; at this point, the policy becomes dynamic. Each physical agent through the incoming EPC information dynamically validates the state transitions of products, not only during the negotiation process. In this proposal, the agents have enough internal beliefs to confirm that the transition was successful. This solution applies dynamic policy during the processes but a prior static policy can be simulated or even applied without the information coming from the visibility frameworks. It can gain knowledge to the agents before a full dynamic implementation of the global policies as demos or trials in order to prepare the final deployment. This also allows defining a procedural deployment of MAS-DUO; it means that initially a static policy can be implanted in order to achieve a strategic alignment but when disturbances arise, the sequential implantation of visibility devices feed the real state of products and resources.

Once the decision to deploy a dynamic policy is taken, the next question is to calculate how many visibility points are needed to cover the requirements, which depend on the accuracy required to provide the decision policy. The MAS-DUO proposal allows adding read points and business steps in a sequential way; it simplifies the deployment. In a real system it is not easy to understand what information is valuable and where the framework needs to improve visibility and can even change; therefore, the question is where the system needs more information feedback. For that reason, the design of the MAS-DUO achieves these requirements in order to simply add more information and sequentially plug new components, as its functionality is extended. One way to know when the system needs new visibility points is by adding measuring ratios of errors by using software techniques. This process can be based on reinforcement learning, in which the final results are compared with the initial prediction of the MAS. In most of the cases, the comparison is studied by analyst techniques like Data Mining but if the error is big and cannot be estimated, the system needs more field information and more visibility. It has already been tested that the performance of this expert system improves when the introspection is bigger (Wooldridge, Weiss et al. 2002); it means ‘dividing complex problem in more sub problems’, in which the information is accessible and the granularity is higher. Therefore, the decision-making systems must analyse when they require more information. This situation coincides with the initial deployments the company directly requires to know how many visibility points such as RFID readers are needed to cover the requirements. In the system proposed, the only solution is to improve the introspection by dividing the problem into more detailed sub-problems by increasing the quantity of states. If there are more states the information granularity is smaller and the details can be identified. This work identifies the states of the

products and resources as the number of visibility points per business steps. Therefore, the only solution is to raise the number of readers but it has negative economic issues or to increase the number of business steps in the operational supply chain. Then, if a task corresponds with a set of sequential state transitions, the reward obtained in more states is segmented and better detailed. Nevertheless, in order to accurately identify where the visibility needs to be improved, the only way is comparing the reward transitions during the tasks. If a product has similar rewards during the state transitions, the decision process is controlled, but when there are big differences in the reward quantity between states, it means that between these transitions there is a lack of visibility (Equation 10). This lack can be treated adding new business steps that make more complex the EPCIS or adding new visibility points or even moving closer the affected visibility points in order to reduce the uncertainty.

Equation 10 The equation to detect the lack of visibility

$$State(p) = \begin{bmatrix} S_{00} & \cdots & S_{0n} \\ \vdots & \ddots & \vdots \\ S_{m0} & \cdots & S_{mn} \end{bmatrix}$$

$$Task = \{S_{00}, S_{12}, S_{23}, S_{31}, S_{33}\}$$

if $T(S_{mn}, S_{xy}) \gg Avg(T(S, S'))$ there is a lack of visibility

Once the number of RFID readers is defined by analysing the problem with a static policy definition, the MAS-DUO is designed to increase visibility in a dynamic way and the agents are dynamically classified according to the information that they are taking from the environment through the EPC vocabulary. The internal structures of the Physical BDI agents automatically update the beliefs that represent the state of the

product; the case of encountering new beliefs as new business points or new business steps the states are updated. For example, in the case of a distribution center, it is easy to know that the system needs visibility on inputs and outputs. However, depending on the particular characteristics of the products, clients, and providers it is interesting to get better visibility on the rejections, at picking zones or at the doors of the automatic warehouses. These flexibility features of the Physical BDI agents allow fast modification of the layouts and configuration at the shop floor.

3.7 Development Methodology

An intelligent product-driven is achieved through customized BDI agents that provide the argumentation of why a division of the agent platform is required. MAS-DUO proposes the customization into the Physical BDI agents to integrate the visibility frameworks data into their beliefs with a decision making process based on the Markov Decision Process, together with aligning the rewards in every decision. Therefore, customized types of agents and their reasoning have been added to the system, which need to refine some of the existing models adapting these new requirements. Besides, before deploying these concepts in real scenarios, a methodology is essential in order to be sequentially set up and to put in common the processes that can be reused (Giret, Botti et al. 2005). The methodology at hand needs to be adapted in order to accomplish deployments with the MAS-DUO requirements. Several approaches of the MAS methodologies have been proposed in the manufacturing research area with common requirements, specifically Botti proposed the next list of requirements (Botti, 2003):

- The definition of autonomous entities to be organized in hierarchy and heterarchy structures; in this case, the entities are the Physical BDI agents and the IS BDI agents.
- The description of routine-based behaviours with a straightforward control task on every resource or service, these are the state transitions and tasks in the Physical BDI agents.
- The methodology should define a development process that is guided by abstraction levels, and should also provide modelling artefacts, tools and guidelines to manage this process. These features are addressed by using the models that are explained in this section.
- The methodology should integrate the entire range of manufacturing activities, from order or booking through design, production, and marketing in order to model the manufacturing company. This MAS-DUO is very focused on shop floor but any information from any company levels can be taken into account.

In the following definitions, this work presents a methodology as a development process divided by steps. This way of work requires approaches split into several models; this idea has also been used in other MAS methodologies as GAIA (Wooldridge, Jennings et al. 2000). The proposed methodology combines concepts coming from Commonkads, GAIA and the Agent Unified Modelling Language (AUML) methods. In GAIA there are concepts that can be used inside the models of Commonkads like roles, permissions, activities and properties, and this combination can be defined by using the interaction diagrams, use cases, and activity diagrams coming from AUML (Bauer and Odell 2005). What

follows is the explanation of the proposed shop floor MAS control methodology from several models.

The complexity of the design of the MAS can be better understood and addressed if the system is subdivided into the following models that are explained in this section. These models have been designed based on the feedback obtained during the deployment in the test scenarios.

3.7.1 Organization model

The organization model is the basis of the MAS-DUO since it includes the division of the DSS and the definition of the physical platform as well as the information system platform. It therefore affects the behaviour of the rest of the models, which are software-based in nature. Moreover, a correct organization model makes MAS-DUO a system easier to develop, to test and to maintain. Then, the shop floor instructions are defined in the task model, which establishes the relationships between the elements of each platform, in which the physical elements and the information systems interact in search of the best-fitted schedule but in separate organizations. The results of this negotiation are calculated by the physical platform but based on modifications of the global policy parameters and the instructions by the IS platform. This organization model allows validating or rejecting the results in an uncoupling system because they only share results and decision parameters.

3.7.2 Agent model

This model specifies the agent characteristics that need to be included during the deployment; the agent features are the reasoning capabilities, the skills (sensors/actuators), the services, the agent classes and the

hierarchies in the physical platform. The agentification process determines the set of agents that adequately represents the shop floor environment. Firstly, the selected agents need to generate enough beliefs, desires and intentions, and adding functionalities like roles, permissions and responsibilities to be representative in the agent system. Secondly, the agentification identifies and shapes the classes of physical agents; based on the generated agents, it will be necessary then to identify the basic functional Physical BDI agent groups that will constitute the agent classes. After that, agents can be aggregated or specialized depending on their features; for example, in an airport there are classes associated to the stairs, the conveyors, the carts, the employers, the trucks, the buses, etc. However, inside every class there are specializations because of its internal features; for example, in the case of the ground handling stairs, the following might be defined as fast stairs, electrical stairs, driven stairs or manual stairs. When the classification and specialization is achieved, the next step in the agent definition is the identification of states, which is automatic with the EPC information. Then, the definitions of the reward state-transition matrixes per agent class need to be manually fed. The reward matrixes produce the specialization of the agent that will offer different features.

3.7.3 Task model

This model describes how to define the tasks that the BDI agents carry out during operations; the tasks cover the initial states, the decompositions of states and the goals or final states. In a MDP reasoning point of view, the tasks describe the series of states that need to be done during the execution of the instructions by the products. The instructions coming from the IS platform only include the final state and the policy parameters. At this point, the MDP framework is useful in the

search of the maximum reward path to cover the state transitions. Therefore, the shop floor planning is a weighted and directed graph where the products select their best reward during operations while the attendant resources also select their best reward (Figure 41). It is a competitive product-driven decision-making where products compete to access resources as services during the supply chain process. The nodes represent the state of the products; the arcs are the available state transitions/actions of the agents and both have a reward function in every action. Nevertheless, during planning the actions have a probability of successful state transaction and the goal is achieved when the product arrives to the final state with the maximum reward. This probability also needs to be included during the definition of the tasks.

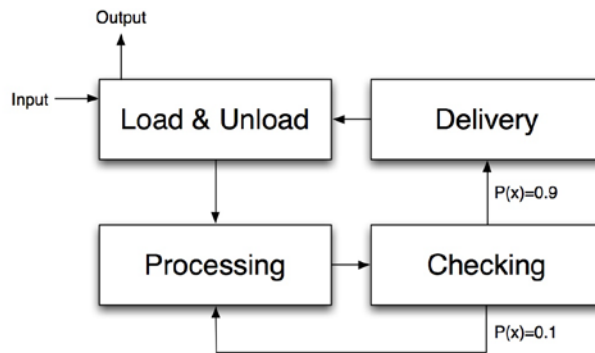


Figure 41 A shop floor weight direct graph

For example, in the airport GH scenario several tasks are related to the incoming Boeing 373; the BDI product agent, which is the specific flight, includes a series of states like landing, in-block, luggage handling, air cargo handling, fuelling, catering trucks, refuelling, ground power, passenger stairs, wheelchair lifts or off-block. In the case of the BDI resource agent, the states are simpler, in most of the cases the business steps only define a brief number of states as: free, busy, in transit or even

breakdown. Therefore, in the case of the BDI products agents, the tasks and the states transitions need special attention during the definition of the agents.

One detected problem during the implementation of the Physical BDI agents was the definition of the tasks in complex processes. A pre-established template of the processes as fix state transitions cannot be defined because there are many variables. These planning responsibilities need to be divided between the products and resources agents; the agents planning depend on their internal features, on the external conditions and on the negotiation taken with the involved agents. The expected construction of the planning is not centralized; in agent terms, the product state transitions are defined during the search for the final state and the best reward, so there is a distribution of the knowledge. In a Markov decision point of view, the last sentence means that it is hard to define a previous full state transition list of a product because the results of the actions are hard to predict in every complex problem. In the logistics center scenario, a pallet can have states such as: stored, in transit, in picking or delivery. However, in the airport GH scenario, the flights have many states that are complex to define and even worse to neatly schedule them.

3.7.4 Learning model

It describes the knowledge needed by the agents to better achieve their goals and to gain knowledge. Once the MDP customization is presented, there are several techniques to resolve the equations based on linear programming or dynamic programming such as value iteration, policy iteration, modified policy iteration or Prioritized sweeping. Then, if the probabilities or rewards are unknown, which is the case of the shop floor

environment, the problem is one of reinforcement learning. The power of reinforcement learning lies in its ability to solve the Markov decision process without computing the transition probabilities; the transition probabilities are needed in value and policy iteration. The reinforcement learning can be combined with function approximation, and thereby one can solve problems with a very large number of states. Besides, the reinforcement learning can also be handily performed within Monte Carlo simulators.

3.7.5 Coordination model

This model describes the conversations between agents that include their interactions protocols and the required communication capabilities; both of them are well defined by FIPA. The FIPA specifications include interaction protocols, speech act theory-based communicative acts and content language representations. In the proposal, every single decision between agents is validated using variations of the contract net protocol defined by FIPA. In the case of communication among platforms, the contents of the message include the obtained reward and the new beliefs from the physical platform, which are sent to the IS platform waiting for a validation. Then, the proposal is negotiated using the market simulations between the IS BDI agents; in the test scenarios there were three IS BDI agents: the ERP agent, the CRM agent and an expert system agent. If the proposal is accepted, the Physical BDI agents execute the plan. Otherwise, if the proposal is rejected the physical agent has to recalculate the plan with new Markov Parameters.

3.8 Implementation

Once the models define a process to design the decision framework in a specific case, this section explains the software development of the MAS-DUO core. After that, the deployment in two different scenarios is presented. In order to develop the MAS, there are several programming languages and software development tools that can be useful in MAS programming. Currently, the MAS-DUO development is being focused on programming languages like Java with a specific framework for MAS, in order to reuse all the existing powerful Java libraries with the agent-based point of view. In the development at hand, the selected MAS framework was JADE with the standard interaction protocols defined by FIPA and the ACL. Software development tools such as: Eclipse, Mysql, Fosstrak and XML java libraries as JDOM, were also used. JADE allows deployment through a FIPA compliant middleware and provides graphical tools to support debugging, which can be controlled via a remote graphical user interface. JADE has been very popular in both academic and industrial experiments as the most complete and popular open-source tool available to develop BDI agents.

Looking at the Java software development, the following UML class diagram shows the software classes that made up the Physical BDI agent (Figure 42). They correspond with the different behaviours as a part of the reasoning algorithm. The main classes that made up the Physical BDI agents are: The EPC class, the PLC class, the BDI agent class and the agent core class. The agent core is the parent class that provides the agent facilities supported by the JADE agent framework. This JADE core includes agent facilities such as: communication protocols, life cycle definition processes, AMS and DF services. Then, the BDI module is the extension of the core agent that includes BDI features; this class defines

the BDI information as vectors in which the EPC and PLC modules feed with the incoming information. In this module, the reasoning algorithm is triggered when new events occur feeding the beliefs vector.

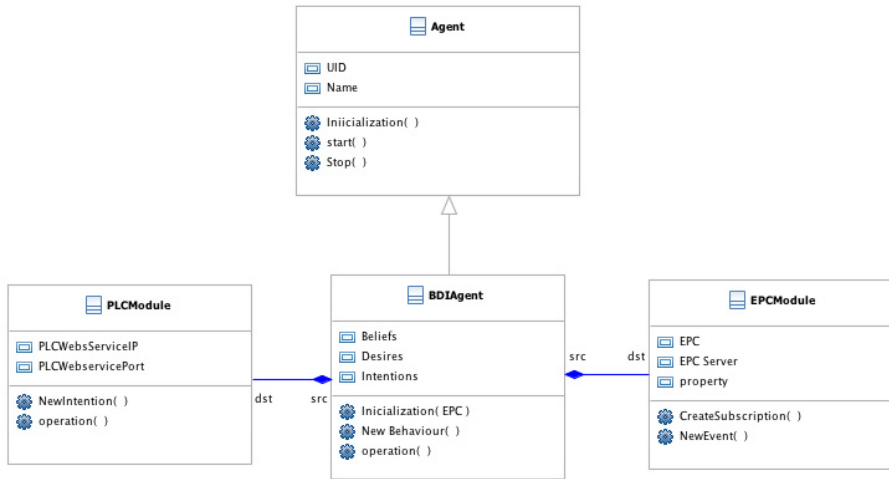


Figure 42 The UML Class diagram of the Physical BDI Agent

Then, the communication with the plant to feed the BDI module is divided into two: the first one is the EPC module, which manages the communication with the EPC services through the implementation of an interface with the specific EPC Fosstrak implementation. This EPC Module feeds beliefs from the EPC repository, which is initialized with the EPC of the shop floor element that wants to be tracked. Then, this module establishes a Web service communication by using the EPC capture interface; then, the Fosstrak EPCIS implementation sends any event of the element to its avatar agent. The second module sends and receives PLC instructions to track the state of the shop floor elements through an interface defined by the specific manufacturer API. This module gets the information feedbacks from the machinery by using a PLC when the RFID technology is not suitable. The PLC Communication module was designed to cover this functionality through

a Web service, which interacts with the ladder programming in order to share variables and function blocks. The PLC has ladder programs as function blocks that are triggered by this web service and controlled by a dispatcher on the PLC Master.

Once the software structure of the BDI agent is described, the next figure sequentially shows the functionality that is taken in account by each software module to achieve the intelligent product requirements (Figure 43). After the initialization, the BDI module receives events coming from the EPC module to build the beliefs. Then, after the reasoning process, the instruction is executed through the PLC module that sends the instruction to PLC equipment.

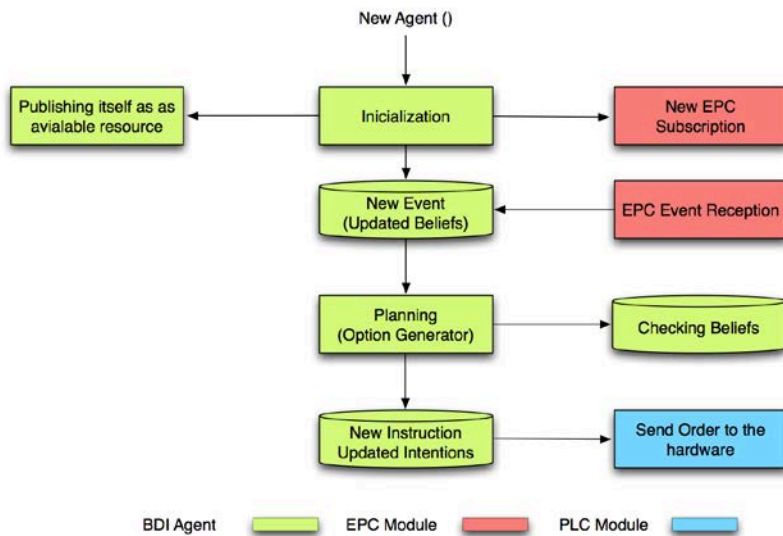
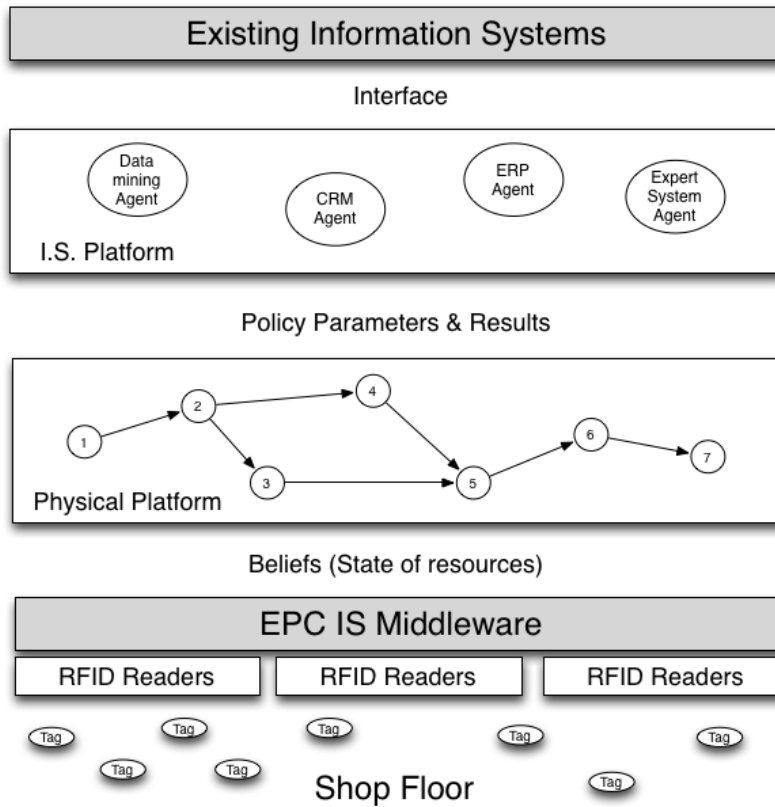


Figure 43 The Physical BDI Agent reasoning

3.9 Key notes

This chapter has detailed a layered agent-based decision-making system, called MAS-DUO, in order to align global/business decisions and

physical disturbances in two different agent platforms. In the physical platform, the Physical BDI agents choose the best action in the spaces of possible state transitions depending on their internal reasoning and on the global alignment parameters. These global parameters are defined by the business intelligence at an upper decision level, called IS platform. The internal reasoning of the Physical BDI agents is divided into the intentions, beliefs and desires; then, the set of sequential intentions constitutes a plan. Therefore, the plans are state transitions based on a MDP where the rewards are calculated taking into account local and global weights. The execution of these plans generates new valuable beliefs depending on the chosen options generated in the selection of actions; then, it generates knowledge to the system in a reinforcement learning process. These new internal beliefs are really useful during the decision process because they are the result of the reasoning process while striving to meet the goal state. The Physical BDI agents constitute intelligent products/resources that publish their services during the SCM, execute instructions depending on their internal reasoning and during the negotiations the collisions are moved upward to the IS platform in search of better solutions. Therefore, the division of the MAS-DUO produces two decision levels: a product-driven negotiation based on services at the operational level and a complex market-driven system in the business intelligence. The next figure shows the MAS-DUO schema, the models that are involved in methodology and what is the implementation selected in each model (Figure 44).



Models	Technologies
Organization	MAS-DUO, Agent platforms
Agent	BDI, JADE, EPC, Fosstrak
Task	MDP, EPC
Learning	Reinforcement, MDP
Coordination	Contract Net, FIPA

Figure 44 MAS-DUO Schema

4 VALIDATION OF THE MAS-DUO: DEVELOPING THE TEST CASES

In order to test the MAS-DUO architecture in real situations, this chapter presents the deployment of MAS-DUO in two different scenarios. The first test bench of the proposed MAS architecture is an application for the assignment/management of the GH resources that attend to the incoming flights at the Ciudad Real Central Airport. Secondly, the proposed architecture is applied to a distribution centre platform in order to test new ideas about the agent-based organization models.

4.1 The airport ground handling scenario

The Ciudad Real Central Airport is a new airport that was looking for tools to help the management of GH resources, which are used to provide service to aircrafts and passengers; this airport started operations in 2008. The airport is focused on the low cost market, which means that it needs to reduce costs associated with ground handling operations, airline management and airport operations. The airport management aims to improve the productivity of the ground handling resources that attend

to incoming flights by trying to reduce delays, overloads and overstaffing, through improving the visibility of resources and their status.

In the case at hand, the target of the MAS-DUO is to allocate, plan and control the previous mentioned resources to assist incoming flights during operations. The MAS-DUO has to communicate the actual status of the resources to the visualization tools that supports validation by the supervisors on the terminals, so the unforeseen changes on the real system will be directly considered in short-term planning.

4.1.1 Tracking ground handling resources (ZigID)

Firstly, the deployment of the visibility framework needs to be addressed on the airport terminals. RFID technology comes up as especially suitable for capturing information from heterogeneous shop floor environments, offering rapid responses and continuous tracking in big spaces but cannot cover big distances. One of the problems of the RFID in big installations is that the reader can communicate to short/medium distance with active (with batteries) tags – less for the more typical case of passive tags (not powered) - while there is not a defined network topology to coordinate several readers. The active RFID is not standardized and the current vendor specs cannot locate resources in big scenarios. This problem can be directly improved using mobile Wireless Sensor Networks (WSN), which abstract the mobility, the topology, the scalability and the communication between the mobile nodes in a network (Sung, Sanchez Lopez et al. 2007). In this case, the aim of the WSN is to cover large distances in the environment while performing identification at specific zones with a flexible data model, which allows new devices or new information to be added dynamically. Besides, the routing information can be added to the submitted identification package that will help the

location of the identifications, apart from more information attached that includes the EPC data. Looking at the visibility point of view, the first main requirement from the airport staff about the WSN is the need for a dynamic increase in the visibility of the environment, which is provided by a step-by-step deployment provided by the communication technology. The areas that need to improve visibility are not known beforehand. This means that the proposed WSN needs a dynamic topology and a data model that can include new requirements or definitions. Other surrounding problems of the WSN are the management of the mobile nodes with low power consumption, fault detection and auto diagnosis.

To face situations such as those described above, the combination of RFID and WSN technologies to track resources in industrial environments, which could be an airport, a factory or a logistic centre, seems to fit the requirements. The proposed WSN is based on the deployment of a Zigbee wireless communication network of RFID readers that is capable of covering medium/long distances within specific identification zones inside the industrial environment. Therefore, this visibility framework proposes the combination of RFID and WSN technologies in a framework, called ZigID, in order to localize and track mobile resources in industrial environments. The feasible applications range includes from airports to factories or logistic centres, in which vehicles, products, pallets, machinery or employees conform the activities. Besides, the use of the EPC data definition allows to the WSN information to directly share the identification between the involved air transport stakeholders in a transparent way. This WSN builds an accessible EPC repository for all the airport agents, which is important in this high-segmented industry.

The communication technology selected is Zigbee, which is chosen due to its characteristics of range, auto routing capability, low power consumption and a sufficient bandwidth. Zigbee is a specification of a set of high-level protocols of wireless communication for use with low-power digital radios based on the IEEE 802.15.4 Wireless Personal Area Networks (WPAN). Its main features are the low power consumption, the mesh network topology, the ease of integration and the cost; the nodes can be manufactured with cheap electronics. Zigbee networks can adopt three alternative topologies: star, tree and mesh. For the situation in hand, the mesh topology is particularly well suited. It allows communications to be maintained even in the case of a Zigbee node breakdown by means of dynamically recalculating new routes, while covering medium/long distances. By using WSN based on Zigbee with RFID reader nodes, the ID information can be routed over large distances but the identifications are located in specific zones. The communication architecture is detailed in the next figure (Figure 45).

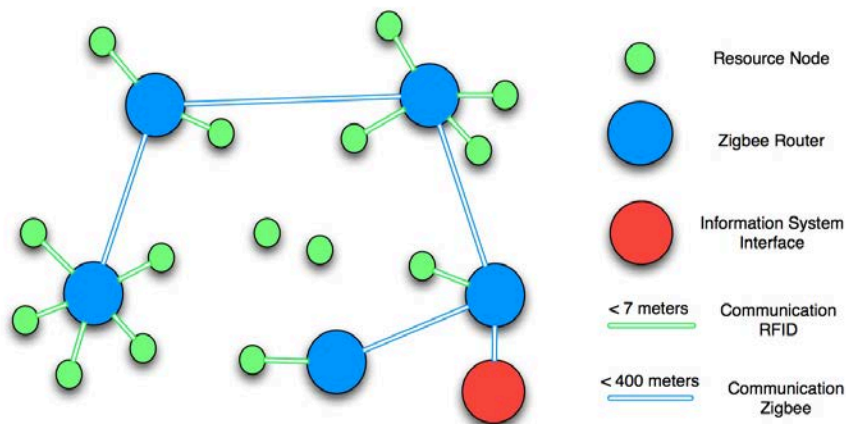


Figure 45 The WSN Architecture in ground handling operations

The benefits of this combination of technologies are:

- The transparent information flow between nodes and their possible configurations.
- The RFID identifies the sensing elements through EPC, which is a standard in all levels of the business model.
- The WSN allows large distances to be covered in a dynamic way while RFID focuses the identifications on specific zones.
- The cost of the RFID end nodes (10-12 \$) is low compared to other existing devices.
- The battery consumption of RFID devices in the mobile nodes is really low compared with other electronic devices like mobile phones or radio devices.
- The possibilities of the auto-routing of packages in case of breakdowns in one of the nodes of the WSN by using mesh topology.

In order to take advantage of the technologies presented above as well as of the conceptual benefits of their combination, a new architecture has been developed with the name of ZigID. ZigID has been divided into three levels to facilitate the distribution of tasks and foster the cooperation among devices (Figure 46). The first of these levels is the end node level, whose features are the small and unobtrusive size, the low consumption features, (enabling battery supply), the possibility of including integrated sensors, the short-range coverage (10-12 m) and the identification capabilities. The second level is router-reader level and its objective is to facilitate the communication and routing of the

information among end nodes and the information systems. The third level is the coordination level, which collects the distributed information across the network and provides an interface with the existing information systems.

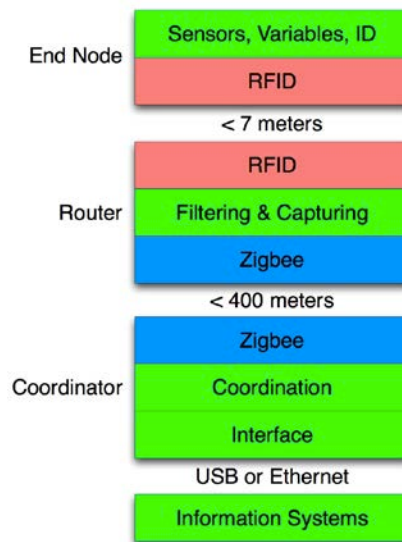


Figure 46 The ZigID architecture

The ID information - captured by RFID readers in the network – is transmitted to the EPC repository; this information package contains variables of the sensors, the routing path and the unique ID of the tracked element as this composes the complete data model in the WSN. In order to coordinate the distributed hardware devices, a data model needs to be defined to describe the information flow that satisfies global requirements. The data model defines the Object Event as the main data-sharing EPC object in the network. Then, the Management Event contains the internal information required to maintain the functionality on the network. The Object Event is serialized in the router node and it is communicated to the rest of the WSN until reaching the coordinator.

The steps of the data in the WSN are shown in Figure 47 and enumerate as (Figure 47):

1. The end node keeps the ID information in the local memory while its sensors take data from the environment.
2. The router-reader node receives the information through the RF connection from the end node (RFID).
3. The router node prepares the Object Event, which is also shown in Figure 47. The Object Event is a software structure that includes an initial version of the EPC event; but it can include more information than the identification. When the reader router node establishes a communication with an end node, the end node sends an initial object that includes the value of the variables of the sensors, the timestamp of the identification and the EPC identification. After that, the reader router node with the information from the end node composes the object event, also adding information about itself, like its location, the ID of the reader, its business process or even other sensing information.
4. The Object Event is transmitted through the network until reaching the coordinator node.
5. The Coordinator receives the Object Event and it sends the data to the existing stakeholders. When the package reaches the coordinator it defines a XML file and transmits the data to the EPC repository, which publishes the information to the interested elements.

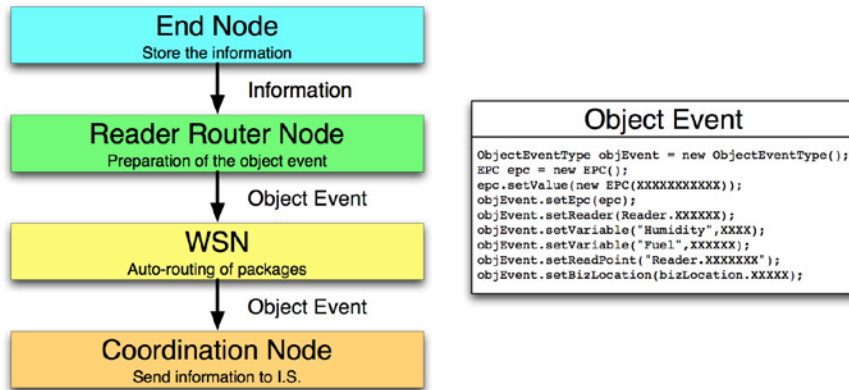


Figure 47 The information flow based on the Object Event

The other type of object defined in the data model is a management package that contains status information about the end nodes and router-reader nodes. Information likes the state of the batteries, problems with the antennas, routing problems, breakdowns on the devices or even Zigbee exceptions (Figure 48). Many of these exceptions are caught by specific functions of the Zigbee stack. With a proper definition of these events, the coordinator can manage different situations, generating alarms or providing specific instructions. These events are constantly checking the “alive time” of each router node in order to examine the WSN.

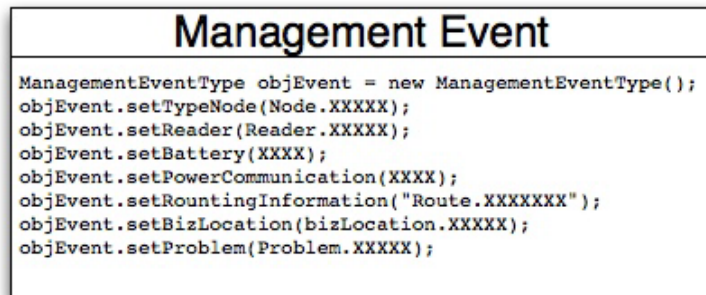


Figure 48 The management event

Once the communication/identification technologies and data modes are defined, the next step is to deploy the WSN in the airport terminal. The design of the terminal is based on a hub with four aircraft parking positions divided in front of the gates (Figure 49); close to these main parking areas, there is a secondary bigger area for long-term parking.

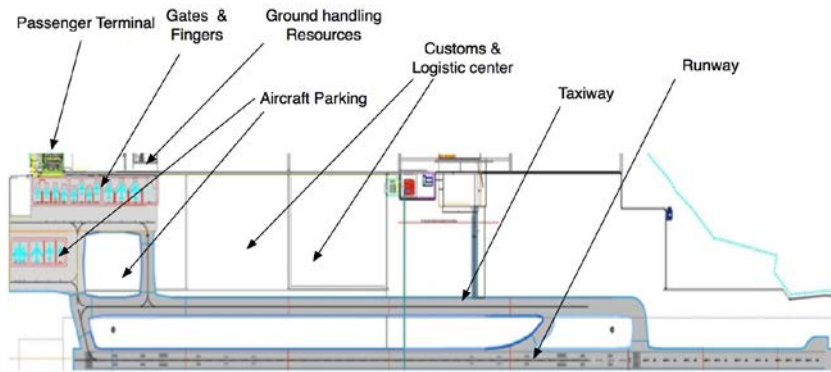


Figure 49 The Ciudad Real Central Airport

The topology of the WSN that was designed for the airport is shown in Figure 50 (Figure 50). The end nodes were attached to the mobile ground handling resources; while router-reader nodes were situated at the doors parking position and in the hangars for GH resources. In this solution, the distance between routers was always less than 400 meters; furthermore, alternative paths have been defined for all routes. The coordinator node is placed in the computer that runs the EPC interface at the management area inside the passenger terminal.

The information regarding the position of the router node on the terminal was determined and programmed beforehand. Each router node has the information regarding its position and the state of the end nodes; busy, free or in transit.

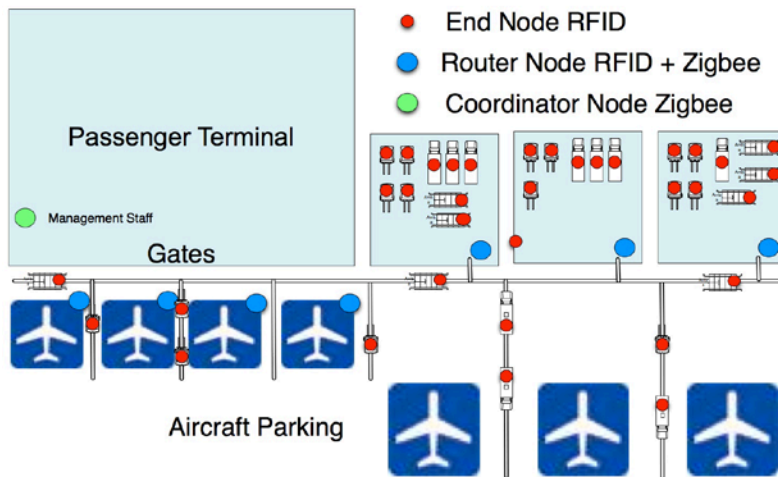


Figure 50 WSN on Ciudad Real Central Airport

The use of the WSN allows the management staff to share data between the involved air transport agents in a transparent way. This airport needs to share information between ground handling companies, airlines, airport suppliers and air traffic control in real time. The feedback of the airport staff was very positive indeed, as they considered that with a brief look at the interface they could get a well-formed picture of the state of all the ground-handling resources at the airport. The development of the WSN was satisfactory in several ways; the existing RFID and Zigbee protocol stacks are standard enough to easily abstract the development of the WSN. Therefore, the work is not as focused in the development of the hardware as well as in its deployment and the definition of the data model for final requirements.

4.1.2 MAS-DUO deployment

At this point, the airport staff wants to include the software system to improve the short-term decision-making in real time and to permanently feed the business expert system with the data coming from WSN. Once

the WSN is deployed and the information is accessible to the MAS-DUO the tests were dispatched. The launched tests, during the demo show, were based on the scheduling of a standard workday in the airport. The tasks of the BDI product agents, which represent flights, corresponds with the definition of processes by the Ciudad Real Central Airport in the attendance of a Boeing B737 in a normal 45 minutes scale. Besides, the probability of successful attending was approached with the information of other Spanish airports like Malaga Airport. The next table shows what is the information that initializes the BDI product agent; in this case, the agent represents a flight (Table 3).

Table 3 B737 Transfer features

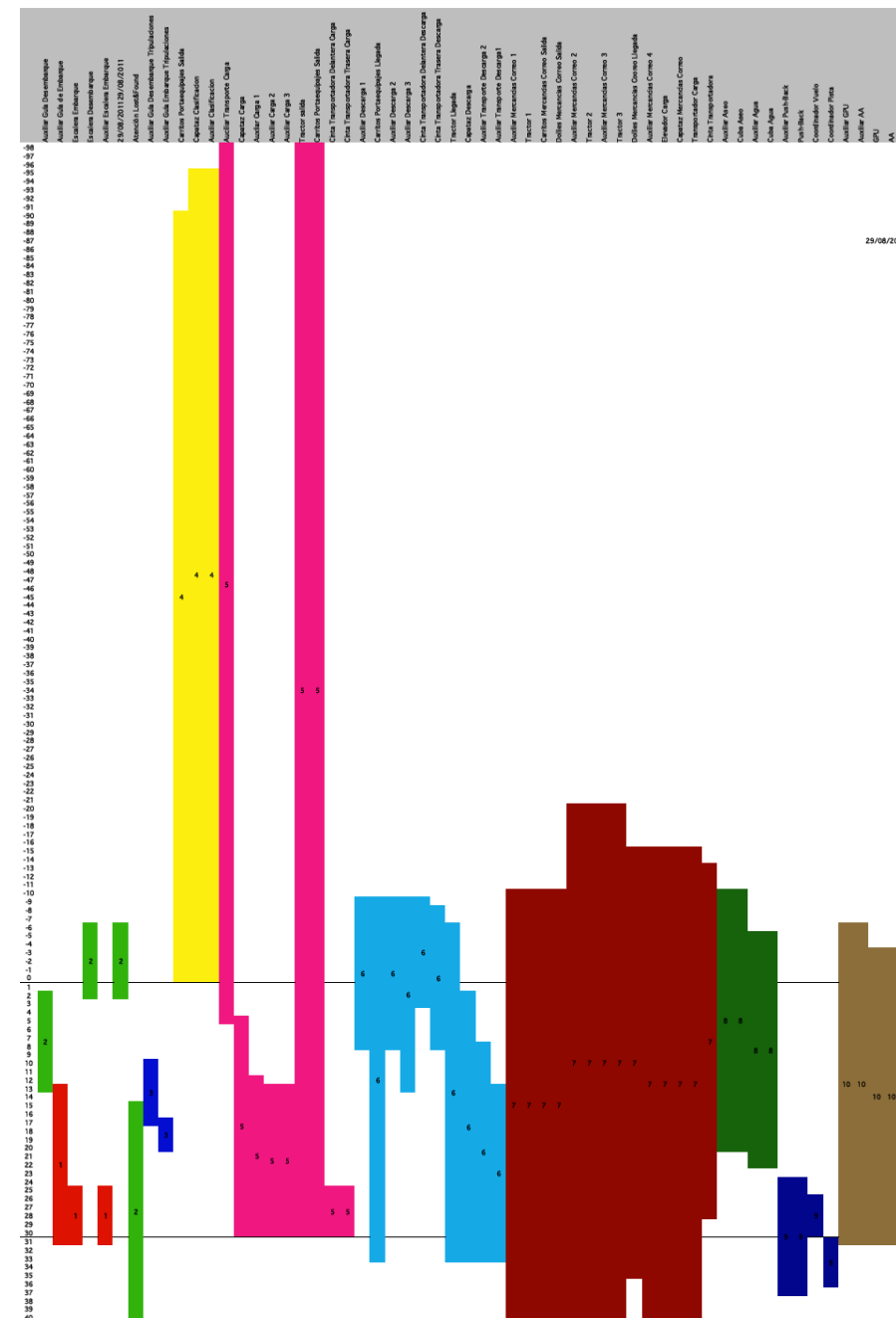
<i>Flight</i>	
Aircraft	B737-800
Transfer time	30 min or 35 min hub
Passenger	189
% Passengers	85 % (160 passengers)
Passenger Ratios	loading – 23 passengers / min (7 min) unloading– 16 passengers / min (10 min)
Baggage/Passenger	1
Baggage distribution	Front – 69 Rear – 91
Kind of hold	Not automatized
Baggage in transfer	20%
Facilities Aircraft	Integrated Stairs
Extra	
Airline	Iberia

Once the features of a common B737 transfer are in the product agent, there is a set of business steps that represent the states of the flight during the transfer. In the case at hand, International Air Transport Association (IATA) defines the states, which are:

1. Organisation and Management System (OMS).

2. Station Management System (STM).
3. Load Control (LOD).
4. Passenger Handling (PAX).
5. Baggage Handling (BAG).
6. Aircraft Handling and Loading (HDL).
7. Aircraft Ground Movement (AGM).
8. Cargo and Mail Handling (CGM).

Then, the next figure shows the business steps by colours, and every single line represents the resources that are need in a time scale, these are the BDI resources agents (Figure 51). In the test cases, the number of the business steps was 10 because of the specific needs of the Ciudad Real Central Airport, not only the 8 steps defined by IATA. Then, this chart also details the time spent by the resources in the execution of the instructions. Getting more details of the next figure (Figure 51); in vertical there is the time scale, in which there are two horizontal lines: the Actual Time of Arrival (ATA) and the Scheduled time of departure (STD). In horizontal, there are the required resources during the transfer, which are using the same colours in the same states. Therefore, if the airport initially has 4 reader points, one per parking area, and the B373 transfers have 10 business steps; the number of possible states per flight is 40. One example state could be: the Iberia Boeing 737 in the state PAX at the parking area 1.



list enumerates the possible type of BDI resource agents and the possible states that are based on the business locations and in the reader points of the GH resources in the terminal:

- The types of BDI resources are based on the roles of the GH resources, each of them with its specific functionality. Some examples are the stairs, the chocks, the auxiliaries or the supervisors.
- The detected business locations per GH resources are: busy, in transit, free and not available. Then, there are 5 reader points on the terminal, which are: hangar, park1, park2, park3 and park4. Therefore, the BDI resource agents have 20 states during the demo show at the airport. The next equation details the states of Physical BDI resources, this states cover the online representation of the GH resources at the Ciudad Real Central Airport (Equation 11).

Equation 11 The resource states in the Ciudad Real Central Airport

$ReadPoint(p)$

$= \{Hangar, Parking1, Parking2, Parking3, Parking4\}$

$BusinessStep(p) = \{Free, Busy, inTransit, notAvailable\}$

$States(p) = \{RP \times BS\}$

$$States(p) = \begin{bmatrix} S_{00} & \cdots & S_{0n} \\ \vdots & \ddots & \vdots \\ S_{m0} & \cdots & S_{mn} \end{bmatrix}$$

Once the states are defined, the reward transitions need to be defined in every transition in order to provide the reasoning able to calculate cost, quality of service or energy consumed in each transition. At this point, the airport management staff fixed the Markov parameters of the reward function during operations and the reward matrixes of the BDI agent; the focus was to improve the delays, the direct costs and briefly the energy consumed. There is not QoS represented on the reward function because initially the airport management staff does not want to give preference to any specific airline as the air traffic management. The reward function is represented in the next equation (Equation 12). At the IS BDI agent platform, three agents made up agent platform: the ERP agent, the CRM agent and an expert system. These agents validate the results of the flight during the transfers; in case of non-validation they propose a new policy, which by default was (A=0.5, B=0.4, D=0.1).

Equation 12 The reward function of the Ciudad Real Central Airport

$$R(s, s') = 0.5 \text{ Time}(s, s') + 0.4 \text{ Cost}(s, s') + 0.1 \text{ Energy}(s, s')$$

With the states and rewards defined, the next step was the preparation of the environment before the real implementation with the RFID readers. In order to achieve a dynamic policy definition, a demo of the application based on the installations of the airport was carried out to calculate the decision. To begin with, the airport staff wanted to know the state of the GH resources and the next instruction of each resource, especially if the resources will be were free or busy. The resources are considered free when they are in the hangars for ground handling resources or “in Transit” when they are on the way. This also provides other interesting information such as:

- Administrative information: the system can control working hours, productivity, statistics of performance, logs, identification of errors, etc.
- Security information: the WSN can generate alarms when the resources are delayed or when there are elements in security zones or in dangerous places such as taxiways or in the path of planes.

The obtained decisions are presented in user interfaces that can run/show the decisions in different ways as well as simulate decisions. This visualization and simulation tool helps the airport management to identify possible conflict situations. The next figure shows the interface of the application during the demo carried out for the airport management staff (Figure 52). This visual software tool has been developed to manage the ground handling resources that assist the incoming flight on the terminal as a supervisor tool. On the left, the interface shows the time information and the free resources that are in the hangar. The application has a player with a time bar; which can be forwarded or rewound to get the history of the information caught by the EPCIS or the next instructions. On the right, there are four pictures with the incoming aircrafts and all the resources that are attending these flights. The resources that have to attend the flight surround the aircraft on a Graphical User Interface (GUI), it means that the resources that are shown in this interface are around the aircraft or they have to be in the future if the supervisor forwards the time. This GUI works like an alarm to the supervisor, which calls the missing resource to support the process. The feedback of the airport staff was very positive indeed. They considered that with a brief look at the interface they could get a well-

formed picture of the state of all the ground handling resources at the airport. This software was tested by the airport staff, who did not take long getting familiar with the application and were very positive in their response to this tool.

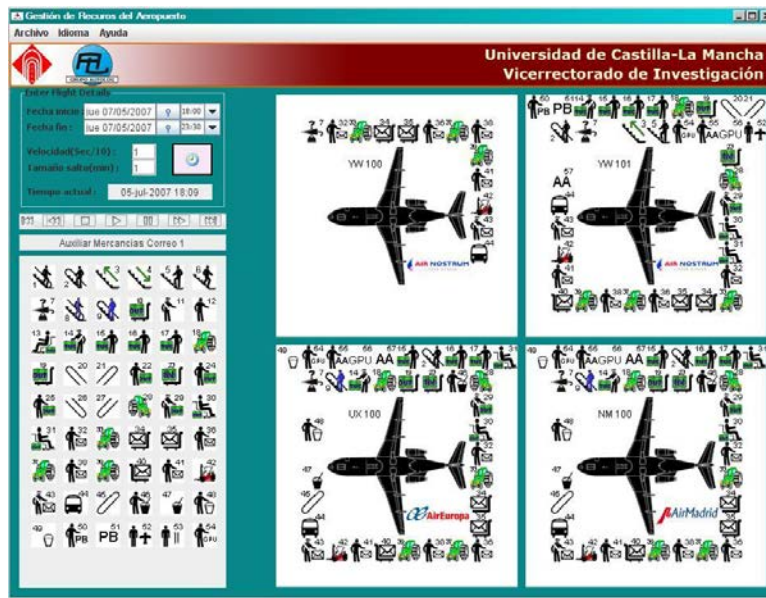


Figure 52 The Ground Handling GUI

The airport GH operations were used as a test-bench to validate the design and development of the MAS-DUO. The system helps the scheduling duties even when disturbances occur, filling the void of classical scheduling tools that relate more to medium and long term planning.

4.1.3 Airport GH Productivity Results

The results are satisfactory compared with another initial dimension realized by a major consultancy in the Ciudad Real Central Airport. This consultancy presents a FIFO schedule, in which the definition of attending teams of GH resources does not allow to divide them into

atomic functional units that attend other flights when the resource is available. The main reason to define GH teams is to simplify the management and the control. Besides, this planning does not take into account any preference or optimization, the team is assigned only depending on the expected arrival time. The comments about our proposal by the airport staff was that this DSS really takes in account the physical details of the terminal, and it allows for focusing an accurate dimension of the resources based on the real needs of the airport while a strategic point of view is also represented in the decisions. The next figure shows the Gantt results during the simulation process by using a GUI representation of the physical agent platform, which obtains the decision policy of every resource during operations (Figure 53).

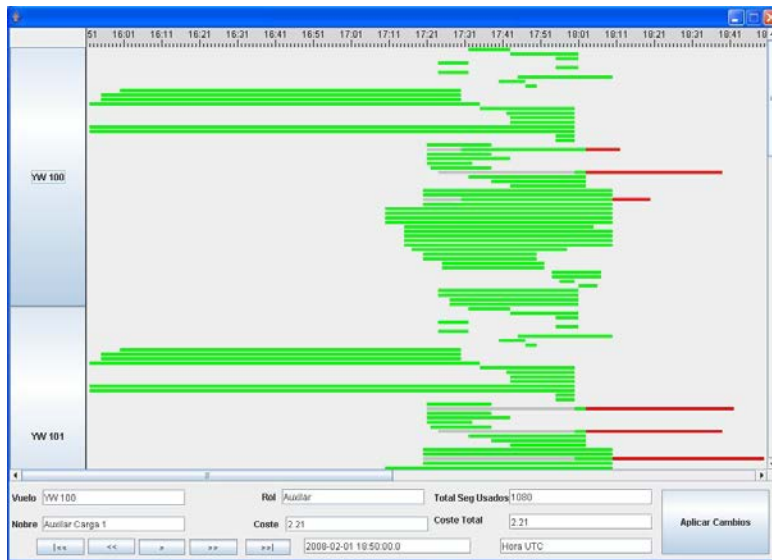


Figure 53 The results in a static policy during the demo show

In this figure, the box on the right shows the flight and every line details the spent time of every GH resource by the time bar during the transfer of a flight. When the colour of the bar is green the resources are working on time, with the red colour the resources are delayed with the expected

time and with the bar in the grey colour the resources are waiting for other processes.

The results only include the material resources and machinery; the human resources were not incorporated during the demo show because the human dimensions were pre-established by air transport organization. Initially, the airport staff defines the next global policy parameter for the dimension $\{A=0,5, B=0,4, D=0,1\}$, it supposes a great dependence on the delays and costs in the decisions. Then, there were launched several simulations with different frequency of flights, from only one flight per hour to six flights per hour. The results are shown in the next figure divided by the frequency of flights per hour, it details the number of resources needed in each case in the horizontal bars (Figure 54).

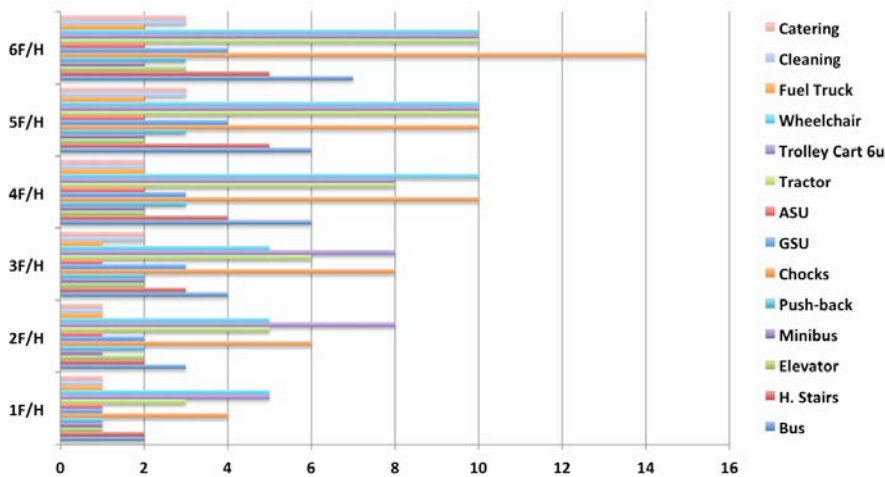


Figure 54 The dimension of resources in a just on time assignment by the MAS

Only looking at the specific situation of four flights per hour, the next chart is only based in this layout since this is the expected frequency of flights in the airport layout (Figure 55). The new results show the outcomes but modifying the policy in order to analyse the results when

the global policy parameters change. In the case at hand, the parameter A was modified with several values, which is the parameter that quantifies the delay in the global policy. The results detail the requirements of resources with several time constraints, which were classified into 4 types, which represents several expected timings in the overall attendant process, these are: 15 minutes of delay, 5 minutes of delay, just on time and 15 minutes of guarantee. These results are also compared with the proposed FIFO decision-maker introduced by the external consultancy.

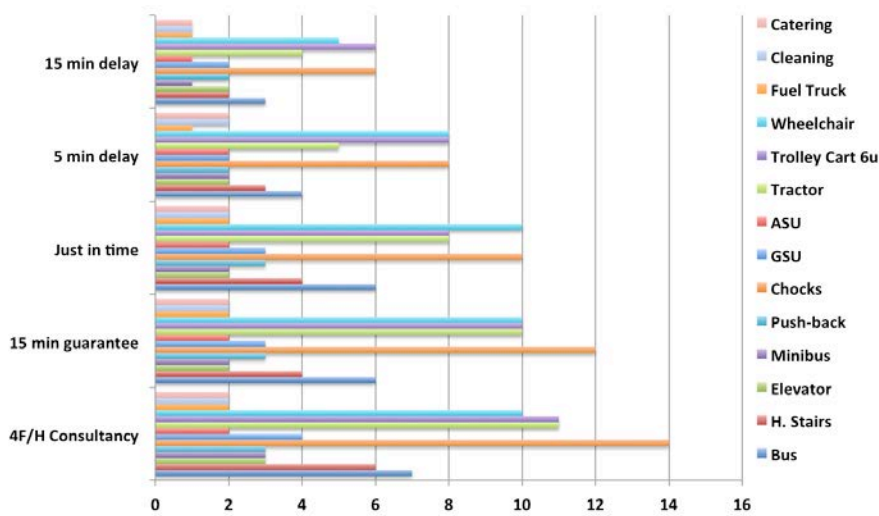


Figure 55 The results by applying different policies

Before the real deployment of the dynamic decision policy, this static policy definition helps the airport management to realize an accurate dimension of the resources required to attend the incoming flights with several configurations by modifying the global policy parameters. Therefore, in order to obtain the improvement percentage of the proposed system against the FIFO, the next table shows the reduction per resource in two cases and the comparison with the FIFO decision-maker (Table 4). The first column indicates the name of the resources,

the second and the third indicate the reduction of resources compared with the FIFO and the fourth and the fifth are the same but detailed in percentage. The analysed cases were the “just on time” and the “five minutes of delay” configurations by modifying the weight of the global parameter that represents the delay. MAS-DUO improves the use of resources in average about 15.80% in a “just on time” attending, and if the airport staff modify the policy playing with five minutes delays, the reduction can achieve 31.25% in front of the FIFO configuration.

Table 4 The reduction of resources compared with a FIFO

<i>Resources</i>	<i>Reduction of Resources Just on Time</i>	<i>Reduction of Resources with 5 min delay</i>	<i>% Reduction of Resources Just on Time</i>	<i>% Reduction of Resources with 5 min delay</i>
Bus	1	3	14.29	42.86
H. Stairs	2	3	33.33	50.00
Elevator	1	1	33.33	33.33
Minibus	1	1	33.33	33.33
Push-back	0	1	0.00	33.33
Chocks	4	6	28.57	42.86
GSU	1	2	25.00	50.00
ASU	0	0	0.00	0.00
Tractor	3	6	27.27	54.55
Trolley Cart 6u	3	3	27.27	27.27
Wheelchair	0	2	0.00	20.00
Fuel Truck	0	1	0.00	50.00
Cleaning	0	0	0.00	0.00
Catering	0	0	0.00	0.00
Average (%)	1.14	2.07	15.89	31.25

These improvements are only getting the static policy as a dimension process in a non-stochastic environment, after that if a dynamic policy is achieved with the EPC information any disturbance during operations will be taken in account. The next results should include the impact of

applying real time field information to the policy instead of only a static policy. However, the impact of reducing the waste of time because of the visibility frameworks needs to be analysed in each case and configuration. The waste time is the time spent in order to identify delays and disturbances from the expected schedules. The feedbacks of the airport staff about adding visibility points were clear; the system will generate the best-fitted plan depending on the disturbances and the global policy parameters will be reflected in the decisions at the shop floor.

4.2 The logistics centers scenario: the Autolog Labs test-bench

The distribution centers operate in a highly competitive environment, which forces them to constantly improve in terms of quality, response, agility and flexibility. In the current economic situation, it is crucial for logistics enterprises to stay in business; moreover, the customer calls for customized products and traceability processes as basic product attributes; products without a manufacturing history are becoming virtually worthless. The complexity becomes especially hard to tackle when dealing with typically big distribution centers where the usual control is too rigid and hard to manage, maintain and update.

The role of this test scenario instead of having pure productivity results as the previous scenario, it studies the improvements and drawbacks in the deployment of the MAS-DUO in a logistic center. This test case analyses the integration of the hardware, the development problems and the specific constraints that comprehend the deployment of MAS-DUO in a logistic center. Specifically, the work that is highlighted here is a flexible way to integrate machinery controlled by PLC and RFID information with this agent-based proposal. This leads to the problem of ensuring an

efficient and flexible access to the relevant information for all parts in logistics centers. In order to be able to experiment agent-based DSS in a manufacturing environment similar to the real facilities, despite being in a laboratory, an experimental platform is necessary to serve as a test bench to introduce ideas and carry out quick tests. This experimental platform has been set up at the Autolog Lab, which is situated at the University of Castilla La Mancha in Ciudad Real. The platform simulates a logistics center made up by a combination of a physical part and a simulated part (Figure 56). The simulated part represents the processes of storage and distribution, while the physical part represents the processes of loading and unloading pallets, which take place in the docks. Previous similar scenarios were based on similar manufacturing test benches as the done by Soylemezoglu et al and Vrba et al (Soylemezoglu, 2005) (Vrba, 2005). A relevant example was done at the Auto-Id Labs in Cambridge, which presents a test bench to experiment with RFID and decision making processes (Harrison, 2004).

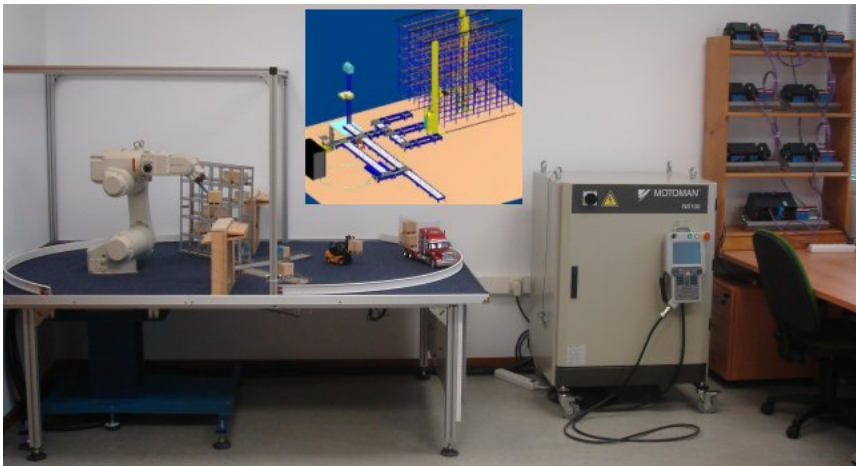


Figure 56 The experimental Platform

Specifically, the figure 57 shows a schema of the physical platform with all its parts (Figure 57). The sixth PLC SIMATIC S7 200 and the PLC MASTER S7 300 controls the model docks; it represents the area for input and output of products where the RFID tagged pallets arrive. In this physical part the human intervention is required at some operations; thus introducing the possibility of error as happens in real facilities. There are two conveyor belts at the docks for incoming and outgoing products respectively. An RFID reader has been installed on the input conveyor to identify incoming pallets. Once the pallet has been identified, the management system transfers to the robot arm, a Motoman HP3, the position where the pallet has to be stored in the physical shelf. As from that point, the pallet appears at the 3D simulation of the warehouse.

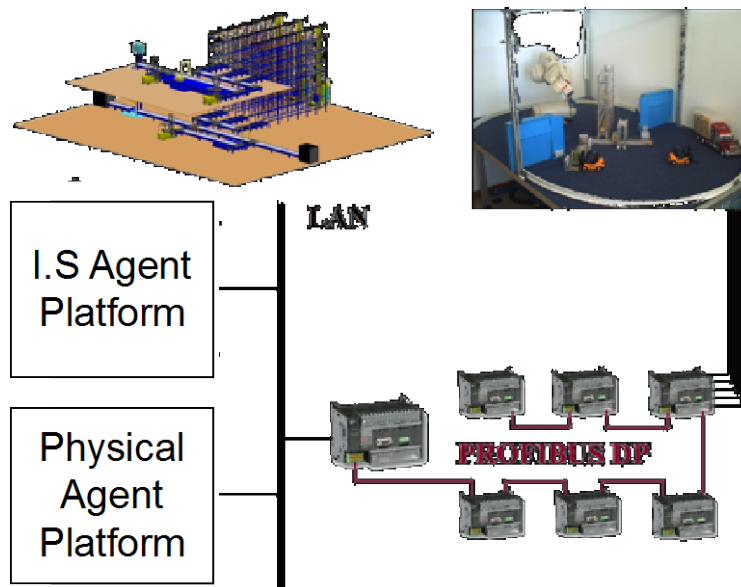


Figure 57 MAS-DUO on the platform

The physical platform, represented in the miniature model, should be understood as the tangible part of the distribution centre, which represents the loading and unloading docks of the distribution centre,

incorporation RFID. Beside the control test bench, the Autolog hardware team also test the RFID hardware as antennas, tags or aerial protocols. Some details of the hardware components that could be interesting to explain before the integration with DSS are: the RFID readers situated on the conveyor belt by the entrance and exit of the products, just before the exit door, make the checking of the loads in the lorry possible. This RFID equipment is connected to the local EPC repository through an Internet connection. A variety of readers has been tested with this platform, such as the Caen C48 and the Skytech M9. A robot takes charge of managing the miniature pallets that enter the warehouse. It is worth noting that there is no direct relation between the storage of pallets on the physical miniature platform and the simulated model, since the existence of a model lies in the complexity of setting up and running the systems. The rack, where the pallets are stored, has 36 positions with replica pallets of 1:20 scale. Then, some miniature machinery such as conveyor belts, automatic doors, forklifts, R/C forklift trucks, a control panel, formed by actuators, light indicators, buttons and a security system, which assist the operation of the platform.

Then, for the 3D modelling and a simulation of the distribution centre, the Grasp10 software from the company BYG Systems has been used. This program is specifically suitable for the simulation of robots and conveyor belts. This simulation has been programmed to accurately reflect a real distribution centre (Figure 58). Grasp10 allows communication with other programs or computers, which are connected to a network via sockets. Thanks to this, the simulation works according to the orders that are sent from the outside: guidelines from the programmable PLCs using the programmed Java interface to share the instructions between both. The instructions needed by the simulation in

order to work are limited to decision points: conveyor belt junctions, inputs from production, reception and formed pallets, location in the warehouse and priority in the elaboration of orders; without forgetting the movement of the conveyor belts.

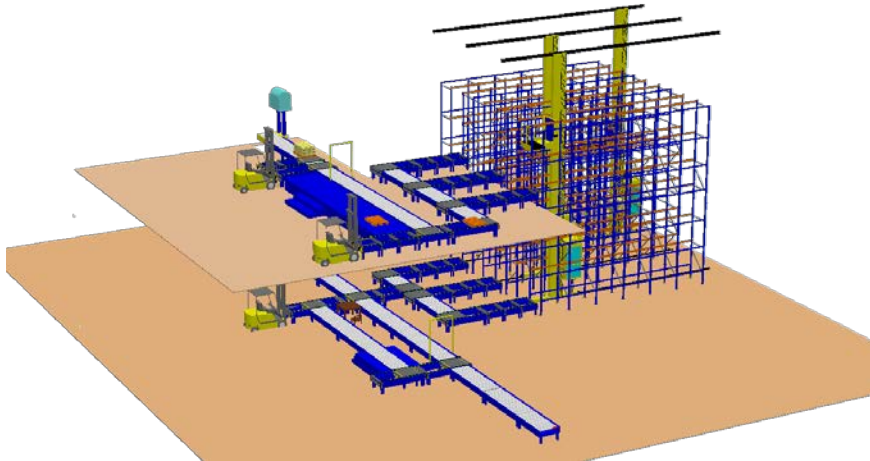


Figure 58 The Grasp 10 Simulation Platform

The simulated plant has three corridors served by three automatic cranes (Figure 59). The system is situated between the factory where production takes place and the docks. The distribution center is divided in two levels: a ground level by the docks for inputs to the warehouse and an upstairs level for outputs and picking operations. The pallets arrive in the warehouse from the docks through “reception” and from the “production” area. Once the pallet is situated over the main input transfer system, it moves over a line of roller tables to an Inputs Identification Point (IIP 1), through a profile gate. If any problem is detected in the identification of the pallet or in its shape, it is sent to “rejection”, where problems are manually addressed. If everything is correct when the pallet is at IIP 2, then it is sent to the shelves by chain tables and the corresponding crane when an output is requested, the pallets are taken

out of the shelves by the cranes that leave them at the chain tables of the upper level. From there, the pallets move to the “elevator”, where they can be transferred to “output” at dock level. If a pallet is needed for “picking”, this operation can be performed when it stops in front of the picking station on the output line of roller tables.

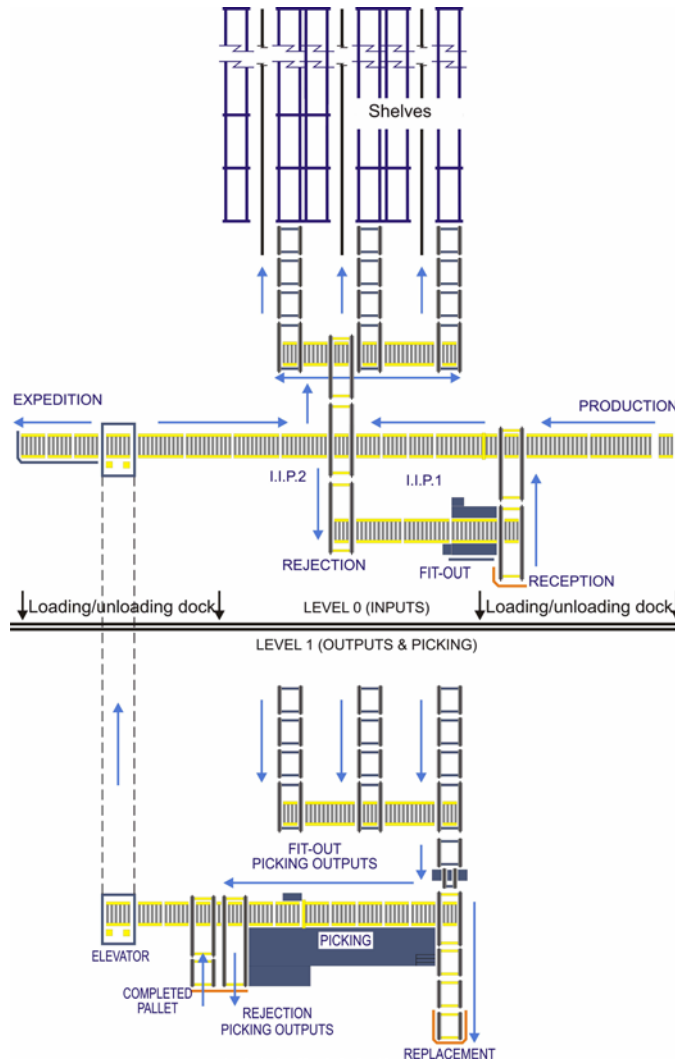


Figure 59 The layout of the simulated logistics center

The implementation of the low-level control system of the experimental platform has been carried out in various parts. These parts have been duly

coordinated. It can be said that the control system divides its work into two branches: firstly, the series of controls and mechanisms that directly interfere with the model (physical or simulated); and secondly an interface with the MAS-DUO, which is detailed in the next subsection. In the low level, the PLC network is one of the series of Siemens SIMATIC s7 PLCs connected through a PROFIBUS-DP network. This test case network consists of 7 PLCs: a SIMATIC s7-300, which is the master device of the network and 6 SIMATIC s7-200s working as slaves in the network. The PLC master takes charge of connecting between the PLC network and the LAN network in which the MAS is deployed. The master is managing the information coming from the management system and sending it to each of the slave PLCs, according to which it corresponds. The PLC 1 carries out the control on the ground floor of the distribution centre and the entrance and the path of the pallets until they enter the warehouse. The PLC 2 controls the first floor of the distribution centre; it manages and guides the pallets to the exits of the warehouse, even during the picking and departure processes from the distribution centre. PLCs 3, 4 and 5 take in charge of each of the three trans-elevators in the warehouse. Finally, the PLC 6 controls the physical miniature model, its entrance and exit of the distribution in perfect coordination with the simulated model. After the control with PLCs is detailed, the next subsection details how to integrate it with the agent-based system.

4.2.1 Connecting MAS with PLC network devices

After the introduction of the simulated logistics center, this section details the interface that has been designed to communicate the Siemens S7 PLC series and the Physical BDI agents through the PLC module. In next developments, the IEC 61499 architecture will be useful to directly

support this interface instead of use the proprietary software of the PLC as the proposed solution to test MAS-DUO. The IEC 61499 is a set of open architectures for distributed and embedded control and automation to develop and deploy distributed systems control systems but currently is not implemented in many of the existing hardware (Lewis 2001). This standard tries to address the next list of requirements for PLCs:

- Portability: the software tools can accept and interpret PLC software components and system configurations produced by other software tools.
- Interoperability: the embedded devices can operate together to perform the functions needed for distributed applications.
- Configurability: the software tools from multiple vendors can configure any device and its software components.

The Siemens S7 model of PLCs offers the option to directly access to the data via the S7-API beans of a Java application. The beans are encapsulated Java component with functions that can be published, in this case this beans can access to the addressable memory of the PLC by using the S7variable classes. These variables can include subscription methods and request about what is the state of specific data blocks or variables in the PLC. Therefore, a Java interface based on this library is implemented in the PLC module of the Physical BDI agent to support this communication (Figure 60). The objective is to be able to send and receive instructions from the Physical BDI agent to its represented physical element through the PLC and vice versa. The physical control maintains the operation and the independence; it is only a way to share variables, not direct instructions. The PLC must only work by using its

function blocks but selected by the instructions of the agent, in other case the results during operations are non-robustness and out of control situations can appear. The current agent technology is not as robust as to be able to directly control machinery as a PLC, a dispatcher program in the PLC is essential to translate the instructions coming from the agent.

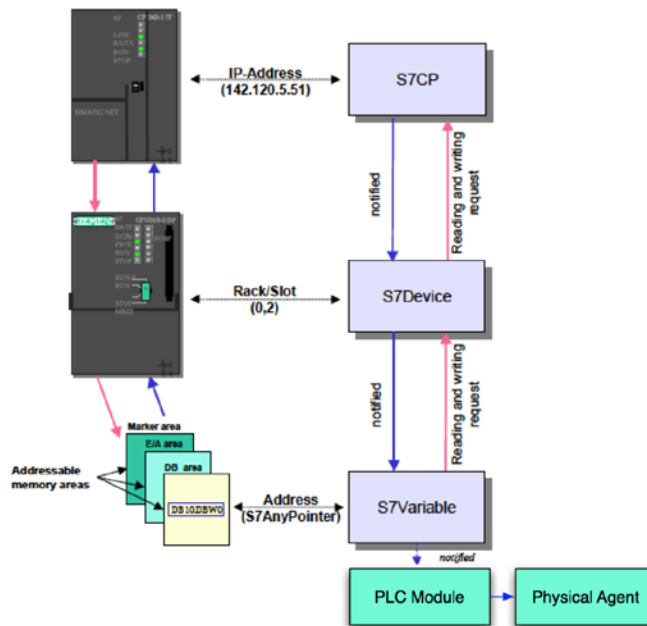
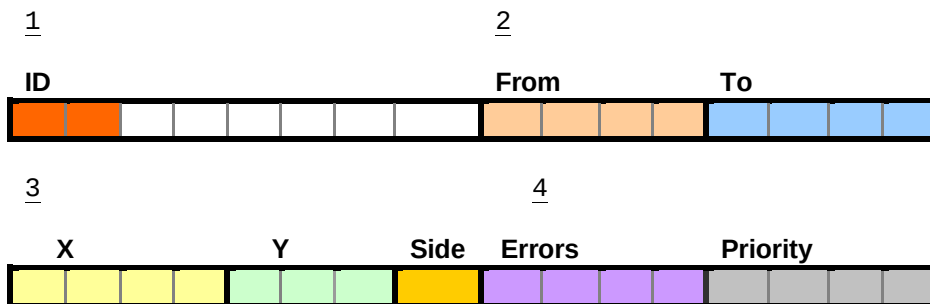


Figure 60 The PLC with S7 Java Beans. Siemens Copyrights

This way of work needs to program the PLC by using divided function blocks and checking these shared variables. The PLC has a series of instructions implemented and divided into specific tasks as function blocks. These function blocks suppose distinct operations that have to be carried out such as: moving products along conveyor belts until reaching a determined part of the warehouse, the control of the pallets when reaching meeting points on the conveyor belt or the movement of the pallets until a cross. The automated activity at the distribution centre can be carried out through simple instructions that will carry out the

machinery of the distribution centre, according to the definition of the function blocks.

Therefore, to preserve the independence of the PLC during operations, the only instruction that the agent can order to the platform control is the moving of pallets, the instruction is not as deep as to control every variable or the specific movement of the shop floor elements. With this way of work, the integration is possible by using common information about the localization of the products but in different control levels. This is similar to the ideas of the EPCIS, in which several business models share data by using a common access model. It is not a pure agentification process where every functional resource has an agent avatar connected to the PLC but is enough to detect the product states with the information coming from the PLC. The PLC sends and receives the information about the new instructions thanks to the information packet sent by the PLC module of the agent. Therefore, a previous common definition of data is required; this data has parameters such as the identification, the origin and the destination of the pallet. These instructions have a determined structure as shown in the previous figure, in which the necessary data appears coded in an efficient way (Figure 61).



Zones Level 0	
	0000
Production	0001
Rejection	0010
Reception	0011
Picking 1	0100
Picking	0101
Picking Rejection	0110
Expedition	0111
Warehouse	10xx

Figure 61 The PLC Message and position codes

In the case at hand, a 4-byte binary code is used, which incorporates the identification of the pallet within the control system, the origin and destination of the pallet, a series of flags equipped for possible errors, which could take place during the fulfilment of the order and a reserved space to define the priority of each order. Specifically, the package contains the next fields:

- The EPC of the product is mapped in the ID field. The relation between the ID and the EPC is stored in the Physical BDI Agent (ID). If the complete EPC is in the message the size of the packet will be very complex to be controlled by the PLC network.
- The initial localization (From) defines the current position of the pallet. The codifications of the localizations are detailed in the table of the figure 61 (Figure 61).
- The target localization of the instruction (To) corresponds with the goal position of the pallet.

- Then, more details about the localization are the position in the warehouse or the specific picking zone (X, Y, and Side).
- The priority assigned by using the decision parameter by the global policy (Priority).

The system works with a series of these instructions managed by a dispatcher program in the PLC Master but previously generated by the agent-based system. This helps the MAS-DUO to carry out its role but being effectively independent of the physical layouts and configurations. It will be decisive when it comes to certain flexible situations where the plant layout is in change. In the case at hand, the interface is in charge of the communication between the Physical BDI agents and the PLC network. The PLC Module of the BDI agent is in charge of translating the intentions as instructions and sends them to the dispatcher program of the PLC, and vice-versa. The PLC module of the Physical BDI agent prepares a package as the represented in the figure 62 by using the state transitions calculated during the reasoning process (Figure 62).

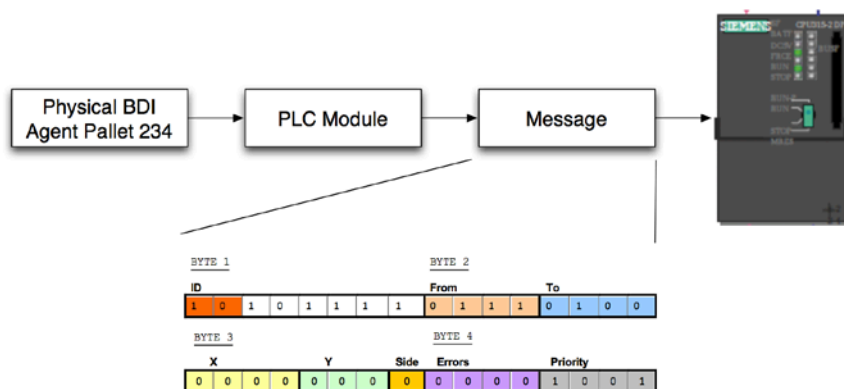


Figure 62 An example of PLC instruction

In the PLC Master, the dispatcher is in charge of collecting these data packets (instructions) that come from the Physical BDI agents and sending them to the slave PLC that it corresponds. In this way, the master device controls and manages the out-coming traffic of instructions through the PLC network. On top of this, if any part that forms the system fails or there is an error, only the affected part would stop working, while the rest of the system would continue working with the received instructions. This is possible due to the automatic distributed control system between masters and slaves. Although there appear to be many advantages, it is worth mentioning the complexity of the coordination and the consistent implementation of the system, given the amount of parts of which it is made up.

Once the interface is defined with the PLC network, the next step is the definition of the states of the Physical BDI agents. In the case at hand, the pallets and the zones are the Physical BDI agents, in which the pallets request for these zones. This is similar with the airport ground-handling test, the flights are the pallets and the resources are the zones. The next figure shows the agentification in the level 0 of the logistics center; the green boxes represent the BDI zones as resource agents, which equipment of the zone that can be busy or free to be able to attend the incoming pallet (Figure 63). The most common example is storing in the warehouse, the way of work consists of: the BDI product/pallet agent request for a warehouse place, this warehouse agent responds with the reward about its attending features, after that the product chose its best option. This is similar to the way of work in the ground handling test scenario. Then, the blue circles are the decision points taken by the PLCs, in which the Physical BDI agents do not directly participate; the Physical

BDI agent sends the instruction and the PLC network executes the function blocks, which define the priority and the destiny of each pallet.

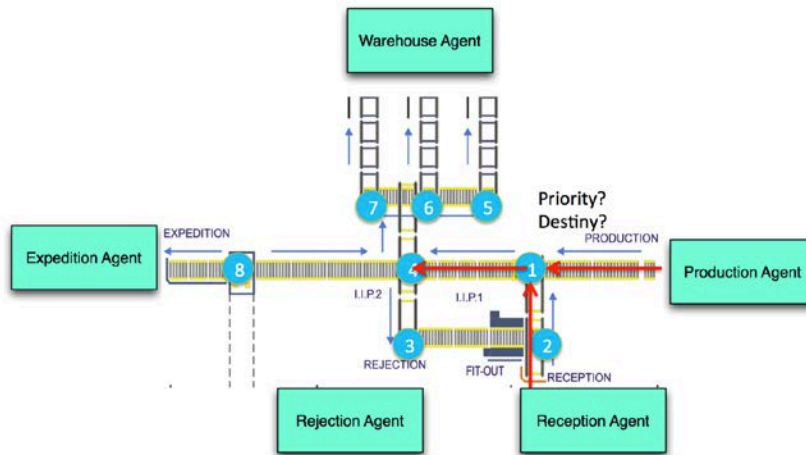


Figure 63 The agentification in level 0

In the case at hand, the agentification comprehends the level 0 and the reader locations and business steps that define the the states of the products/pallets are:

- The reader locations in the level 0 of the simulation are: the production, the reception, the rejection, the warehouse and the expedition. All of them are positions of the logistics center in which a RFID reader has been located.
- The business steps of the pallets are assigned, free, picked and not available. The pallet can be assigned to an order as busy or assigned, not assigned to any order as free or composed by several products during the picking process as picked or even not available if there are problems with the pallet.

Therefore, with the 5 readers locations and 4 business steps, the pallets have 20 states at the level 0 of the simulated logistic center. After that, the state transitions define the flow of pallets in this level; the most common instructions of pallets are movements from the production and the expedition to the warehouse. The next equation shows the most common instructions for pallets (Equation 13); the task 1 represents the incoming of pallets from production to the warehouse and the task 2 details the expedition of pallets from the warehouse.

Equation 13 The incoming pallet instructions

$$Task1(p) = \{Production_{free} Warehouse_{free}\}$$

$$Task2(p) = \{Warehouse_{free} Expedition_{assigned}\}$$

During the demo shows of the MAS-DUO in the simulated logistic center, the IS platform included the ERP as the ERP agent.



Figure 64 ERP Hermes

Olimposoft developed the “Hermes” ERP adapted to the requirements of the MAS-DUO; Olimposoft is a local company that developed this ERP for WMS in Autolog Labs. The previous figure shows the initial GUI of the system (Figure 64).

4.2.2 Application example

This section details a daily operation at the logistics center and the generated information flow between elements as an example to explain the way of work. When a new pallet arrives, it enters the system via an RFID reading, controlled by its agent, so a new element is automatically added to the distribution centre as a new Physical BDI agent. The EPC informs the beliefs of the agent that the pallet is the reception and a new state is achieved. With this information, the IS BDI platform produces a determined movement in the warehouse as a tactical instruction; a command is generated with the necessary instructions in order to carry out the operation. Then, the IS BDI platform generates the instruction to the corresponding BDI product agent (pallet agent). The Physical BDI agent generate new intentions by using the reasoning process and the communication protocols, these intentions are sent to the PLC module implemented in Java, which is in charge of translating the order to the format of the PLC control system.

The data packages reaches the PLC Master, which then sends it to the slave PLC to which each case corresponds. Once it reaches the PLC in charge of carrying out the control of the operation, it arranges the information in its corresponding location on the internal map (in its memory) of the warehouse and proceeds with its execution. For example, an order is generated because a pallet has entered the entrance of the physical platform to be stored. After reaching the PLC in charge of

directing it towards the warehouse, this order passes each of the positions of memory, which represent the real route of the pallet. When the pallet is moving along a belt and reaches a junction, the PLC is responsible for reading, on the corresponding data of the pallet in question, its destination, so that the PLC orders the pallet to advance in one direction or another. The variety of situations that a pallet can find itself in and in which action must be taken is rather diverse. Examples of these are simple crossings, double crossings, height restrictions, and the capacity of a variation of routes according to parameters. For example, if a pallet that is going to enter the warehouse fails to meet the dimension requirements; it is redirected to a preparation area where the pallet error can be corrected and its Physical BDI agent is informed.

Once the pallet reaches the destination defined by intention of the agent, or the instruction has been executed, the instruction is returned to the PLC master and this is returned to the PLC module of the Physical BDI agent, which will return it to the specific agent, which updates its beliefs. Therefore, the agent can plan and work whilst taking into account that this operation has been carried out until the result is sent. On top of this, given that the orders have a reserved space for errors, the management system is informed in case an error occurs during the operation. If the order requires the participation of more than one PLC to carry it out, when a determined slave PLC finishes its task, the order is returned to the PLC master, which in turn, redirects it to the next slave PLC, until the action is executed. The case of the warehouse is clear to denote the effectiveness of the Physical BDI agents. In order to optimize the elevator movements the incoming and outgoing of pallets has to be intermixed. At this point, the reward of the Physical BDI agents in the elevator increases when there is a sequence of input and output orders.

This situation is transparent to the IS platform but the optimization is guaranteed as the Physical BDI agents improve the reward during the plan.

The description of what has been developed in the laboratories of the AUTOLOG group has been outlined. The installations and workings of a controlled distribution centre have been described. A PLC network controls this test bench where the PLCs are responsible for the low-level instructions of the plant. However, this PLC network has dynamic interfaces with the agents through the implementation of the EPC module in order to share information when new events occur; this is a great contribution to the current systems and allows defining uncoupled control strategies. Furthermore, with this test bench it is possible to represent different logistics processes as picking while testing new control ideas as MAS.

4.3 Discussion

The productivity results have been obtained from the airport GH scenario because it is a pure resource scheduling in shop floor operations; therefore, it is easier to get valuable data as times, delays or costs. In the distribution center scenario, the option was to analyse the impact of including specific PLC industrial integration software as a part of the visibility system, apart from RFID. From a management point of view, the next list enumerates the most important results:

- A 15% reduction in the number of shop floor resources during operations. A current way of work in the shop floor is to make teams as units that cannot be divided to assist products in order to simplify the management; it supposes that free resources inside

a team cannot be reassigned to other processes. The MAS-DUO allows dividing these teams in functional units because it simplifies the control.

- The definition of a decision policy to every resource, which is obtained through the reward function. It means that any resource knows what to do in every situation.
- The modification of the decision policy by the higher levels of the company is represented in every operation decisions.
- The independence between platforms, which in most of the cases are even maintained by different companies, is achieved because they only share policy parameters, simple instructions and results. There is not coupling between physical layouts and information systems.
- The fast re-planning in case of deviations from the original schedule. The identification of deviations with RFID is automatic, which means that the errors and out of control situations are quickly treated.
- This can be a first or initial approach to a service-oriented decision-making where the companies can compete in a shared business environment where the contracts do not need to be previously defined.

The results are satisfactory in both scenarios: airport ground handling and logistics centers. The appendix 1 details the productivity results in the airport GH; however, the results are especially positive if the human

resources are involved in the shop floor decision process, not only the machinery and equipment. The schedule of only material resources in small scenarios does not produce income in the first three years; the project needs almost five to be profitable. When equipment, operators and machinery are included, the financial results are positive within three years even in small shop floor environments

Looking at the specific scenarios, the management staff remarks on the decrease of the wasting time of resources and the constant control of the situation, in which any modification from the initial plan is located, even though the disturbances only affect in the future, is a competitive advantage. However, two main problems arose during the development and deployment in the test case:

- The need of expert software engineers with a background in MAS and agent frameworks as JADE or Cougaar. It is specially denoted in the budget, in which this kind of expert developers increases the financial plan. The deployment of MAS is based on a learning process spent by the developer team in order to quickly adopt these requirements into sometimes not well-defined standards, which force to increase the time and the effort. This could be one of the main reasons why the development of agent-based technologies in industry seems to be far. However, similar concepts such as the Web services, which are really adopted by the industry, can include similar functionality but reducing costs in a initial step before the full application of the agent-based solutions.
- The second problem is the integration of the MAS-DUO with the shop floor equipment; the development of the specific

software that feeds information from the PLCs or industrial elements depends on each manufacturer. The integration with the RFID visibility frameworks is already provided but when the information feedback is directly connected to a specific machine the problems arise. This development requires specific integration depending of the manufacturer. In this work, an example with the Siemens Simatic S7 series is detailed. The programming of the PLC is divided into tasks as available actions in function blocks, but a communication with the agent using the specific interface defined by the company needs to be addressed. The PLC receives the message as triggers to launch its specific instructions and vice versa while the agent discovers the state of the instruction through with a specific API. Many of these problems will be fixed with fully interface communication based on standards such as the IEC 61499 architecture.

There is another secondary problem but only existing in the initial deployment, this is the definition of initial knowledge required for every resource. This knowledge consists in the fulfilment of the reward transition matrixes per every shop floor resource. However, although the initial fulfilment is a hard work the reinforcement learning techniques help this process during operations.

4.4 Key notes

This chapter details the deployment of the MAS-DUO in two real scenarios based on the models proposed in the methodology. Each scenario has specific features that need to be addressed individually; however the agent model represents a common abstraction of the problems. The development of the agent model includes the agent

classification and the identification of the agent states, which suppose an abstraction of the problems to the MDP states that define the tasks and instructions of each product/resource. Once the agent model is defined, which is successful in both cases, the application of MAS-DUO in the test scenarios increases the productivity, defines the action policy to the ground handling resources, aligns the shop floor actions with the upper information systems and produces fast re-schedules in case of disturbances.

Then, specific conclusions arise in each scenario. In the case of the airport, this ground-handling scenario provides a scheduling problem where constant disturbances arise during operations. The productivity results are satisfactory and the airport terminal is a perfect test bench to deploy a WSN to track the shop floor resources. In the logistics center scenario, apart from the schedule system, it also introduces the Machine-to-Machine (M2M) problem. At this point, the MAS-DUO needs to include flexible communication with machinery through specific equipment such as PLC. This is an overrun in the deployment and specific interfaces need to be addressed. The customization of the MAS-DUO system to each problem supposes the major drawback; in both scenarios, the necessity of expert programmers to include specific interfaces in the agent-based facilities or to deploy a WSN in the shop floor is essential.

5 CONTRIBUTIONS, CONCLUSIONS AND PERSPECTIVES

The implementation of a framework as the proposed introduces the automatic communication of the BDI agents with the visibility frameworks that can be reused in new developments in order to reduce the impact of the software costs. This proposal can be a suitable first step in the search of a business as usual integration of the RFID with artificial intelligent techniques, allowing direct decision-making. In this work, a new MAS architecture based on a division of the organization model, following the real workflows on the companies is introduced as MAS-DUO. The physical and information system agents are treated separately, facilitating both the understanding of the system, the design and the development. This separation makes it possible to reflect the strategic policies on the physical decisions at the same time as the upper information system receives results about the physical disturbances. The results during the test case were satisfactory; there is an increase in productivity and the scalability of the system is demonstrated. This chapter examines if the proposed architecture and the argumentation of its division in two agent platforms achieves proper results. It also explains

the benefit of applying BDI customization in physical agents to integrate the visibility frameworks into their beliefs and the improvements of an interaction protocol based on sharing the Markov reward parameters as global policy parameters.

The first finding is that the system can work in two configurations and both are useful and complementary for shop floor companies. The first one, which is called static policy decision process, do not include visibility framework, only the MAS. The static policy decision process allows simulating long processes of the shop floor schedule in order to generate dimension of resources with specific conditions; it helps to identify possible out of control situations and to obtain the decision policy of every shop floor element at every moment. All these features allow defining long planning on the shop floor. Also, the dynamic policy configuration can add control in case of the disturbances during operations; this case is much focused in short planning and daily operations. This second decision process level requires a visibility framework to feed information about disturbances; the main example in visibility framework is the EPC Global standard. This also was analysed as a good sequential approach to a complete product-driven decision process, in which at first, there is a static policy phase but during operations there can be added new visibility points where a lack of visibility is detected. Therefore, the sequential phases of a MAS-DUO configuration allow to dynamically add new visibility points (Figure 65).

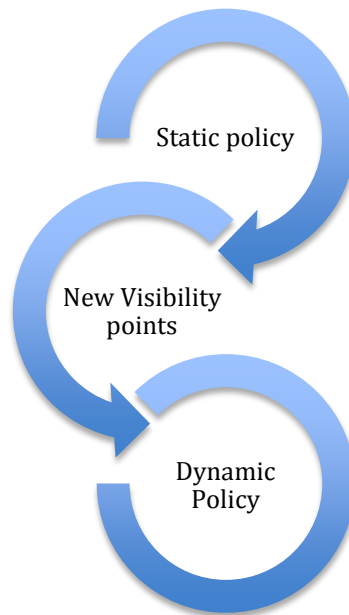


Figure 65 The MAS-DUO configurations

Apart from the sequential implementation of the system, the MAS-DUO have a great scalability. The results have been obtained in small test scenarios or even in simulated installations in order to reduce cost during the test of the system, which can be extrapolated to higher and more complex problems. The scalability of the MAS systems allow them to be easily deployed in bigger systems because the interaction protocols are the same and the computational requirements are not high. Therefore, in bigger deployments, the MAS-DUO maintains the way of work and the operational costs but it resolves higher complexity during operations.

The next sections present the contributions together with the feedbacks from the involved companies based on the analysis of the results and the conclusion obtained during the development of this thesis.

5.1 Contributions

Contribution 1. Division of the agent space. The main contribution of this work is the division of the agent space in two layers in order to allow the definition of the Physical BDI agent, which is a proposal of the intelligent product vision. This product-driven decision process generates accurate decisions by using the information coming from the visibility frameworks and the own knowledge of the agent; therefore, the main objective is achieved through the MAS-DUO (Figure 66). Besides, the global policy parameters calculated by complex strategic systems align these decisions in a product-driven negotiation.

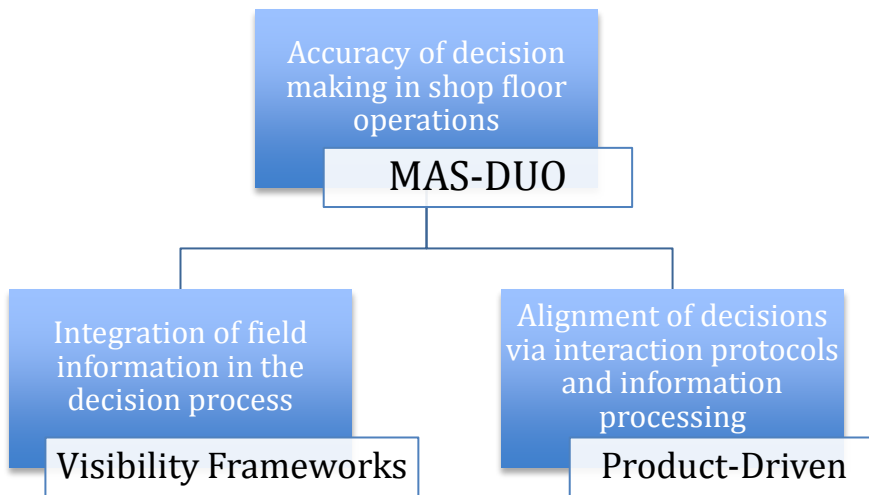


Figure 66 The initial objectives and the contributions

RFID provides the basis that supports the rest of the decision-making done by the agent-based and Markov features. At this point, the accuracy in the visibility is essential because it allows to define precise planning. The time, in which RFID detects any disturbance, is almost online and the MAS directly controls and plans the new schedule. The wasted time is

minor and the productivity directly increases depending on the number of resources, the disturbances or the complex of the system. Once the system has confirmed that every resource is controlled, the decision-making system can include new techniques as EPCIS to share this information and to abstract upper levels about physical layouts. The integration of the field information in the decision process has been detailed with RFID but further specific developments are required with the PLC information feedbacks (Figure 65); the deployment of standards as IEC 61499 is normal in companies.

Contribution 2. Integration of the MDP in the shop floor decisions.

This filed information allows reflecting the physical states of the shop floor elements into Markov states and the instructions in state transitions, which simplify the implementation of a shop floor control and maintain the independence between physical actions and strategic decisions. The Markov framework and the specific agent interaction protocols define an environment able to support online negotiation, which supposes the second sub-objective (Figure 66). In this distributed decision-making, the ERP does not centralize the control in every single step at the shop floor operations, which was a problem detected in the analysed scenarios. Instead, the division of the decision in two layers divides the responsibility and the functionality but maintains a global point of view. The information coming from visibility frameworks through the EPC definition provides advantages to the following agent-based decision processes, these benefits are:

- The identification of the product states, which correspond with the Markov states, that comes from the business step and from the physical location of the product in the EPC events.

- The capacity of the Physical BDI agents to generate new focused beliefs in the search of the goal state, which is useful to the reward function in order to calculate the rewards and prioritize the instructions.
- The selection of the best instruction depending on the beliefs and the global policy parameters defined during the state transitions by using the Markov mathematical framework.
- There is no need of a supervisor agent or similar control elements; the global policy maintains the correct alignment during the distributed decision.

In MAS-DUO, the uncoupling between the physical elements and the information system is clear. The physical changes and the layouts are not taken in account by the strategic levels; this allows achieving the proposed flexibility claimed in the shop floor planning. The information flows are:

- The orders and global policy parameters from the information system platform reach the physical platform.
- The analysed results of applying these instructions by the physical agents are sent to the information system platform.
- The confirmations of the execution or the redefinitions of new policy parameters from the information system platform reach the physical platform.

Contribution 3. The basis of a dynamic and efficient outsourcing.

Other contribution could be the idea of a fully distributed control where

the companies can compete against others in order to sell their services. The shop floor provider companies can publish their facilities and the respective buyer agents have the available information to always choose the most suitable resource. These strategies allow participating in the business to SMEs that cannot compete during the definition of contracts as full service providers. In this case, the contracts do not need to be prefixed; the companies can include their own specific services during operations depending on the situation. MAS-DUO can define a share space where companies compete by publishing their services by using the agent interfaces. This supposes a win-to-win situation where the companies compete and the supplier selects the best fitting service or product in every situation in a “dynamic outsourcing”. For example, this idea is helpful in airport ground handling where prefixed contracts do not initially allow small companies to participate.

Contribution 4. Need of expert programmers. The main problem discovered during the development was the needed of expert programmers in the agent-based topics and PLC integration. It is hard to find programmers in this new area, in which there is not background in the current software development companies. Looking at the current control and decision software in the industry, the introduction of MAS is slow and the impact of the MAS in the industry is brief. Perhaps it is because the next steps of the software paradigms are moving to the Web services in the cloud-computing, which are similar to the agent-based ideas but with Web facilities and fast integrations.

5.2 Conclusions

The basis of this thesis is the division of the control problem into two layers, which allow defining service interfaces between two independent

expert systems. The proposal can satisfy the current shop floor requirements and provide successful deployments; it has been called MAS-DUO. This division allows the physical platform to define a product-driven scheduling based on a proposal of intelligent product, which better fits with short term planning requirements. In addition, an upper platform, called information system platform, provides a decision-making that supports more complex and strategy decisions by using reasoning processes such as the market simulations. Then, a service-oriented interface between platforms achieves maximum cohesion and minimum coupling, which supposes flexible advantages. The implementation/technical benefits of applying this division in a new product scheduling definition are:

- The uncoupling between the physical details of the shop floor and the high level of the decision-making make possible to get the maximum cohesion between the operational levels and the strategic levels of the SCM. This supposes a flexible benefit during operations, in which different companies can play during the decision-making by publishing their own services in a dynamic outsourcing.
- The stand-alone operations of the shop floor elements are achieved in case of errors or breakdowns; the independence of each platform is not affected in MAS-DUO. Both platforms maintain the independence and decision capacity but an alignment decision based on global policy parameters is highlighted.
- The service-oriented definition allows achieving flexibility in the definition of tasks, states and even in visibility points. In the

proposal, this redefinition is business as usual and does not suppose high costs or modifications. The automatic identification of the intelligent product states allows to dynamically add new visibility.

- The existing information systems in the company are maintained by developing agent interfaces, which only supposes a service interface between the existing information systems and the MAS-DUO.

Apart from the technical benefits, there are other advantages in the management point of view. The obtained gains are:

- The operation responsibility is clearly split in two: the providers of hardware in the plant and the providers of the software. Besides, most of the information system elements can be preserved, which is important in complex and large operations where a full control migration can be dramatic.
- The definition of policies that define the product and resource actions in every situation is addressed, which defines a complete plan for every shop floor element during operations.
- The constant control that arises in case of disturbances, in which the new schedule reflects the modification because the MAS with the RFID feedbacks automatically does the calculation of the future assignments and collisions.
- The ROI is very positive in big industries or when the products have a high value; however, the results are also effective in small scenarios when the system also manages human resources.

- This way of work also provides new market opportunities even for SMEs, the provider contracts do not need to be closed before the operations. During operations, the providers publish their offers and the company selects the best one.

Nevertheless, even with all these features, there is the lack of results of MAS in the current industry, which is mainly based upon the lack of software background in agent-based system and visibility frameworks. Currently, the companies are not ready and it supposes more costs, but new interface technologies such as the Web Services, which are usual in the industry, can offer the basis to achieve an intermediate step in the search of a complete distribution decision process in which companies can offer their processes as services. Resteasy is one of these Web service technologies and there are interesting studies that are working in this area as the Web of Things. The other way to solve this lack of knowledge in the agent-based by the companies is to provide existing frameworks to the developments that simplify their work during the development. This is the case of the MAS-DUO, the proposed solution tries to offer a framework that reduces this impact during the development.

At least, this agent-based solution for distributed decision-making supposes great advances during shop floor planning but the usual implantation of these ideas seems to be far from the industry. A middle step base on Web service looks a proper way in the search of a complete agent-based system in manufacturing/logistics control.

5.3 Perspectives

This section also includes the future issues that could be trend topic in the agent-based manufacturing/shop floor control. Several lines of

researches have been detected as future trend topics in shop floor control, most of them related with agent-based control and its integration with field information. These topics are appearing based on the emerging of new technologies that are able to reach the current requirements in shop floor operations, the detected lines in this work are:

- The inclusion of agents in the decision-making provided from different companies is only mentioned but further analysis will be useful in the search of an online outsourcing. This new concept allows defining flexible subcontractors adapted to possible not defined situations during operations.
- The definition of standard interfaces of communication between shop floor elements such as PLCs, robots or any automated equipment and the existing information systems. The Internet of things could be an interesting point of view to the factory of the future; many of these interoperability concepts could be reused. In fact, the IEC 61499 is a very good initial approach.
- The research in the partially observable Markov decision process framework, which can be useful when the states of the products/resources cannot be observable. It is interesting when there is a lack of visibility points and it is not possible to add new reader points.
- New self-diagnosis methods in out of control situation created by the agents during the decision making process; the control of deadlocks or breakdowns is essential.

- A Web adaptation of the planning MAS in order to integrate the agent-based facilities with Web interfaces, which are usual in the current industry. The current Web services do not implement agent features like director facilitator or agent management but are well adopted by the industry; these ideas will reduce the software development impact. For example, the Web adapter of the EPC Repositories that supports fast integration of the EPCIS based on Web services with other systems by using JavaScript Object Notation (JSON); it is currently in development by the Auto-ID labs (Guinard, Trifa et al. 2010).
- The definition of algorithms based on the EPC information that allow discovering critical paths and KPIs during operations.

6 APPENDIX. ROI

This appendix studies the ROI in the application of MAS-DUO in the ground handling operations at the Ciudad Real Central Airport. This study and its results can also be extrapolated to similar scenarios. In the case at hand, the ROI is calculated in the application of the MAS-DUO in GH operations, which in fact, it is a generalization of new visibility frameworks and distributed information systems technologies at the logistics operational levels. This study is based on a demo show in the airport ground-handling scenario. This ROI does not include the lost opportunity cost, the improvement in customer relationship, the productivity in auditory and the green sustainability; it is a pure productivity ROI. The next equation shows how to calculate the ROI, which is enhancing for three years (Equation 14). Therefore, the next required parameters that have to be calculated are the gain and the cost of the investment.

Equation 14 The ROI Equation

$$ROI = \frac{(Gain\ from\ Investment - Cost\ of\ Investment)}{Cost\ of\ Investment}$$

The cost of the investment in this IT projects includes several lines, in the case at hand hardware, software, end-user labour and training are scratched during the implementation and deployment. The next table shows the costs during the development, the deployment and during the annual recurring (Table 5). The total cost of the project is around €160,000.00.

Table 5 The cost of the MAS-DUO project

<i>Organization Total</i>			
	<i>One Time</i>	<i>Annual Recurring</i>	<i>3-Year Total</i>
<i>Client Hardware (WSN)</i>	€ 8,600.00	€ 1,000.00	€ 11,600.00
<i>Server Hardware</i>	€ 2,000.00	€ 250.00	€ 2,750.00
<i>Sub-Total</i>	€ 10,600.00	€ 1,250.00	€ 14,350.00
<i>Client Software</i>	€ 25,000.00	€ 5,000.00	€ 40,000.00
<i>Server Software</i>	€ 50,000.00	€ 10,000.00	€ 80,000.00
<i>Sub-Total</i>	€ 75,000.00	€ 15,000.00	€ 120,000.00
<i>One-time Implementation Labor/Services</i>	€ 3,000.00	€ 1,000.00	€ 6,000.00
<i>Annual On-Going Labor/Services</i>	€ 0,00.00	€ 1,000.00	€ 3,000.00
<i>Incremental Help Desk Calls</i>	€ 3,000.00		€ 3,000.00
<i>IT Training</i>	€ 8,000.00	€ 1,200.00	€ 11,600.00
<i>Sub-Total</i>	€ 14,000.00	€ 3,200.00	€ 23,600.00
<i>End-User Labor</i>	€ 1,500.00		€ 1,500.00
<i>End-User Training</i>	€ 1,500.00		€ 1,500.00
<i>Sub-Total</i>	€ 3,000.00	€ 0,00.00	€ 3,000.00
<i>Total</i>	€ 102,600.00	€ 19,450.00	€ 160,950.00
	<i>One Time</i>	<i>Annual Recurring</i>	<i>3-Year Total</i>
<i>Hardware</i>	\$10,600.00	\$1,250.00	€ 14,350.00
<i>Software</i>	\$75,000.00	\$15,000.00	€ 120,000.00
<i>IT Labor, Services, & Training</i>	\$14,000.00	\$3,200.00	€ 23,600.00
<i>End-User Labor & Training</i>	\$3,000.00		€ 3,000.00
<i>Total</i>	\$102,600.00	\$19,450.00	€ 160,950.00

Especially denoted is the high cost of the software developments instead of the hardware and the training. The hardware has usual elements that

are manufactured and assembled by several industries that provide the solution as a complete product. Then, the employers rapidly assume the RFID technology; their way of work is almost not modified. The RFID is an awareness technology in which the use is transparent; the only training is to manage the software application by the management staff and supervisors. The shop floor elements only have to cross or pass through the specific RFID point gates.

However, looking at the cost distribution chart (Figure 67), which is the next bar chart, the software needs to assume the main part of the budget. The specific deployment of agent systems requires expert developers in the area; these experts need to integrate it with the ERP systems, feed the information about reward and deploy the communication with the RFID and the specific hardware.

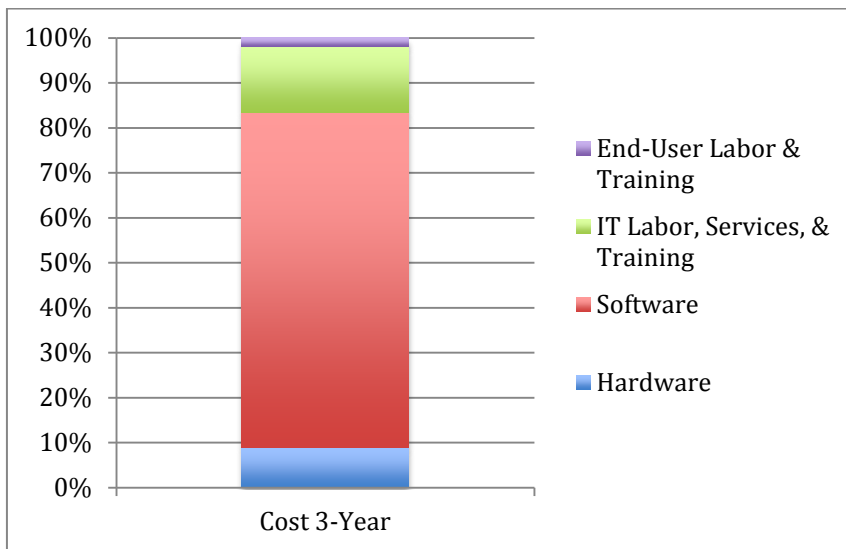


Figure 67 The 3-Year Cost Distribution

Once the cost of the investment is calculated, in the search of the ROI, the next step is to estimate the gain. The Ciudad Real Central Airport

gave the cost information during the purchase of resources. These costs were calculated with eight years depreciation in the amortization of the resources. During the test cases, there was only calculated the productivity improvement in the equipment. However, these results can be extrapolated to human resources, who were not previously included on demo show, both with a discount of the 15% looking at the obtained results. The next two tables show the cost reduction for equipment as well as for human resources, in both cases using the amortization per year in case of material resources and the year salary in the GH personal (Table 6 and Table 7).

Table 6 The cost of the GH resources

<i>Ground Handling Resources. Cost per year</i>			
	Quantity	Cost	Total
Bus	7	€ 10,000.00	€ 70,000.00
H. Stairs	6	€ 4,000.00	€ 24,000.00
Elevator	3	€ 3,000.00	€ 9,000.00
Minibus	3	€ 6,000.00	€ 18,000.00
Push-back	3	€ 6,000.00	€ 18,000.00
Chocks	14	€ 500.00	€ 7,000.00
GSU	4	€ 4,000.00	€ 16,000.00
ASU	2	€ 4,000.00	€ 8,000.00
Tractor	11	€ 1,200.00	€ 13,200.00
Trolley Cart 6u	11	€ 400.00	€ 4,400.00
Wheelchair	10	€ 300.00	€ 3,000.00
Fuel Truck	2	€ 11,000.00	€ 22,000.00
Cleaning	2	€ 2,000.00	€ 4,000.00
Catering	2	€ 2,000.00	€ 4,000.00
	Total		€ 220,600.00
	Gain (15%)		€ 33,090.00

Table 7 The cost of the GH human resources

<i>GH Human resources, Cost per year</i>			
	Quantity	Cost	Total
Ramp Foreman	4	€ 48,300,00	€ 193,200,00
Ramp Agent	7	€ 13,000,00	€ 91,000,00
Platform Agent	7	€ 13,000,00	€ 91,000,00
Flight Coordinator	13	€ 21,000,00	€ 273,000,00
	Total		€ 648,200,00
	Gain (15%)		€ 97,230,00

Looking at the results of the cost in the project, the cost of human resources supposes almost three time bigger than the physical resources, which directly affect the ROI. Based on the calculated data, the ROI in three years is estimated in two cases:

- Only with material resources: the estimated gain in three years is € 99.270,00 and the ROI is negative within three years. With this configuration the breakeven is in 58 months.

$$ROI = \frac{(99,270,00 - 160,000)}{160.000} = -38\%$$

- Including both, equipment and human resources: the total estimated gain is € 390.960,00. In this case, the ROI is positive, the project is viable and the breakeven is in 20 months.

$$ROI = \frac{(390,960.00 - 160,000)}{160,000} = 144\%$$

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8 PUBLICATIONS AND RESEARCH RESULTS OF THIS THESIS

ISI Journals

- García Ansola P., De Las Morenas J., García Higuera A., Otamendi J., "Distributed Decision Support System for Airport Ground Handling Management by using WSN & MAS". Engineering Applications of Artificial Intelligence. Edit. Elsevier.
- De Las Morenas J., García Higuera A., García Ansola P., "Product Driven Distributed Control System for an Experimental Logistics Centre". IJICIC: International Journal of Innovative Computing, Informatics and Control. Edit. IJICIC
- García-Ansola P., García A., De Las Morenas J., García-Escribano J., Otamendi J., "ZigID: Improving visibility in industrial environments by combining WSN and RFID". Journal of Zhejiang University-SCIENCE A.

- Otamendi F.J., García A., García-Ansola P., "New business opportunities along the air tourism supply chain: The combination of identification technologies and just-in-time". The African Journal of Business Management.

Journals

- García-Ansola P., García A., Pastor J.M., Otamendi J., "Agent-based Decision Making Process in Airport Ground Handling Management". Int. Journal of Logistics Research. Edit. Springer.

Book Chapters

- Andrés García, Javier García-Escribano, Javier de las Morenas, Pablo García, "Libro Blanco Sobre la Automatización de la Trazabilidad de Productos en Fabricación y Logística 2005-2010". Edit. SETRA, Julio 2011. ISBN-13: 9788-84-615-2419-8.

Patents

- García A., García P., "*Administración Inteligente de la Red Eléctrica en el usuario final (AIRE)*", Patente de invención presentada por la UCLM en la Oficina Española de Patentes y Marcas el 25 de Mayo de 2010 con referencia ES1874.10 y número de solicitud. P201030790.

Research Stays

- Institute for Manufacturing (IFM). University of Cambridge (UK). Aero-ID Project. Supervised by Alan Thorne (Co-director of Auto-ID Cambridge).

International conferences

- García-Ansola P., García-Higuera A., De Las Morenas J., García-Escribano J., "Decision Making Platform Supported on ICTs: Application to the Management of Ground Handling Operations at an Airport", Invited Session "Industrial applications of enterprise models and integration". 18th IFAC World Congress. Milano, Italy. August 28 - September 2, 2011.
- De Las Morenas J., García-Higuera A., Encinas JC., García-Ansola P., Blanco J., "*MAS-Based Control Architecture Applied to an Automated Logistics Centre*". 13th International MITIP Conference on The Modern Information Technology in the Innovation Processes of the Industrial Enterprises. Edit. MITIP & Tapir Academic Press. Org. NTNU. Trondheim, Norway. 22-24 June 2011. ISBN: 978-82-519-2816-8.
- García P., De Las Morenas J., García A., Otamendi J., De La Fuente M., Ga-Escribano J., "*Wireless Sensor Networks and RFID for Improving Information Visibility in Airport Ground Handling Management*". IFAC-IMS, 10th Workshop on Intelligent Manufacturing Systems. 1 - 2 de julio de 2010, Lisboa (Portugal). Pp. 201-206. ISBN: 978-3-902661-77-7.
- García P., García A., Encinas J.C., De las Morenas J., "*Experimental platform for the analysis of the RFID enhanced agent based management system of a distribution center*". Proceedings of the 11th International Conference on The Modern Information Technology in the Innovation Processes of the Industrial

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- García-Escribano J., García P., García A., De la Fuente M., Pastor J.M., "*RFID System for Airport Control on a MAS-DUO Agents Platform*". 4th European Workshop on RFID Systems and Technologies. RFIDSysTech'08. Freiburg, Germany. 10-11 June, 2008. ISBN: 978-3-8007-3106-0.
- García P., García A., Pastor J.M., "*RFID in Physical Platforms of Agents: Application in Airport Management*". FLINS: 8th International Conference on Computational Intelligence in Decision and Control. Madrid, Spain. 21-24 September, 2008. Pp. 1045-1050. ISBN: 978-981-279-946-3.
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- De las Morenas J., G^a-Escribano J., García A., Encinas J.C., García P., Zangróniz R., "*Diseño y aplicación de un Lector RFID online en un centro de distribución real*". 3as Jornadas Científicas sobre RFID, Bilbao 25 a 27 de noviembre de 2009. ISBN: 978-84-613-5221-0.

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- Garcia P., Poyatos M., García A., Pastor J.M., "*RFID en sociedades de agentes físicos: Aplicación para la gestión aeroportuaria*". 2as Jornadas Científicas sobre RFID, Cuenca 5 a 7 de noviembre de 2008. ISBN: 978-84-612-7225-9.
- Sánchez J.M, García A., Pastor J.M., García P., "*Análisis y planificación de recursos aeroportuarios mediante la tecnología RFID*". 1as Jornadas Científicas sobre RFID, Ciudad Real 21 a 23 de noviembre de 2007. ISBN: 978-84-612-0542-4.